

Upgrading the Joint Neutrino Oscillation Analysis of T2K and Super-Kamiokande Experiments via Surfing in a High-dimensional Detector Uncertainty Space

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Stony Brook University

PhD Thesis Defense • Dec. 10, 2025



Stony Brook
University



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Neutrino Oscillation

1. So what is neutrino?

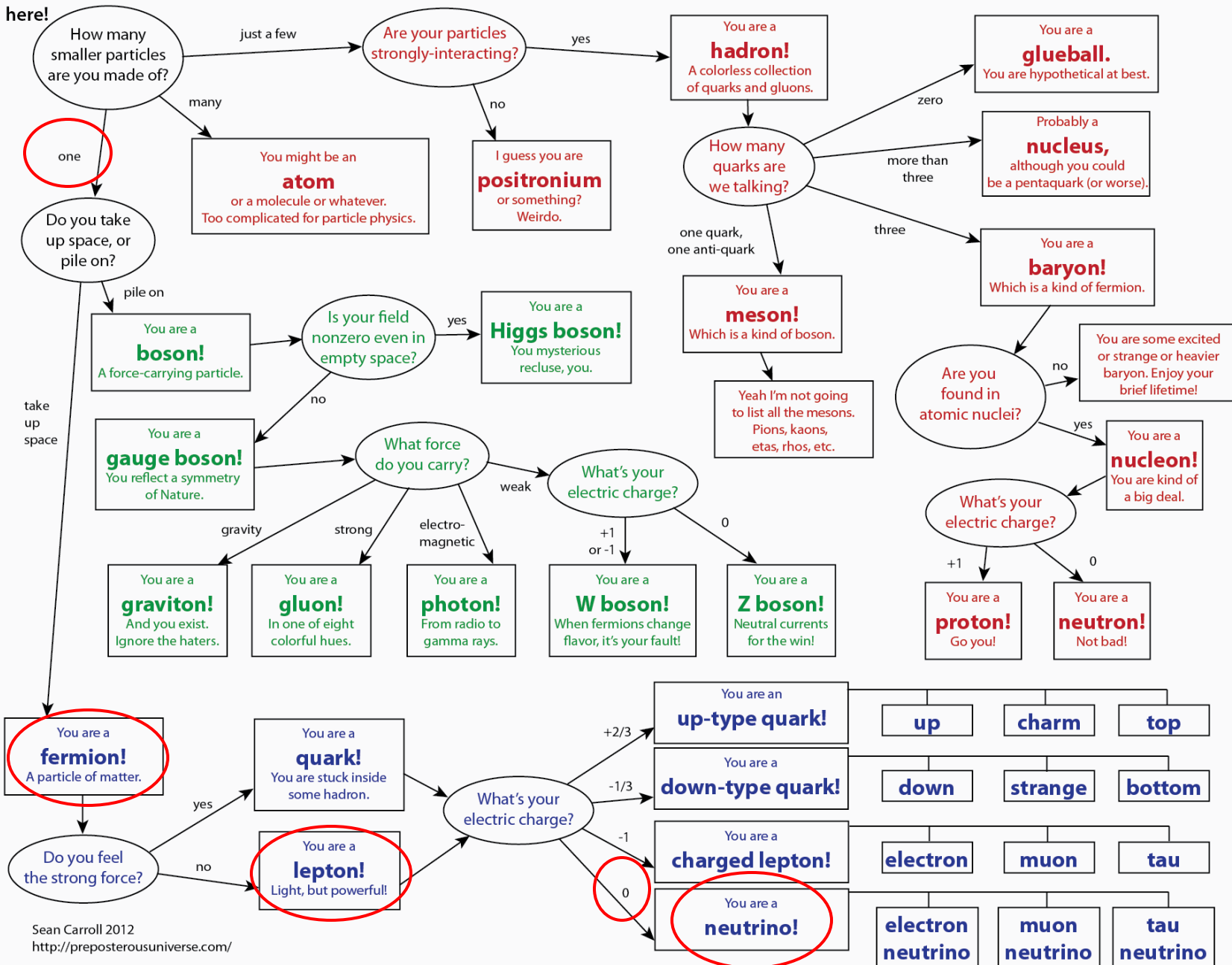
2. Why do neutrinos oscillate?

What Particle Are You?

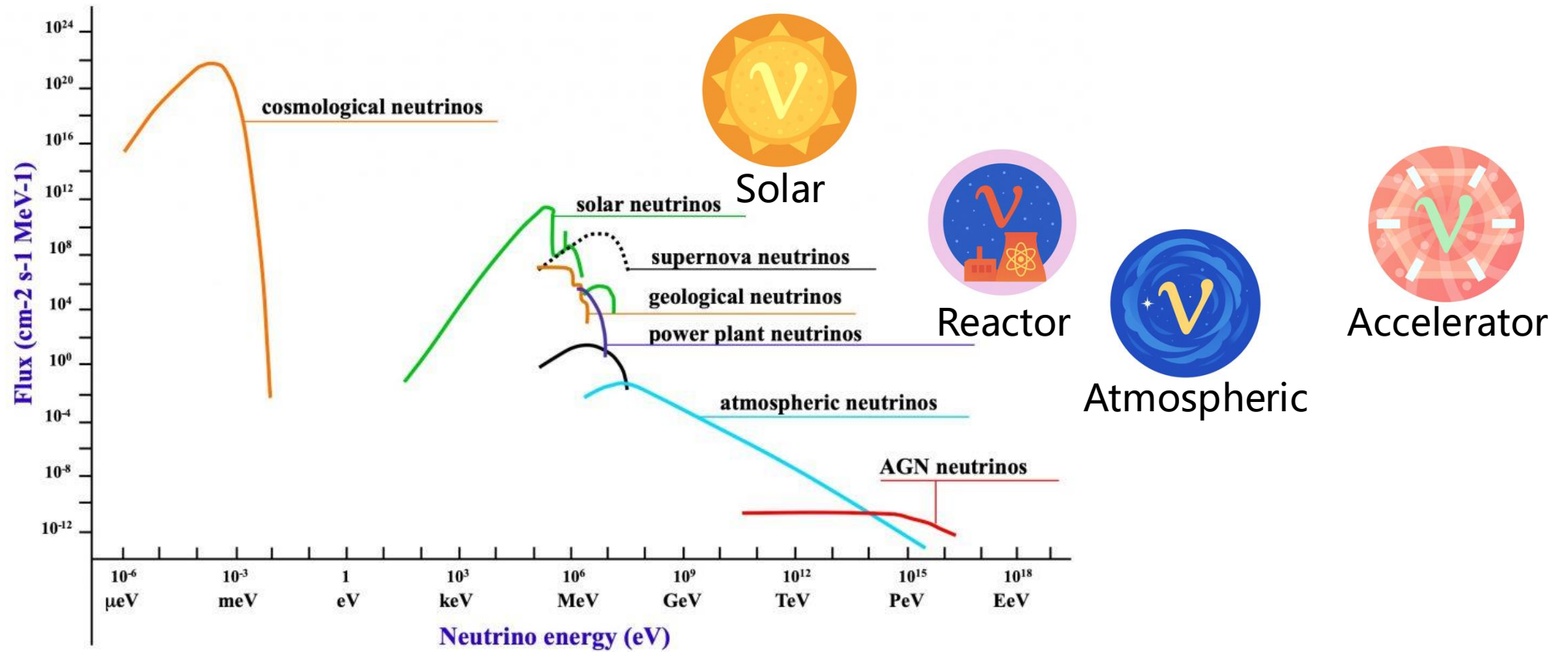
(Standard Model particles only! Dark matter and other exotica not welcome.)

Color code:
 elementary fermions
 elementary bosons
 composite particles

Start here!

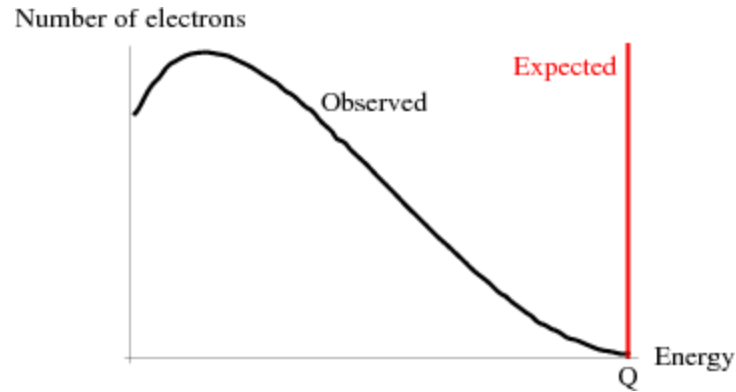


Neutrino sources

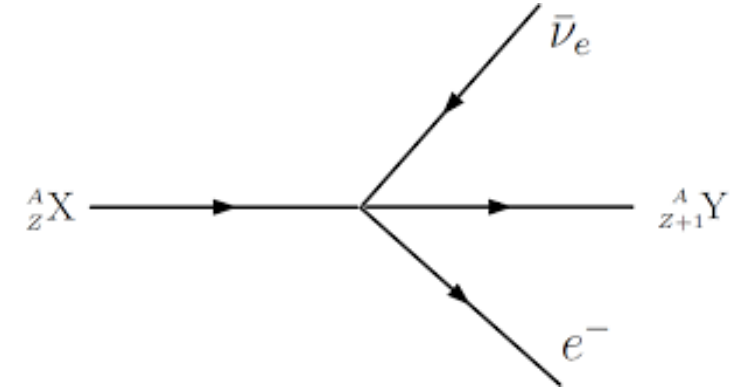


The history of neutrino

1930, Pauli



1933, Fermi



1937, Majorana



Neutrino = Antineutrino ?

$\nu = \bar{\nu}$?

1956, Reines and Cowan

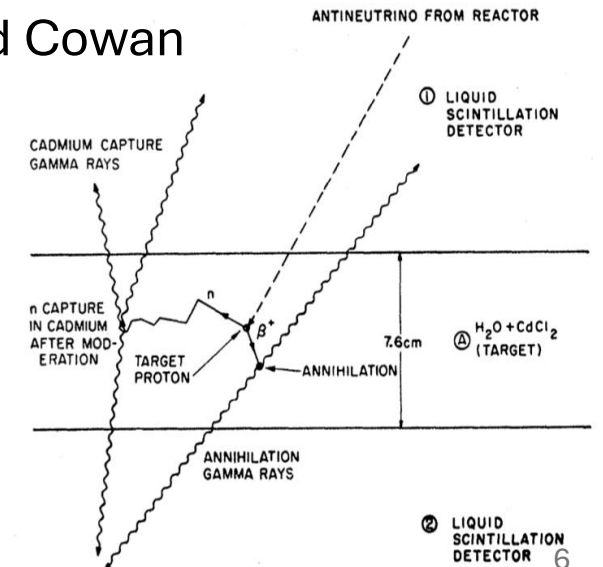
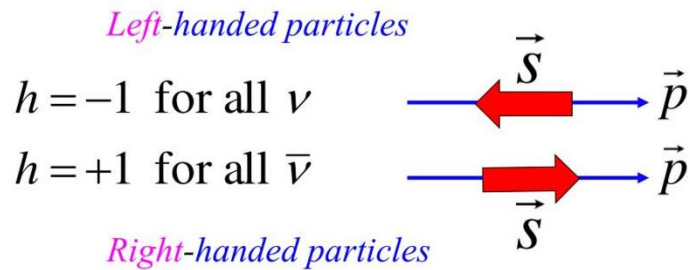


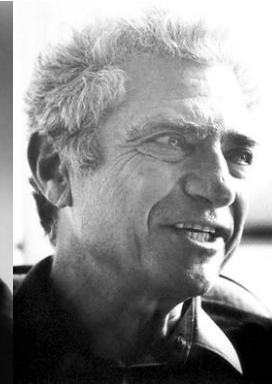
FIG. 1. Schematic diagram of antineutrino experiment.

The history of neutrino

1958, Goldhaber



1962, Lederman, Schwartz and Steinberger

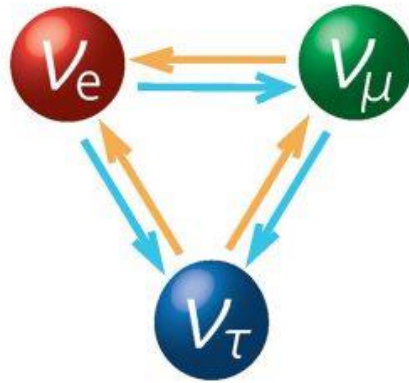


2000, DONUT experiment



Standard Model of Elementary Particles and Gravity

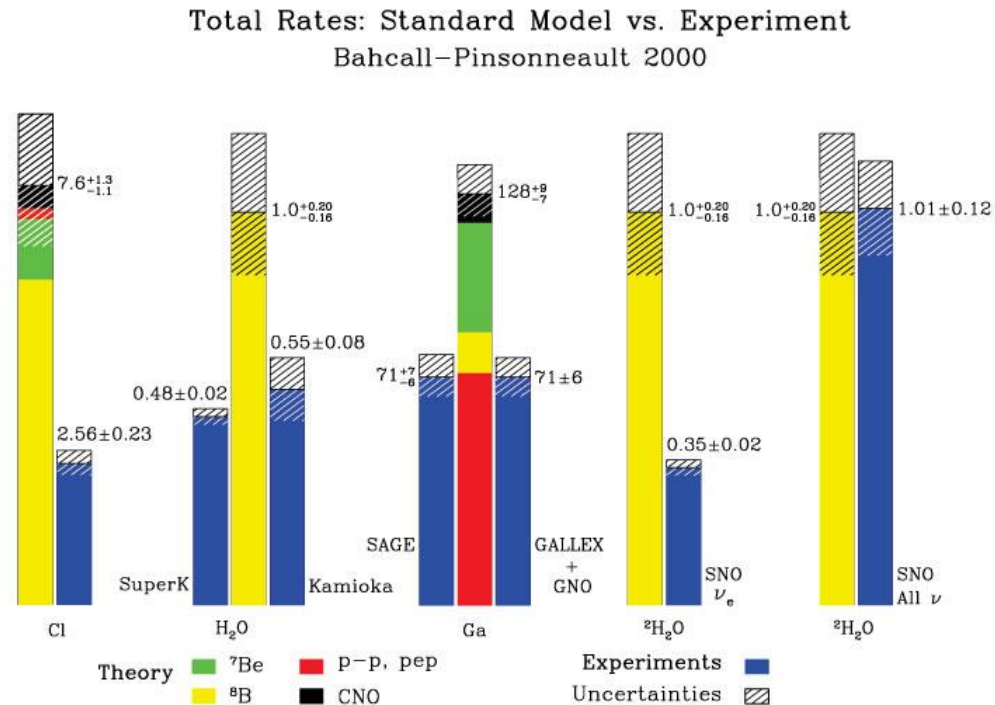
three generations of matter (fermions)						interactions / force carriers (bosons)		
I			II			III		
mass	$\approx 2.2 \text{ MeV}/c^2$	$\approx 1.28 \text{ GeV}/c^2$	$\approx 173.1 \text{ GeV}/c^2$	0	$\approx 124.97 \text{ GeV}/c^2$	0		
charge	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{2}{3}$	0	0	0		
spin	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	0	2		
	u up	c charm	t top	g gluon	H higgs	G graviton		
QUARKS	$\approx 4.7 \text{ MeV}/c^2$	$\approx 96 \text{ MeV}/c^2$	$\approx 4.18 \text{ GeV}/c^2$	0	SCALAR BOSONS	HYPOTHETICAL TENSOR BOSONS		
	$-\frac{2}{3}$	$-\frac{2}{3}$	$-\frac{2}{3}$	0				
	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1				
	d down	s strange	b bottom	γ photon				
LEPTONS	$\approx 0.511 \text{ MeV}/c^2$	$\approx 105.66 \text{ MeV}/c^2$	$\approx 1.7768 \text{ GeV}/c^2$	$\approx 91.19 \text{ GeV}/c^2$	GAUGE BOSONS VECTOR BOSONS			
	-1	-1	-1	1				
	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1				
	e electron	μ muon	τ tau	Z Z boson				
	$< 1.0 \text{ eV}/c^2$	$< 0.17 \text{ MeV}/c^2$	$< 18.2 \text{ MeV}/c^2$	$\approx 80.39 \text{ GeV}/c^2$				
	0	0	0	± 1				
	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1				
	ν_e electron neutrino	ν_μ muon neutrino	ν_τ tau neutrino	W W boson				



**But neutrinos oscillate,
or change type, as they travel**

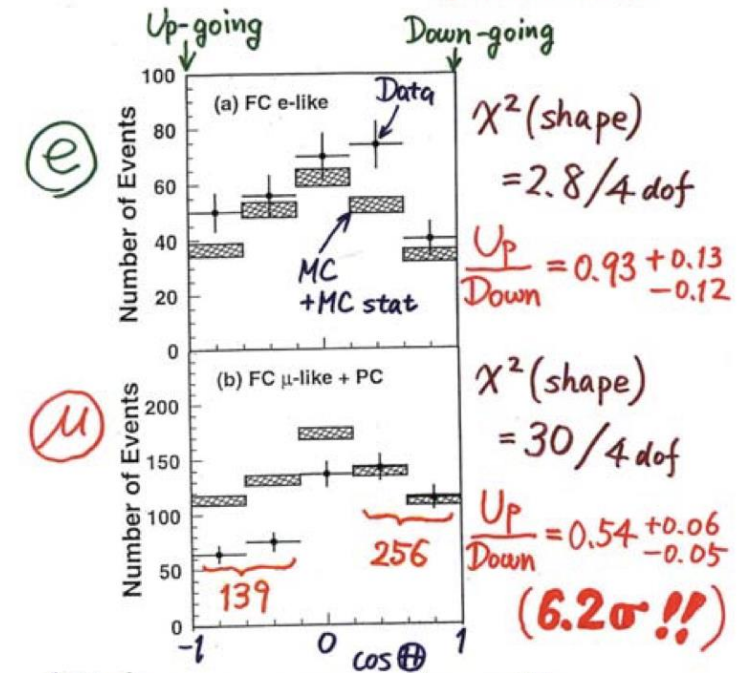
Neutrino oscillation

Solar neutrino problem



Atmospheric neutrino anomaly

Zenith angle dependence (Multi-GeV)



* Up/Down syst. error for μ -like

Prediction (flux calculation $\lesssim 1\%$
1km rock above SK 1.5%) 1.8%

Data (Energy calib. for $\uparrow\downarrow$ 0.7%
Non ν Background < 2%) 2.1%

Neutrino oscillation

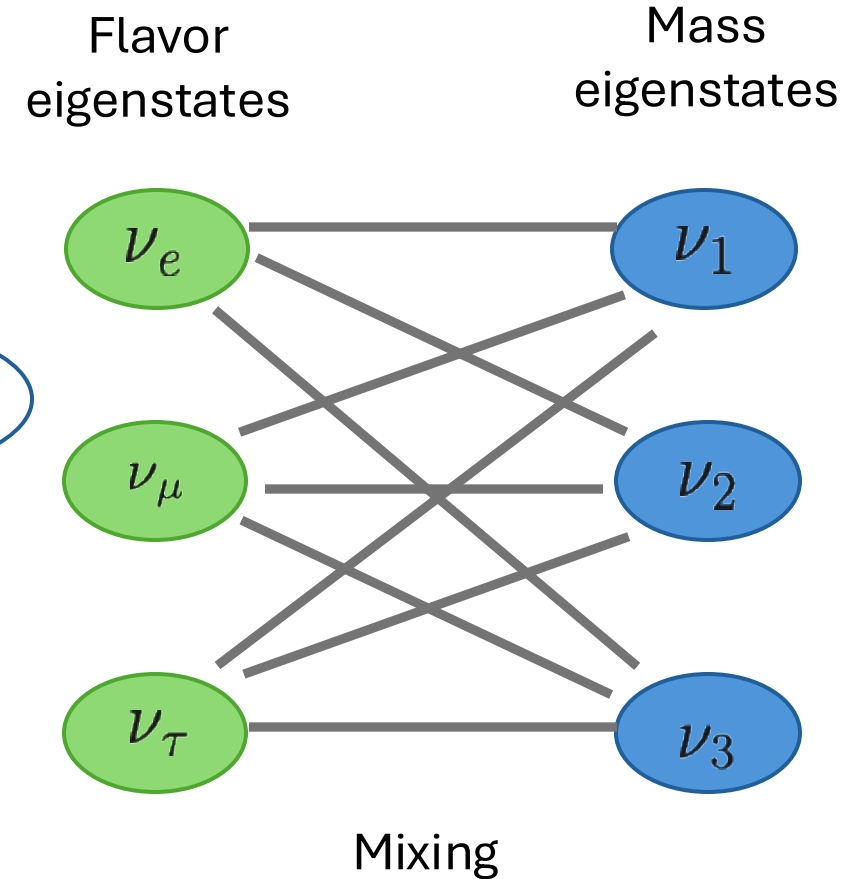
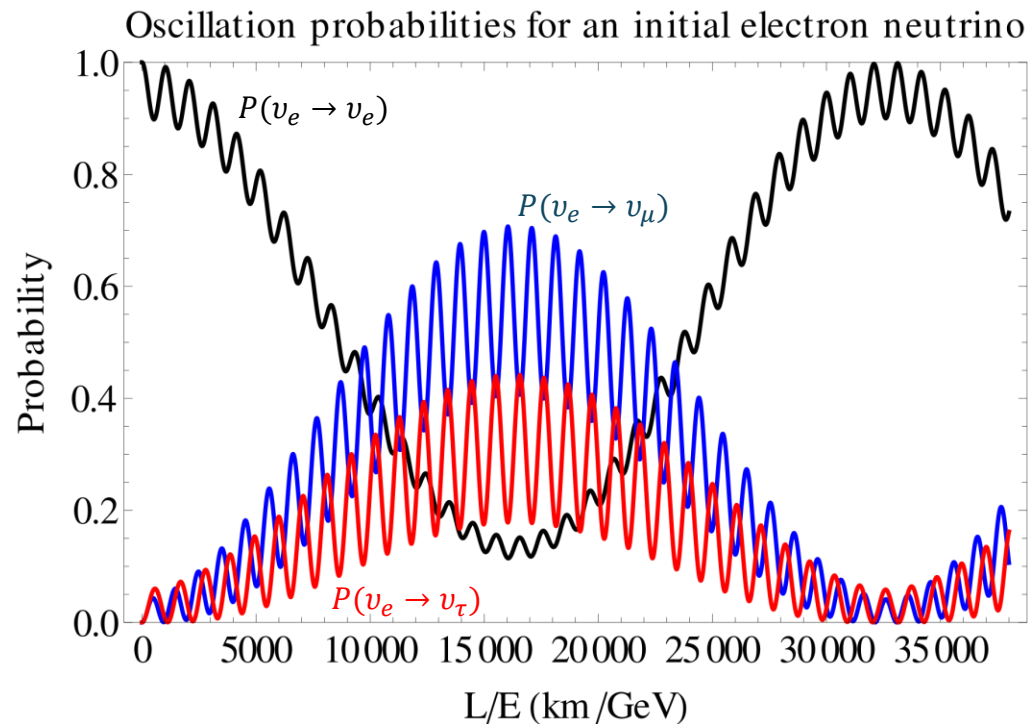
$$|\nu_\alpha(t)\rangle = \sum_k U_{\alpha k}^* |\nu_k(t)\rangle$$

$|\nu_k(t)\rangle = e^{-iE_k t} |\nu_k\rangle$

$$P_{\nu_\alpha \rightarrow \nu_\beta} = |\langle \nu_\beta | \nu_\alpha(t) \rangle|^2 = \sum_j \sum_k U_{\alpha k}^* U_{\beta k} U_{\alpha j} U_{\beta j}^* e^{-i(E_k - E_j)t}$$

$$= \sum_j \sum_k U_{\alpha k}^* U_{\beta k} U_{\alpha j} U_{\beta j}^* e^{-i \frac{\Delta m_{kj}^2 L}{2E}}$$

$E_k - E_j \approx \frac{m_k^2 - m_j^2}{2E} = \frac{\Delta m_{kj}^2}{2E}$
 $L = ct$



Neutrino oscillation

$$P_{\nu_\alpha \rightarrow \nu_\beta} = |\langle \nu_\beta | \nu_\alpha(t) \rangle|^2 = \sum_j \sum_k U_{\alpha k}^* U_{\beta k} U_{\alpha j} U_{\beta j}^* e^{-i(E_k - E_j)t}$$

$$= \sum_j \sum_k U_{\alpha k}^* U_{\beta k} U_{\alpha j} U_{\beta j}^* e^{-i \frac{\Delta m_{kj}^2 L}{2E}}$$

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$

Pontecorvo–Maki–Nakagawa–Sakata,
PMNS Matrix

$$\mathbf{U} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta_{\text{CP}}} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta_{\text{CP}}} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} e^{i\eta_1} & 0 & 0 \\ 0 & e^{i\eta_2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$\theta_{23} \quad \Delta m_{23}^2$

$\theta_{13} \quad \delta_{\text{CP}}$

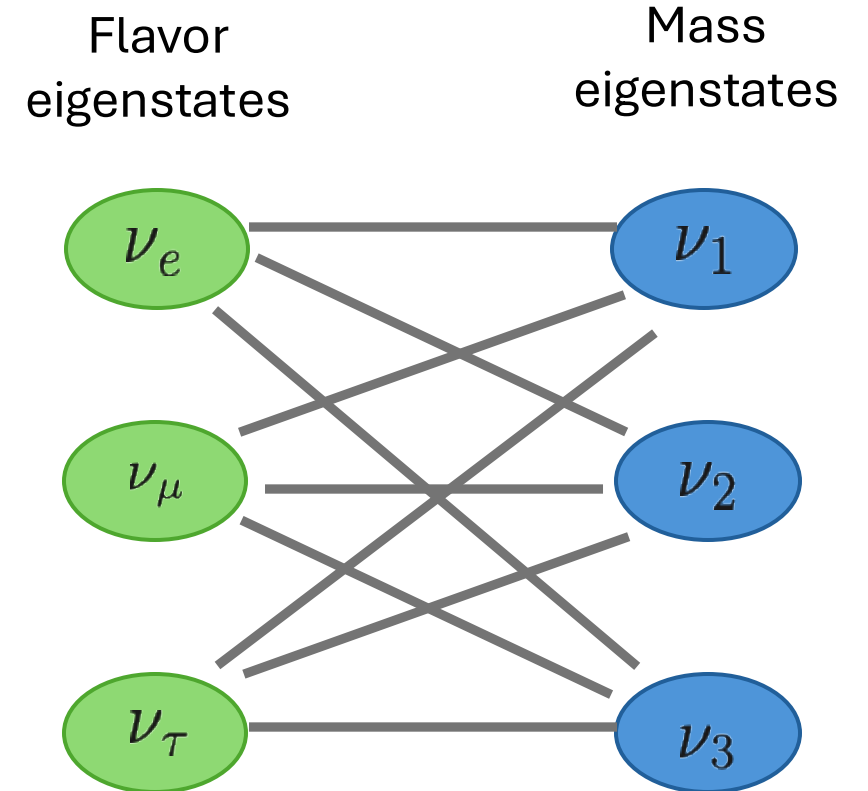
$\theta_{12} \quad \Delta m_{12}^2$

$\eta_1 \quad \eta_2$

Mixing, $\mathbf{U}$

$$c_{ij} = \cos(\theta_{ij})$$

$$s_{ij} = \sin(\theta_{ij})$$



Neutrino oscillation

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta_{CP}} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta_{CP}} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

θ_{23} Δm_{23}^2

θ_{13} δ_{CP}

θ_{12} Δm_{12}^2



Atmospheric



Accelerator



Reactor



Solar

$$\theta_{23} \sim 45^\circ$$

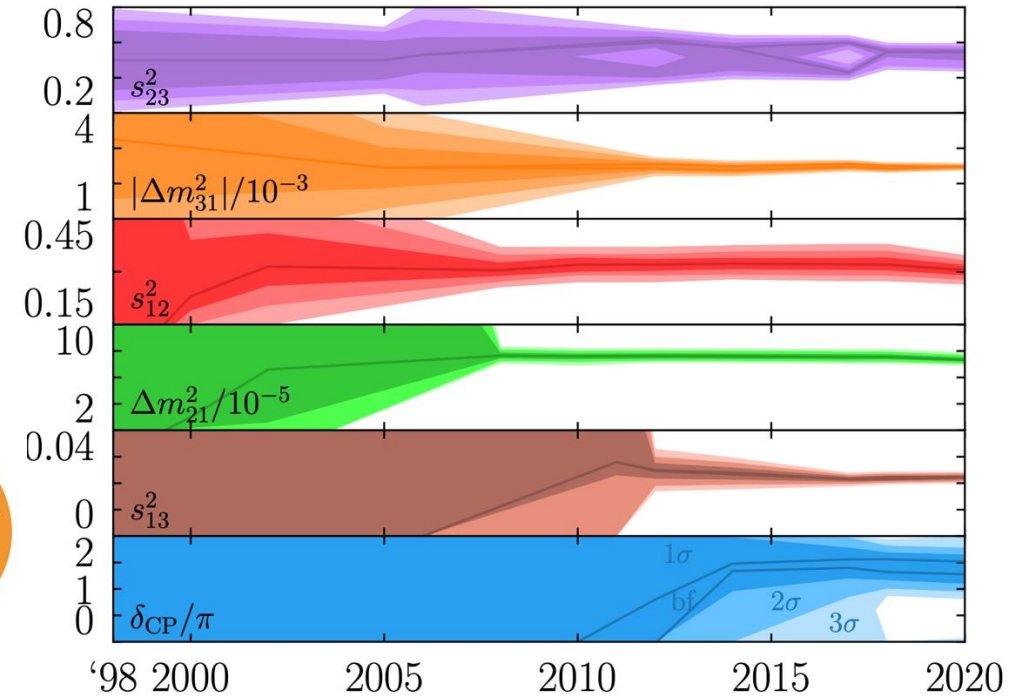
$$\Delta m_{23}^2 \sim \pm 2.5 \times 10^{-3} \text{eV}^2$$

$$\theta_{13} \sim 8.5^\circ$$

$$\delta_{CP} \sim 1.21$$

$$\theta_{12} \sim 33^\circ$$

$$\Delta m_{12}^2 \sim \pm 7.5 \times 10^{-5} \text{eV}^2$$



Snowmass NF01 report
P.B. Denton et al
Ref: arXiv:2212.00809

Open Questions

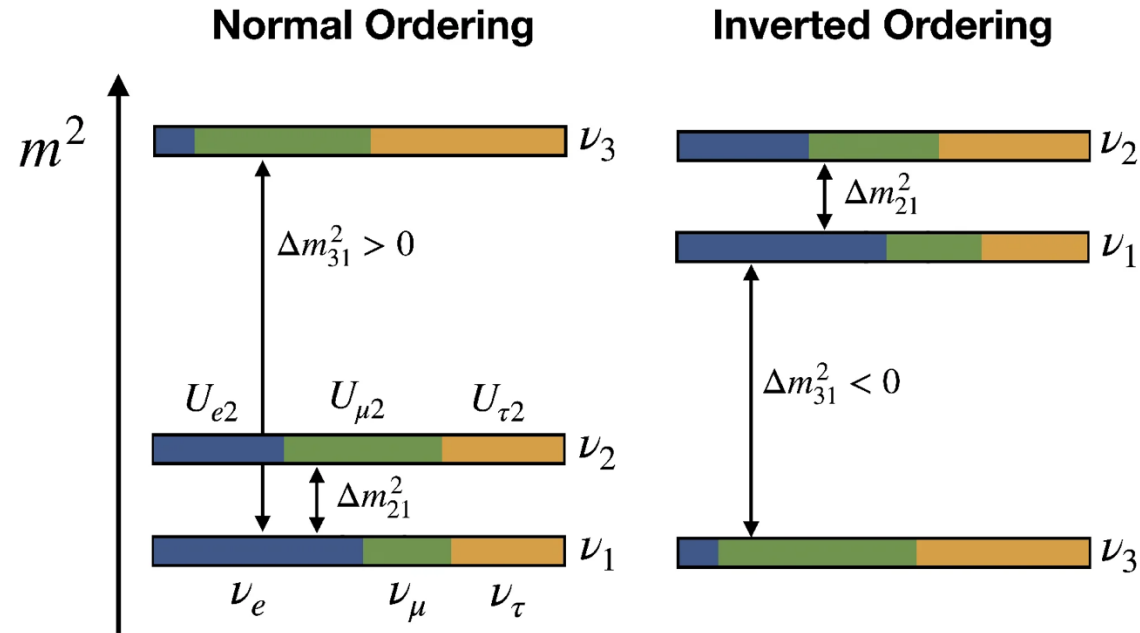
- Is the θ_{23} mixing maximal?
- Which neutrino is the lightest?
- Is CP violated in leptons?
- What is neutrino's absolute mass?
- Are neutrinos their own antiparticles?
-

Open questions



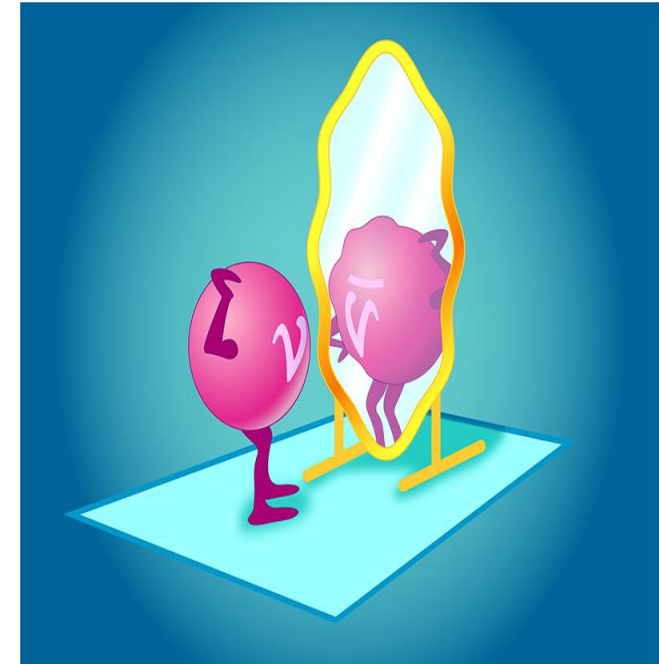
θ_{23} Octant

Which flavor dominates in ν_3 ,
 ν_μ or ν_τ ?



Mass ordering

Which mass is the lightest,
 ν_1 or ν_3 ?



CP Symmetry

Do neutrinos and anti-neutrinos oscillate differently violating the CP symmetry?

Joint Analysis

1. The T2K and SK experiments
2. Why not go for a joint analysis?
3. How to build a joint analysis?
4. Results?

The T2K and SK experiments

$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

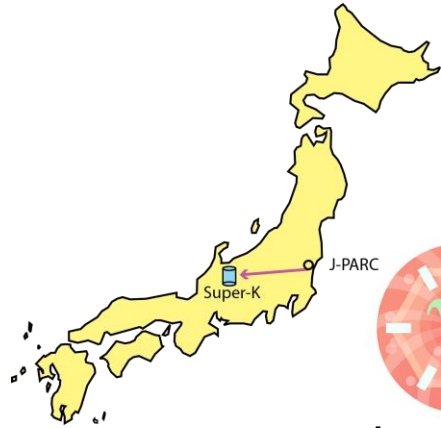
$$\hookrightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$$

$$\hookrightarrow e^- + \bar{\nu}_e + \nu_\mu$$



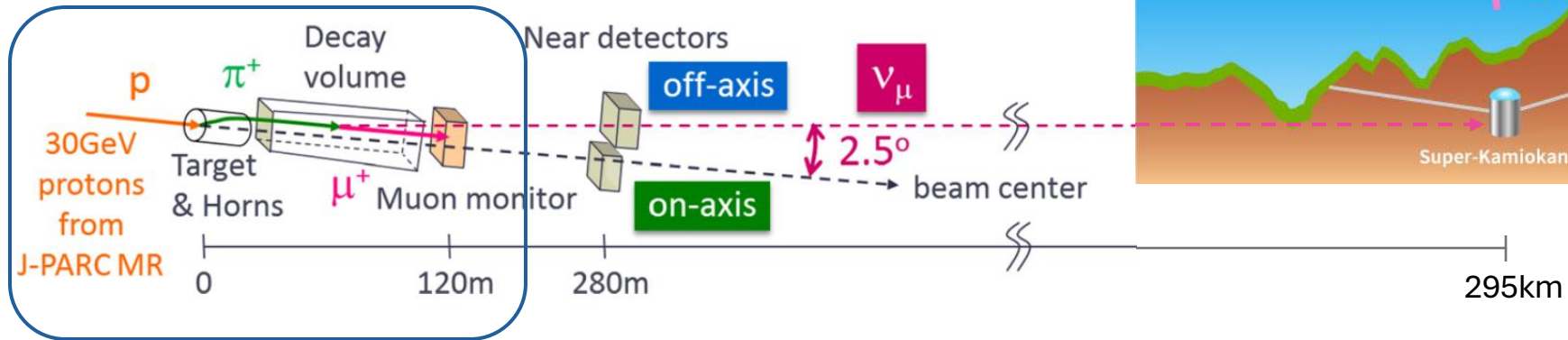
Atmospheric



Accelerator

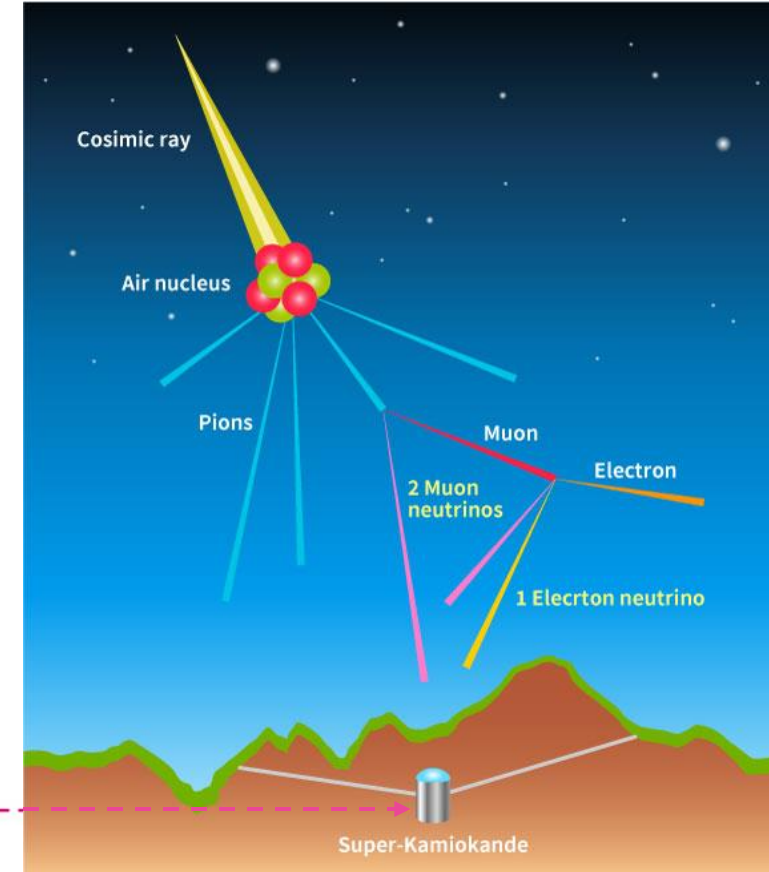
$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$$

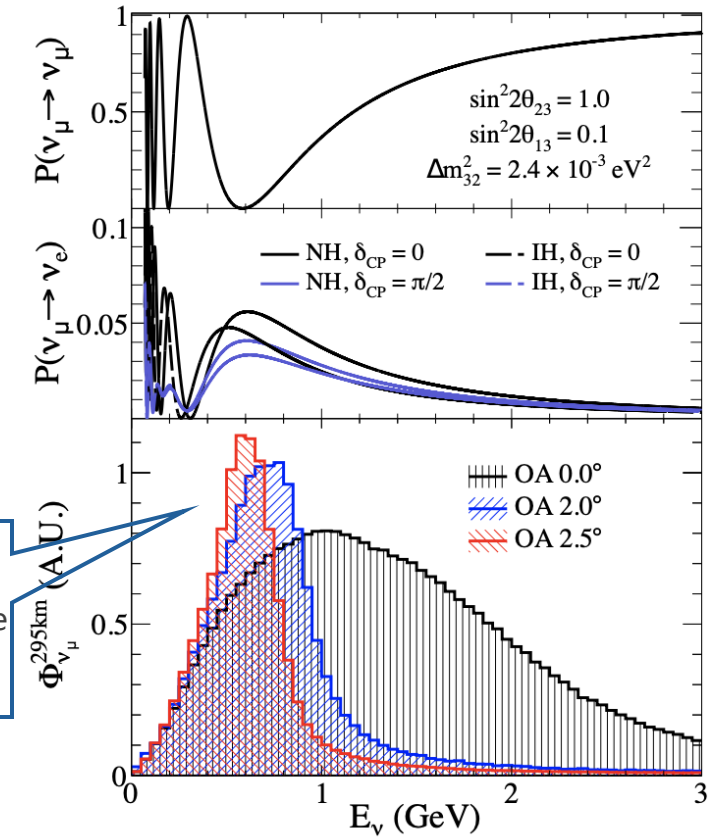
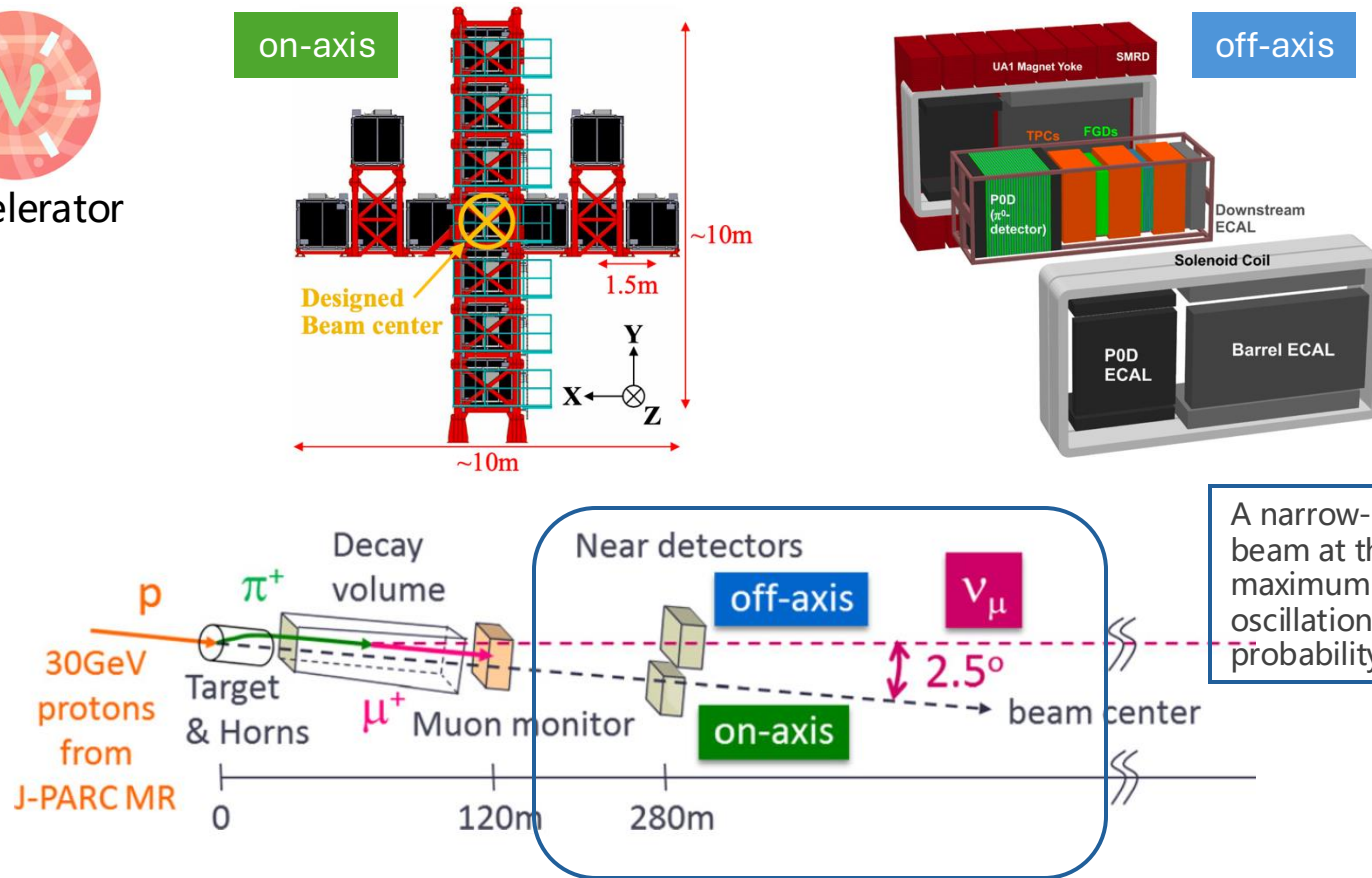


Neutrino beam

- Electromagnetic horns focus and select hadrons:
 - Forward horn current (FHC): neutrino mode
 - Reverse horn current (RHC): anti-neutrino mode



The T2K and SK experiments



- ND complex • Near Detector provides valuable constraints on:
- neutrino beam flux
 - neutrino interaction cross-section

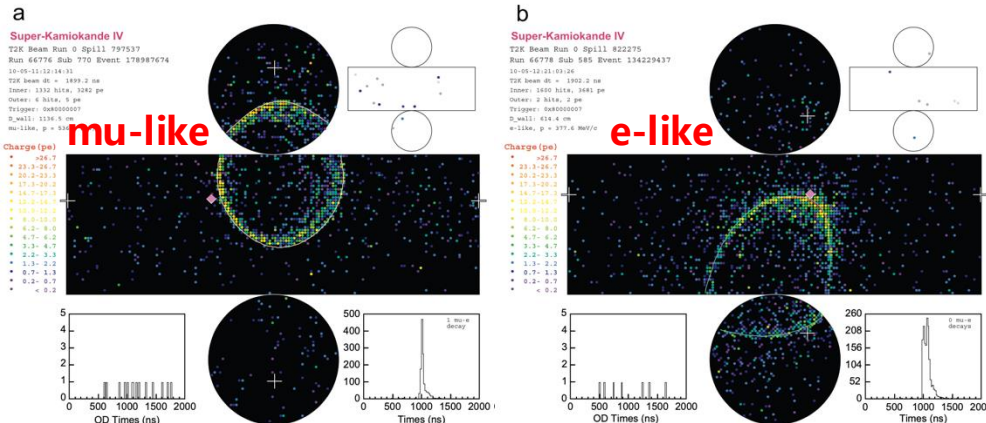
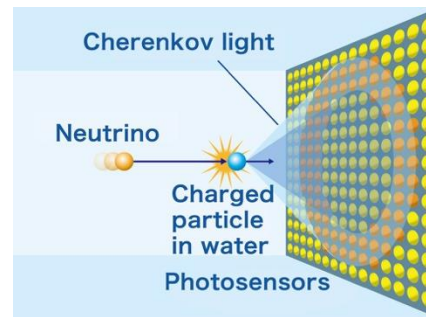
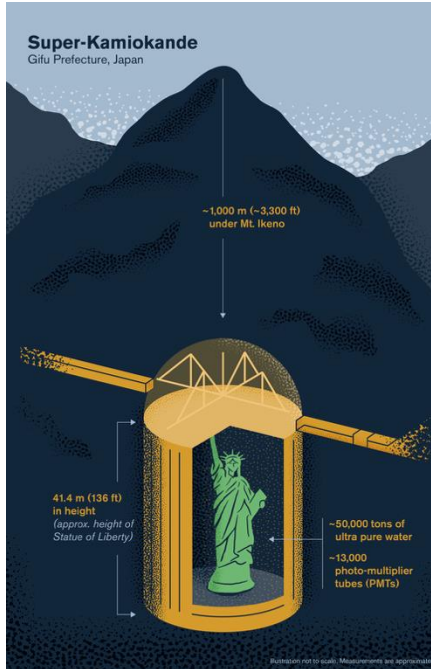
The T2K and SK experiments



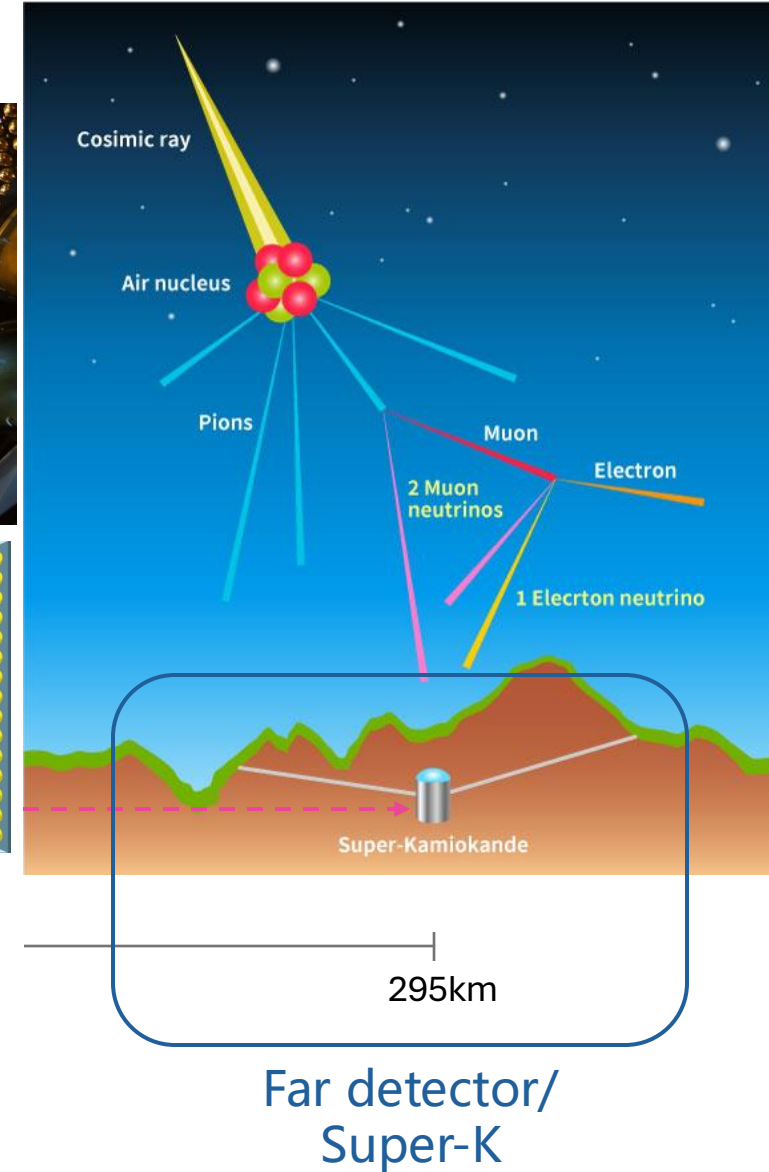
Accelerator



Atmospheric



- Far detector utilizes Cherenkov ring information of charged particles to reconstruct events

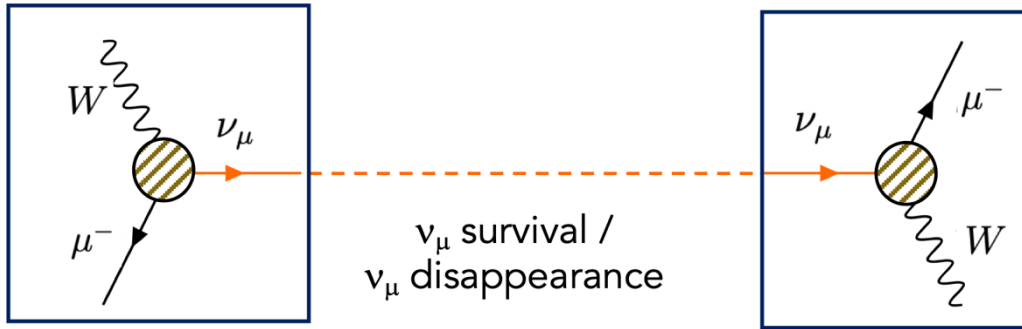




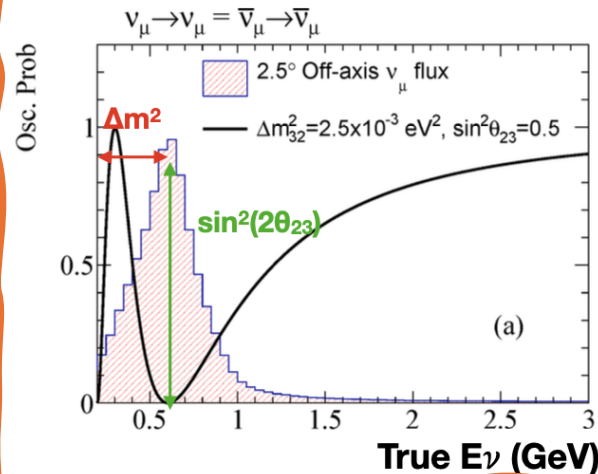
Accelerator

T2K physics

ν_μ disappearance

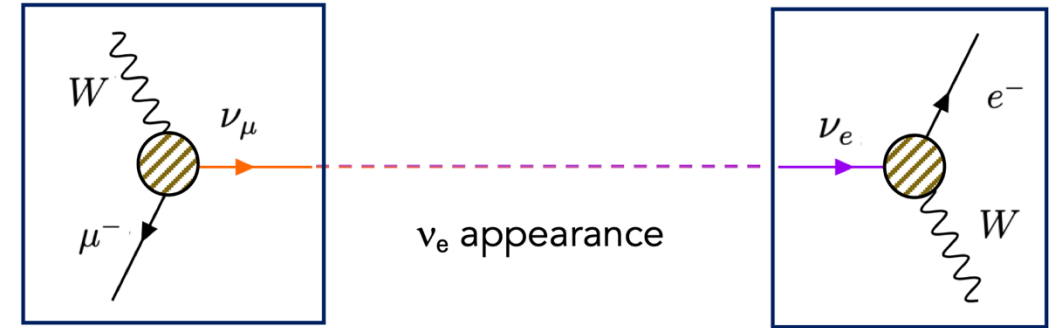


$$P_{\nu_\mu \rightarrow \nu_\mu} \approx 1 - \cos^2 \theta_{13} \sin^2 2\theta_{23} \sin^2 \frac{\Delta m_{32}^2 L}{4E_\nu}$$



- Sensitive to :
 - θ_{23}
 - $|\Delta m_{32}^2|^2$

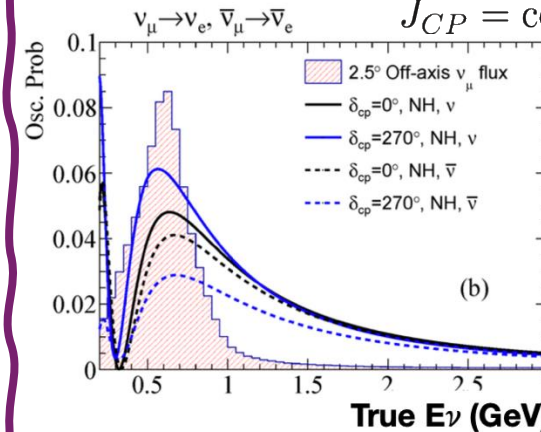
ν_e appearance



$$P_{\nu_\mu \rightarrow \nu_e} (\bar{\nu}_\mu \rightarrow \bar{\nu}_e) \approx \sin^2 2\theta_{13} \sin^2 \theta_{23} \sin^2 \frac{\Delta m_{32}^2 L}{4E_\nu}$$

$$\mp \frac{\Delta m_{32}^2 L}{4E_\nu} J_{CP} \sin^2 \frac{\Delta m_{32}^2 L}{4E_\nu}$$

$$J_{CP} = \cos \theta_{13} \sin 2\theta_{12} \sin 2\theta_{23} \sin 2\theta_{13} \sin \delta_{CP}$$

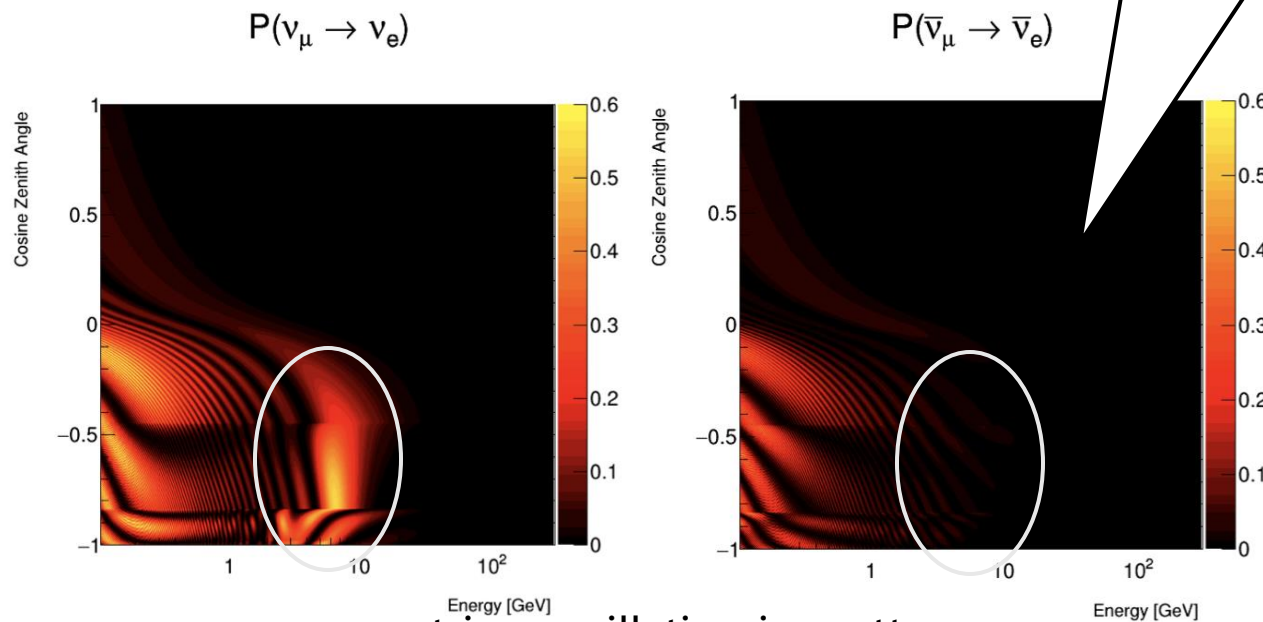


- Jarlskog invariant J_{CP}
- Sensitive to :
 - δ_{CP}

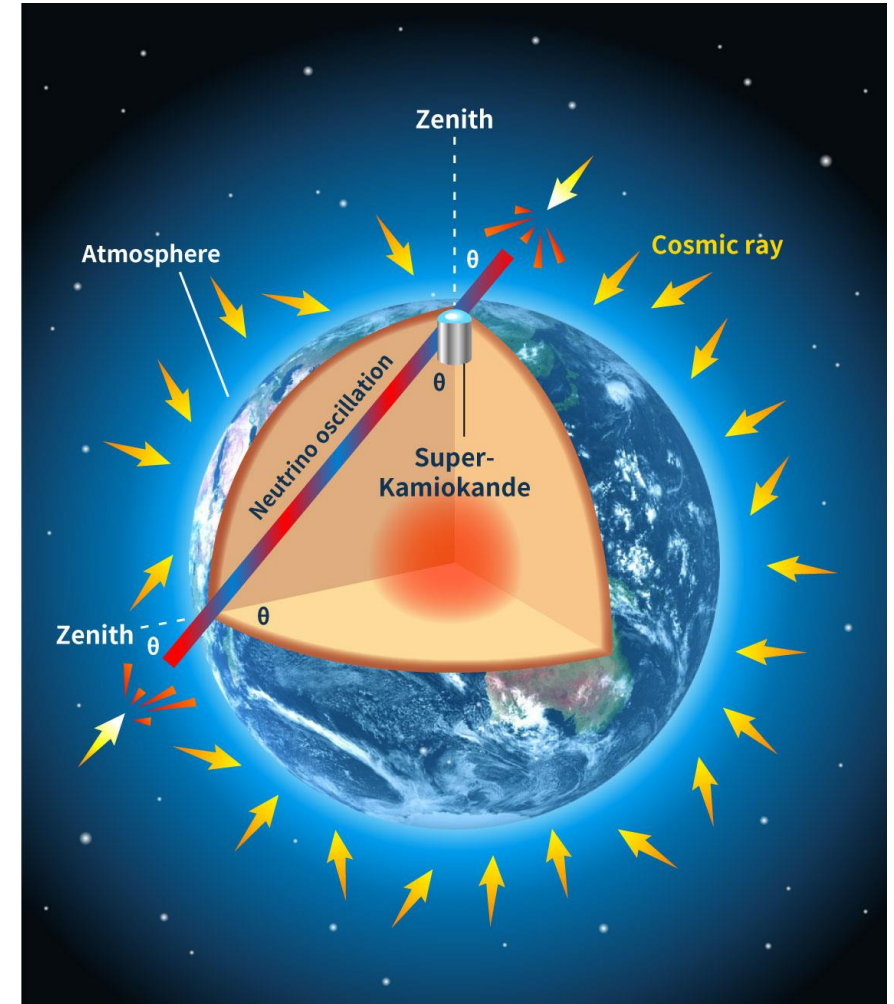


SK physics

- Upgoing neutrinos experience stronger **matter effects** on oscillation
- The resonance on neutrino/antineutrino depends on mass ordering
- Sensitive to :
 - mass ordering



neutrino oscillation in matter
(for normal mass ordering)



Joint fit of T2K and SK



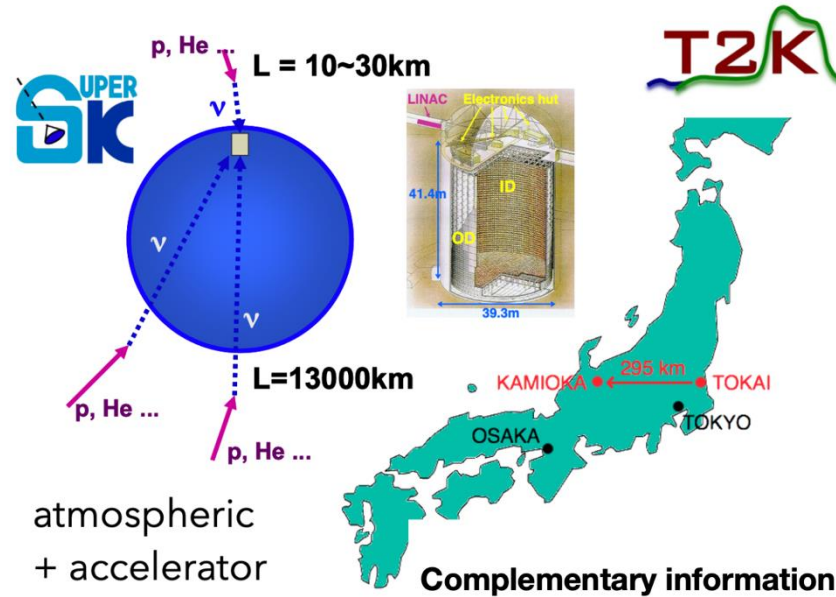
Accelerator



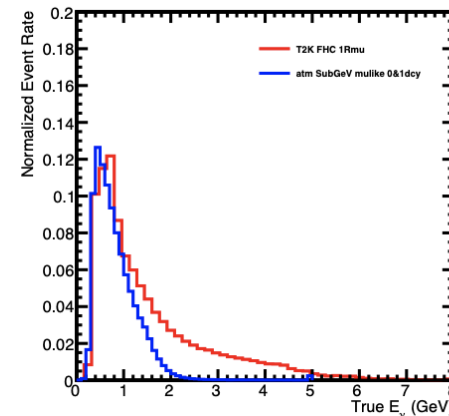
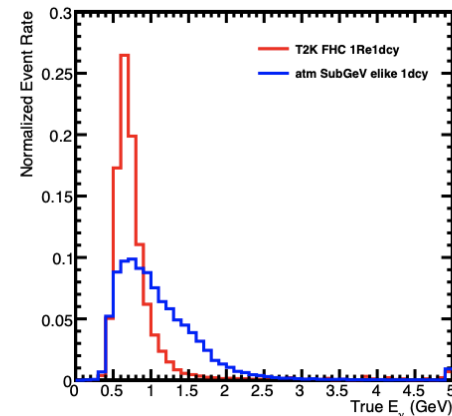
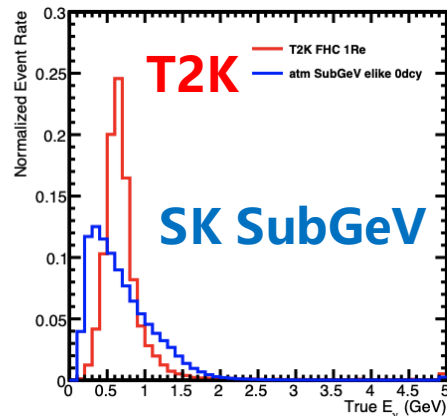
Atmospheric

- Why joint fit?
- How to build a (Bayesian) joint fit?
- What is the result?

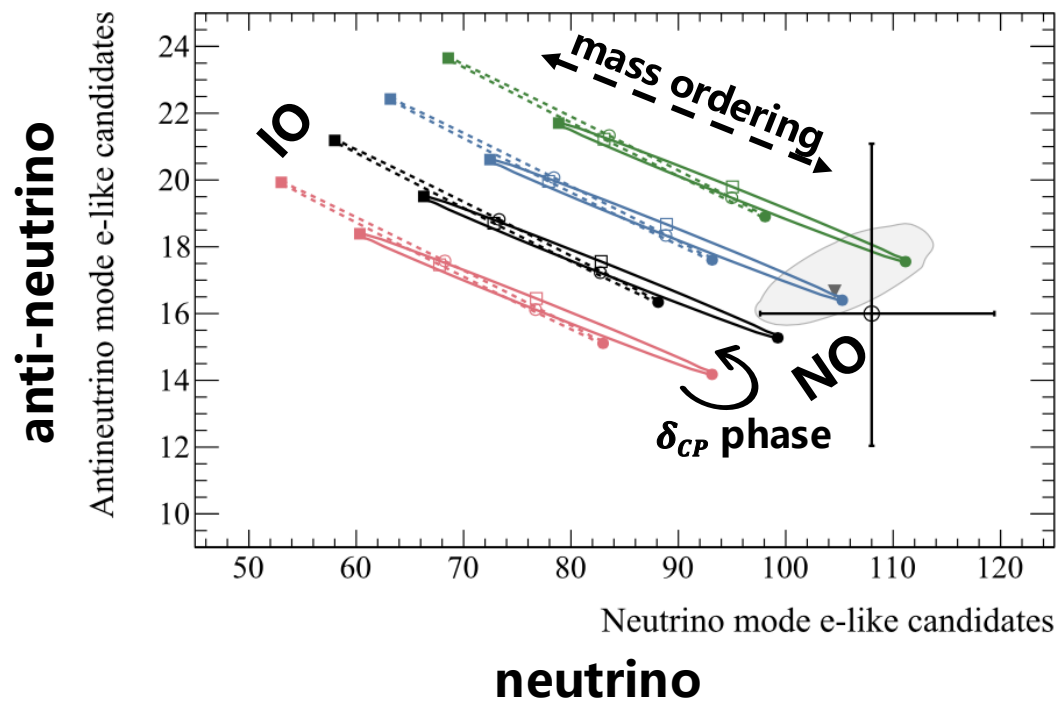
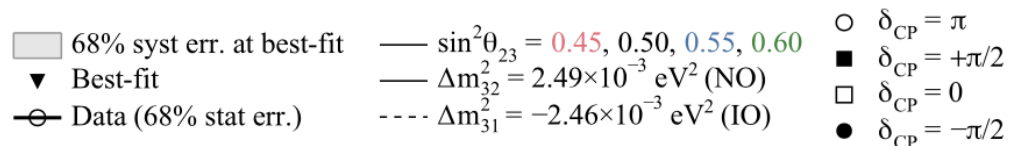
T2K-SK complementarity



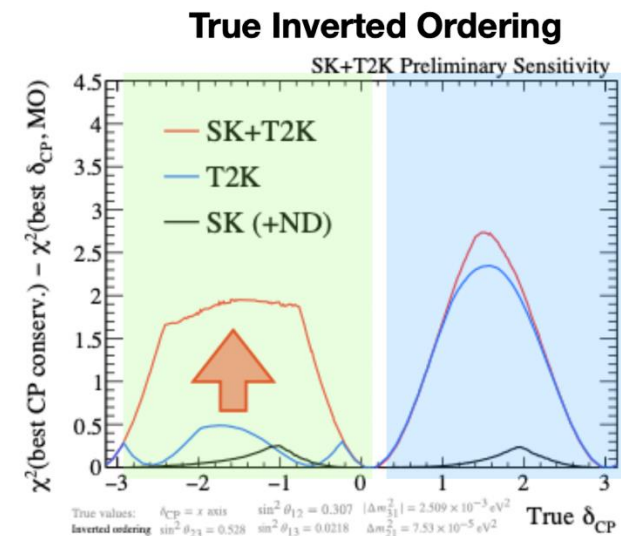
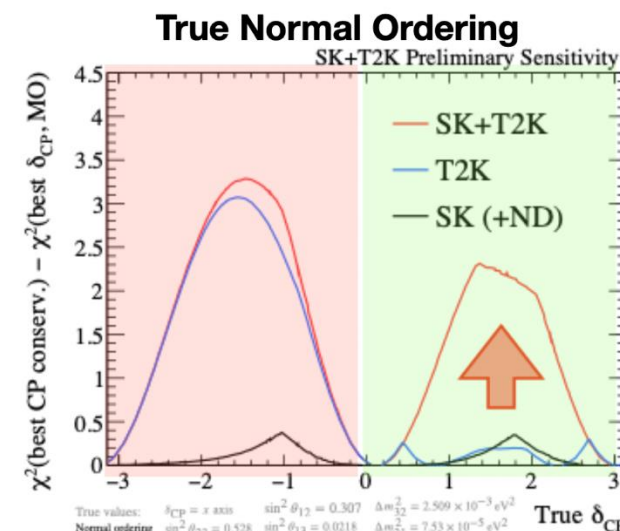
- The two experiments share the same Super-K detector
- Some samples of the two experiments overlap in similar energies
 - Similar event selection cuts
 - Similar neutrino interaction modes
 - Leveraging T2K ND constraints
- Increased statistics



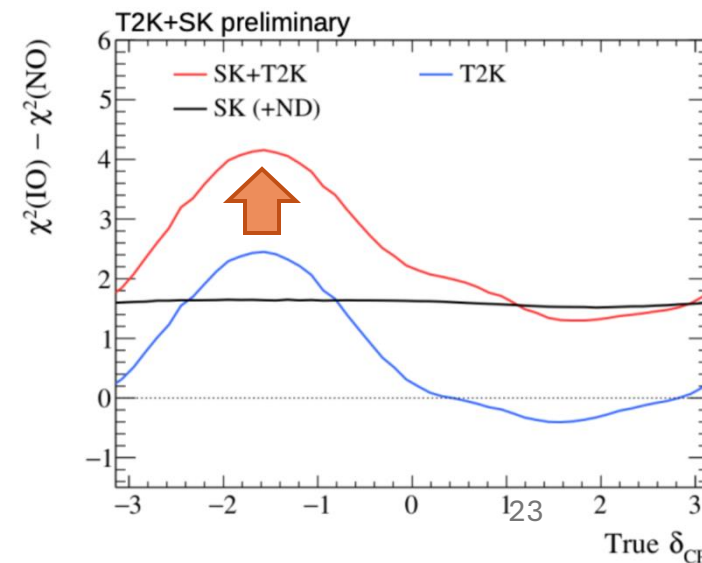
T2K-SK complementarity



T2K δ_{CP} -MO degeneracy



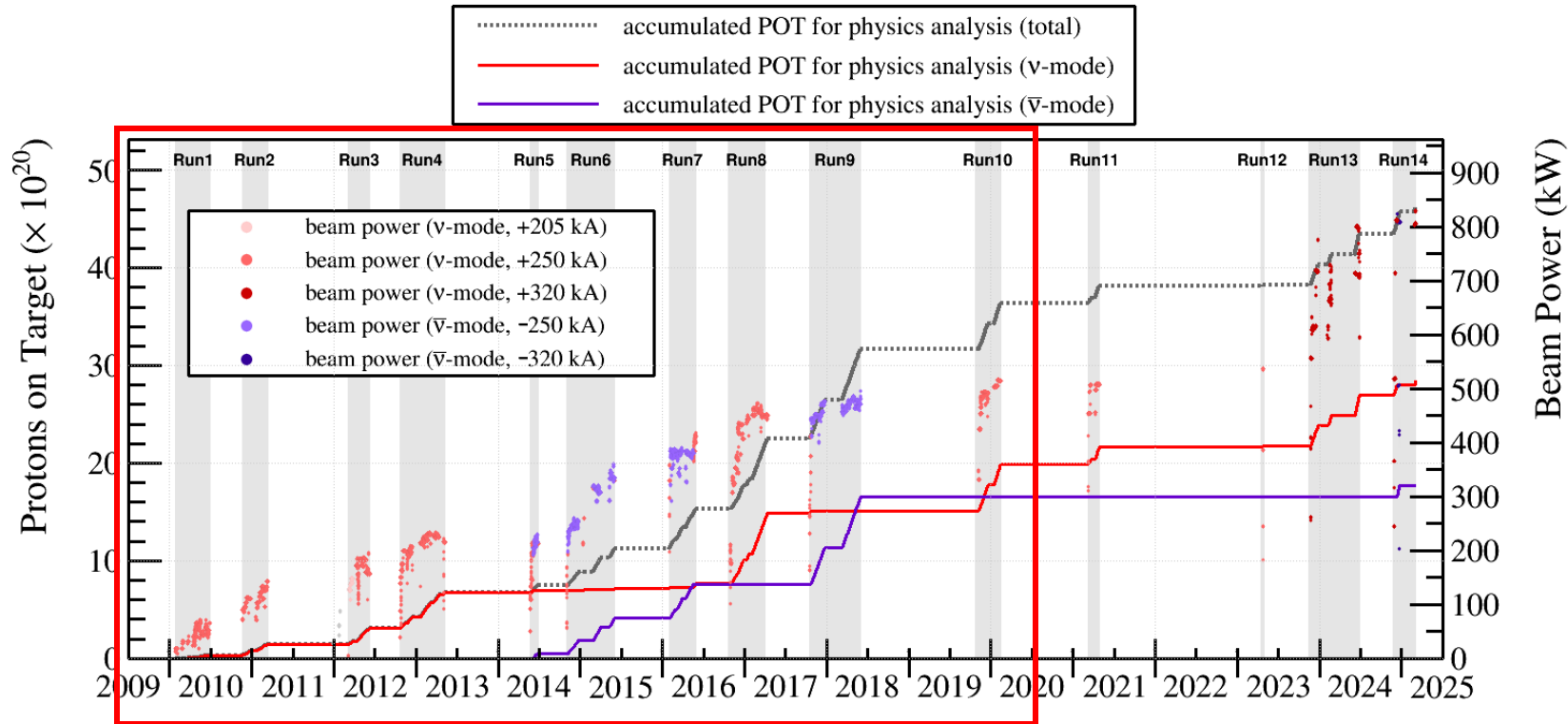
Breaking
the
Degeneracy





Accelerator

Far detector samples: T2K

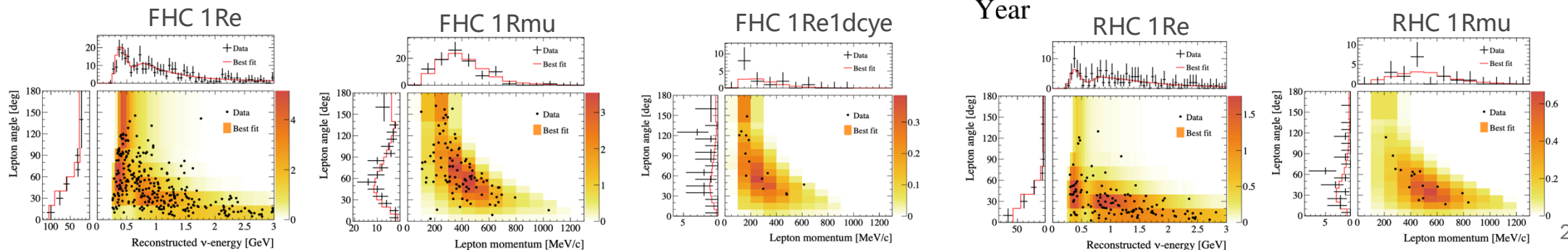


T2K data as in Phys.Rev.D 108 (2023) 7, 072011

- Neutrino mode: 19.7×10^{20} POT
- Anti-neutrino mode: 16.3×10^{20} POT

5 Samples: sub-GeV to O(1) GeV

- FHC: 3 samples
- RHC: 2 samples

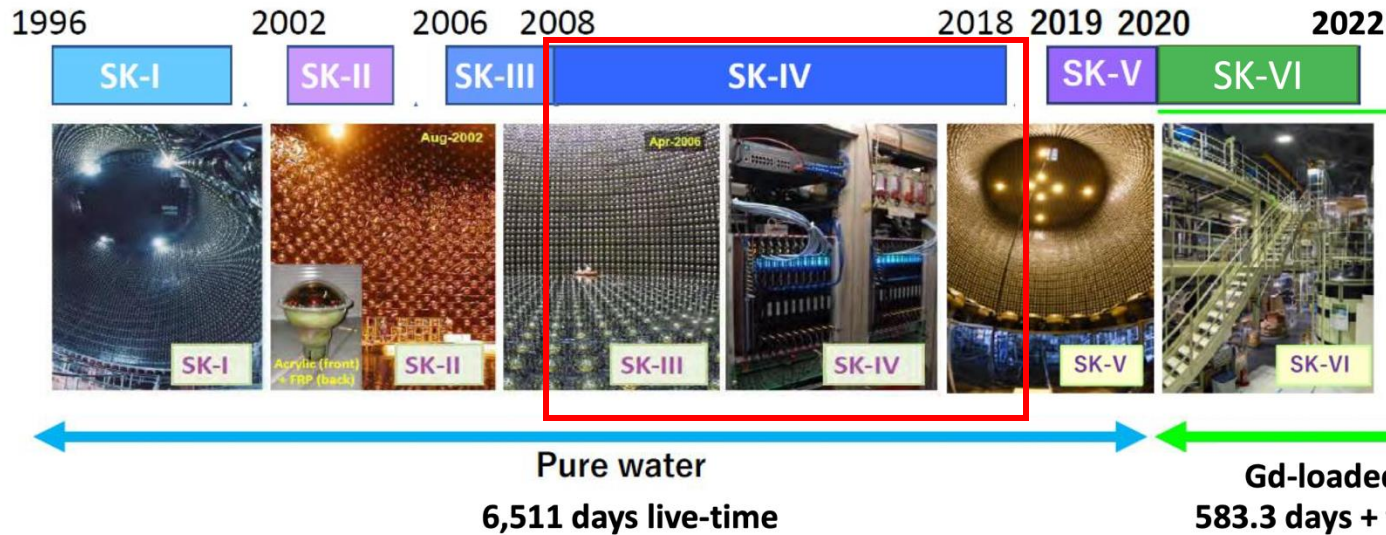




Atmospheric

Far detector samples: SK

Gd concentration at SK-VI:
0.011% in weight.

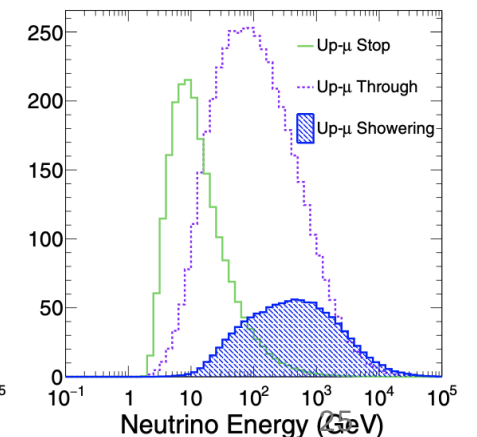
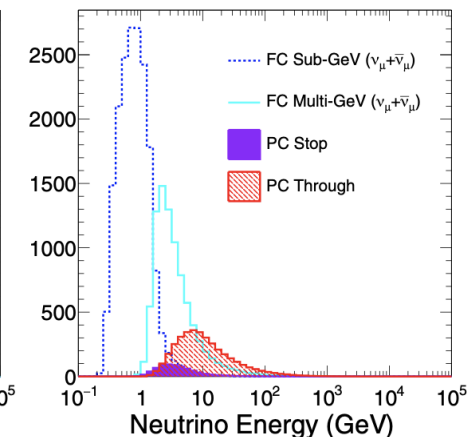
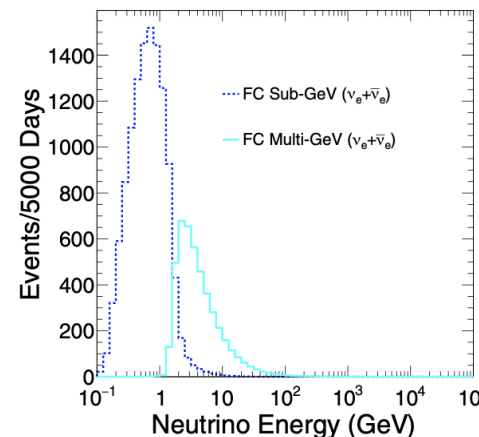
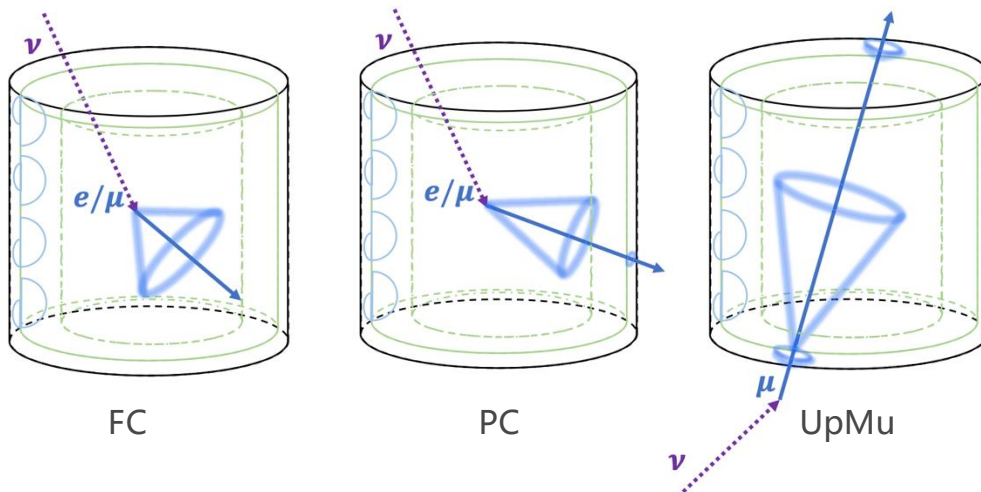


SK-IV data (18 samples) before Gd-doping, as in PTEP 2019 (2019) 5, 053F01

- SK-IV: 3244 days (2008-2018)

18 Samples: sub-GeV to O(10) TeV

- Fully Contained:
 - ✓ Sub-GeV samples: 6
 - ✓ Multi-GeV samples: 7
- Partially Contained: 2
- Up-going Muon: 3





Accelerator



Atmospheric

Let's build a Bayesian analysis!

Posterior:
our knowledge of
the parameters after
taking experiments

$$P(\theta, \alpha | \mathbf{D}) =$$

Likelihood:
Given the values of
parameters, the
probability to
observe $\mathbf{D}$

Prior:
Our previous
knowledge of the
parameters before
taking experiments

$$L(\mathbf{D} | \theta, \alpha) \pi(\theta) \pi(\alpha)$$

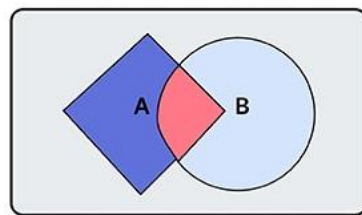
$$\int L(\mathbf{D} | \theta, \alpha) \pi(\theta) \pi(\alpha) d\theta d\alpha$$

θ : parameters of interests/oscillation parameters

α : nuisance parameters/systematic uncertainty parameters

$\mathbf{D}$: data/experiment observables

Visual proof of Bayes' Theorem!



$$P(A) = \frac{\text{blue diamond}}{\text{total area}}$$

$$P(B) = \frac{\text{light blue circle}}{\text{total area}}$$

$$P(A|B) = \frac{\text{red intersection}}{\text{light blue circle}}$$

$$P(B|A) = \frac{\text{red intersection}}{\text{blue diamond}}$$

$$\frac{\text{red intersection}}{\text{light blue circle}} = P(A|B) = \frac{P(B|A) * P(A)}{P(B)} = \frac{\frac{\text{red intersection}}{\text{blue diamond}} * \frac{\text{blue diamond}}{\text{total area}}}{\frac{\text{light blue circle}}{\text{total area}}} = \frac{\text{red intersection}}{\text{light blue circle}}$$

Evidence:
 $=P(\mathbf{D})$,
Normalization factor
invariant to changes in
parameters

Let's build a joint analysis!

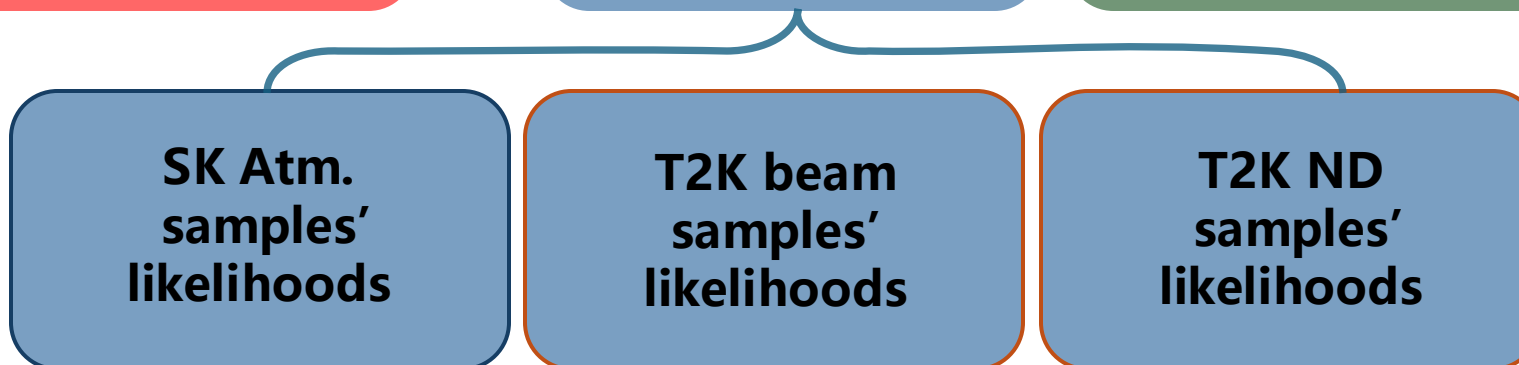


Accelerator



Atmospheric

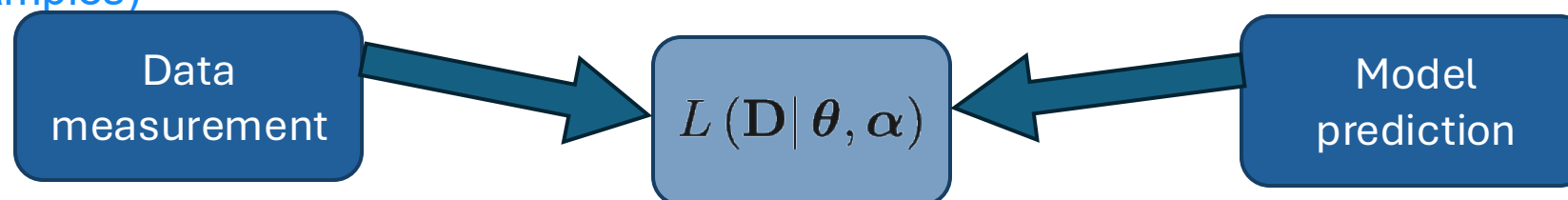
Target posterior



SK-IV (18 samples)

T2K Run1-10 (5 samples)

- Flat priors for $\sin^2\theta_{23}$, $|\Delta m_{32}|^2$, δ_{CP}
- Gaussian distributions as nuisances' priors



Let's build a model!



Accelerator



Atmospheric

Model
prediction

$$N^{\text{pred}}|_{\theta, \alpha} = \underbrace{\Phi|_{\alpha}(E_{\nu})}_{\text{neutrino flux}} \times \underbrace{P_{\nu_{\alpha} \rightarrow \nu_{\beta}}|_{\theta}}_{\text{oscillation probability}} \times \underbrace{\sigma|_{\alpha}(E_{\nu}, \vec{x})}_{\text{cross-section}} \times \underbrace{\epsilon|_{\alpha}(\vec{x})}_{\text{detector response}}$$

Flux model

:

T2K beam
Flux model

No correlation yet
for the systematic
uncertainties
across experiments

SK atm.
Flux model

- Beamline simulation tuned with external hadron production experiments

- 2011 Honda flux model tuned with cosmic ray flux experiments' results

Let's build a model!



Accelerator



Atmospheric

Model
prediction

$$N^{\text{pred}}|_{\theta, \alpha} = \Phi|_{\alpha}(E_{\nu}) \times P_{\nu_{\alpha} \rightarrow \nu_{\beta}}|_{\theta} \times \boxed{\sigma|_{\alpha}(E_{\nu}, \vec{x})} \times \epsilon|_{\alpha}(\vec{x})$$

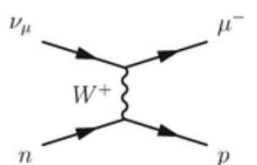
cross-section

Cross-section
model

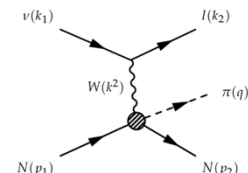
Low energy: T2K samples + SK Sub-GeV samples
T2K cross-section model with extra parameters

High energy: SK multi-GeV samples
SK cross-section model with modifications

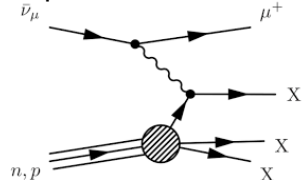
CC-Quasi-elastic



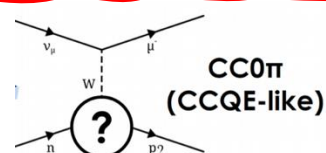
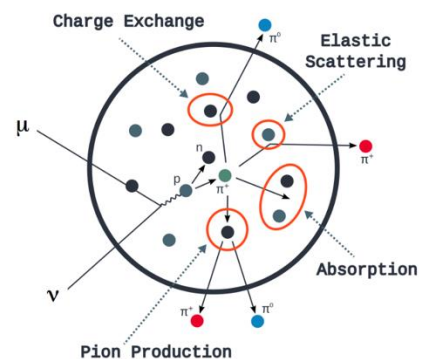
CC-Resonant 1π



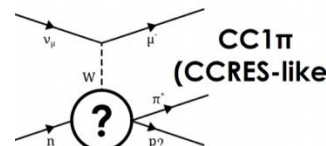
Deep inelastic scattering



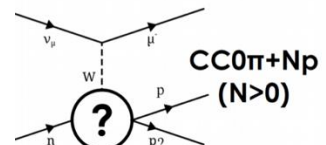
Nuclear
state + effects



CC0 π
(CCQE-like)

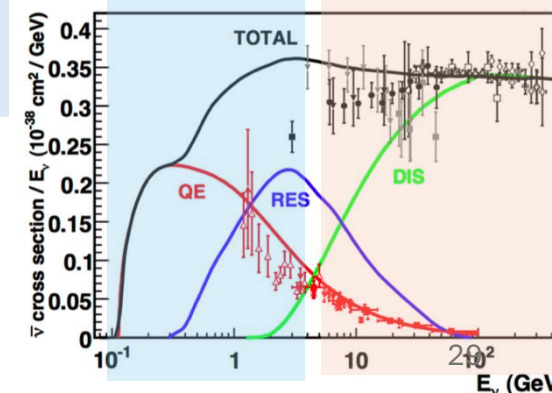
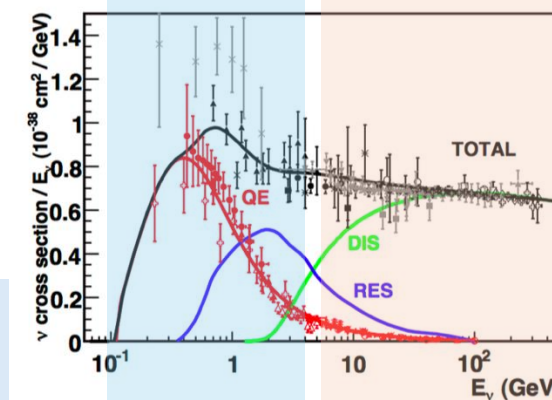


CC1 π
(CCRES-like)



CC0 π +Np
(N>0)

ND data used
to constraint
the low energy
cross-section
model



Let's build a model!



Accelerator



Atmospheric

Model
prediction

$$N^{\text{pred}}|_{\theta, \alpha} = \Phi|_{\alpha}(E_{\nu}) \times P_{\nu_{\alpha} \rightarrow \nu_{\beta}}|_{\theta} \times \sigma|_{\alpha}(E_{\nu}, \vec{x}) \times \boxed{\epsilon|_{\alpha}(\vec{x})}$$

detector
model

:

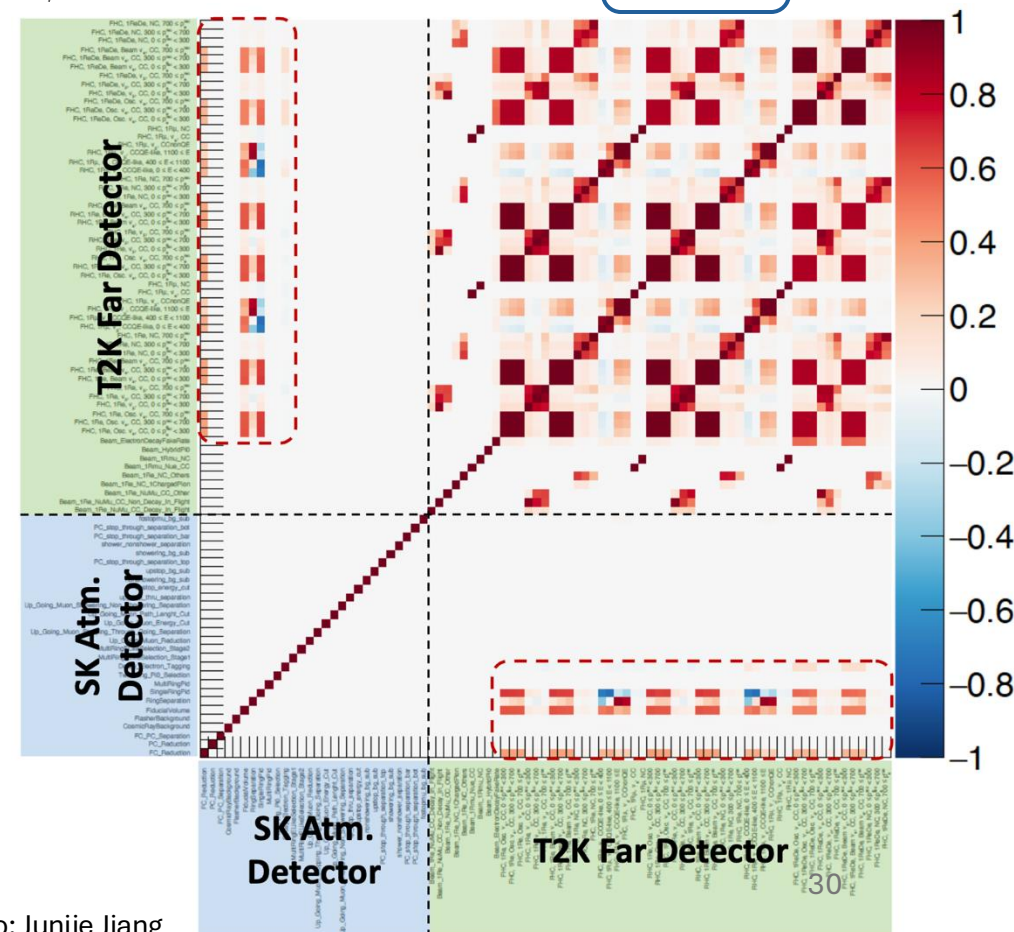
correlation for the
systematic
uncertainties
across experiments

T2K
Far detector
model



SK atm.
detector
model

detector response



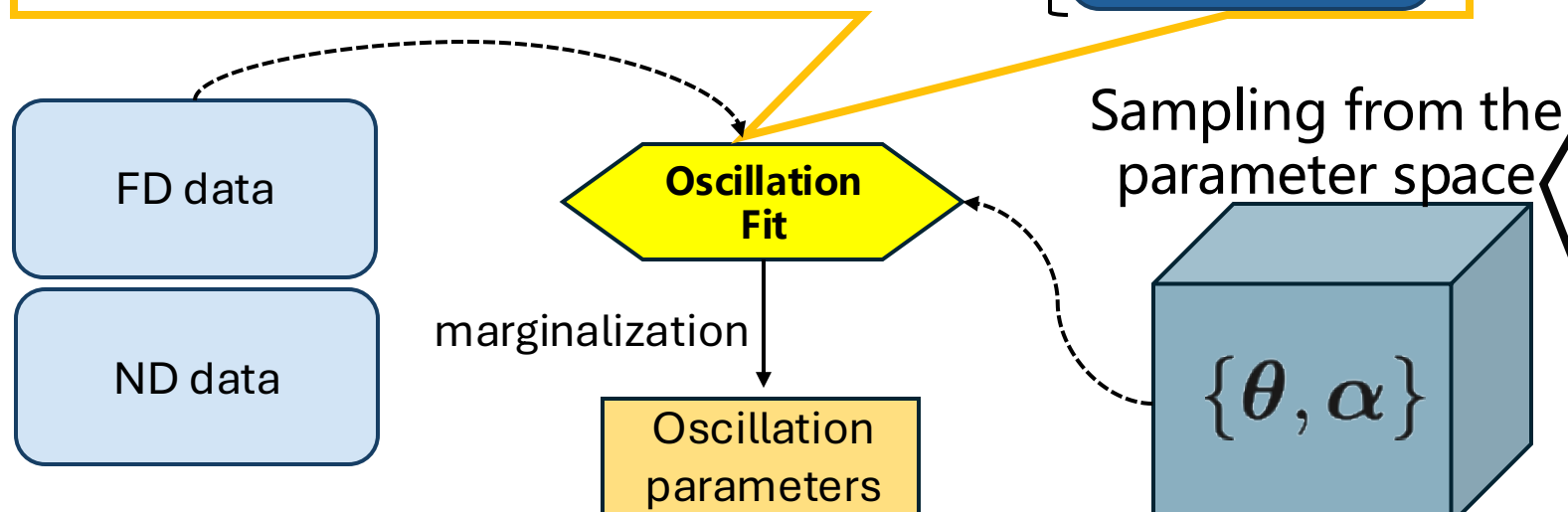
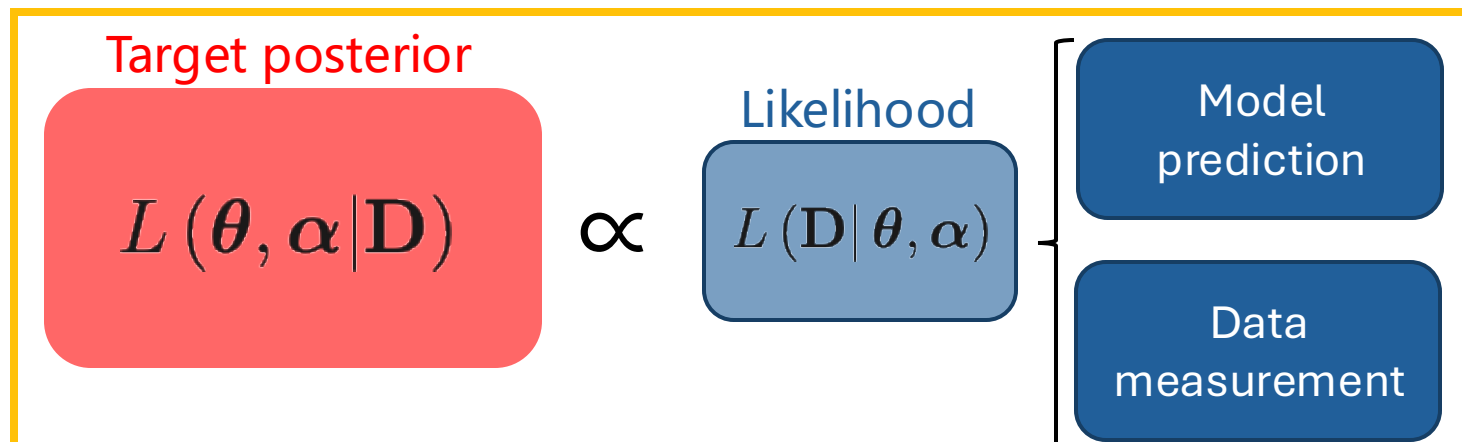
Let's fit the model!



Accelerator



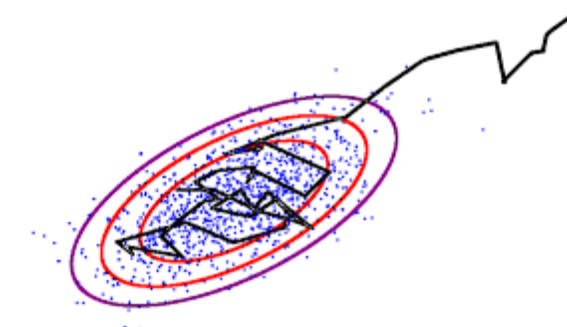
Atmospheric



Markov Chain Monte Carlo MCMC

- Metropolis-Hastings algorithm:

A semi-random walk through the parameter space



- Fitter: **MaCh3**

Other fitters: P-Theta, Osc3++



Accelerator



Atmospheric

Joint fit results

- joint fit
- single-experiment fit

Joint fit results

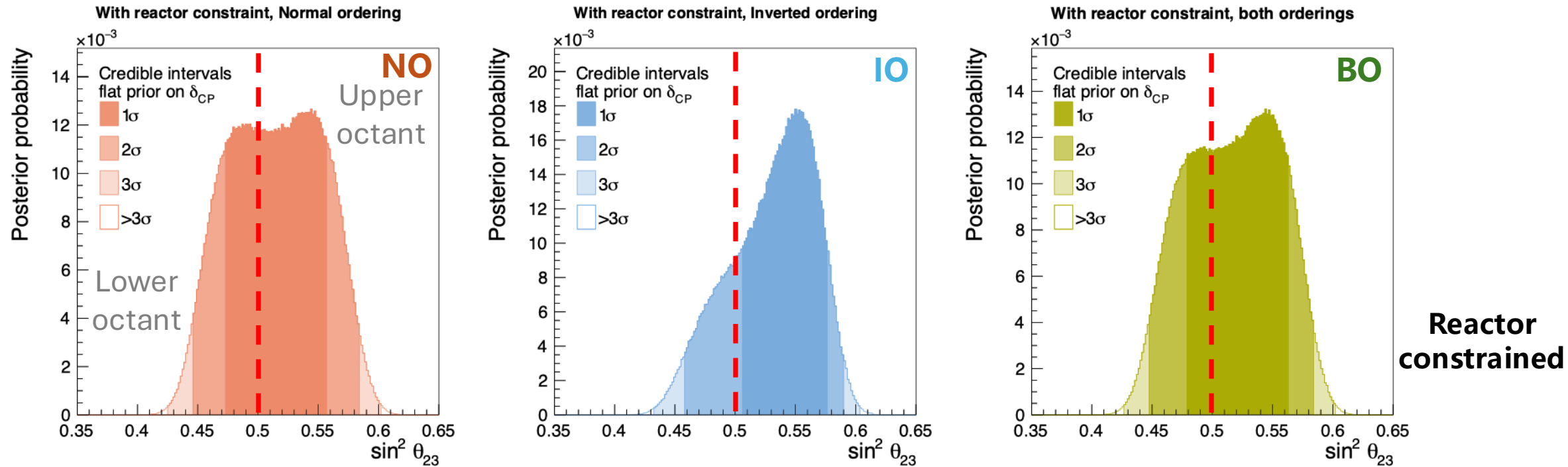


Accelerator



Atmospheric

$\sin^2 \theta_{23}$ Posterior Probability Distribution



- Slight preferences for upper octant of $\sin^2 \theta_{23}$

Joint fit results

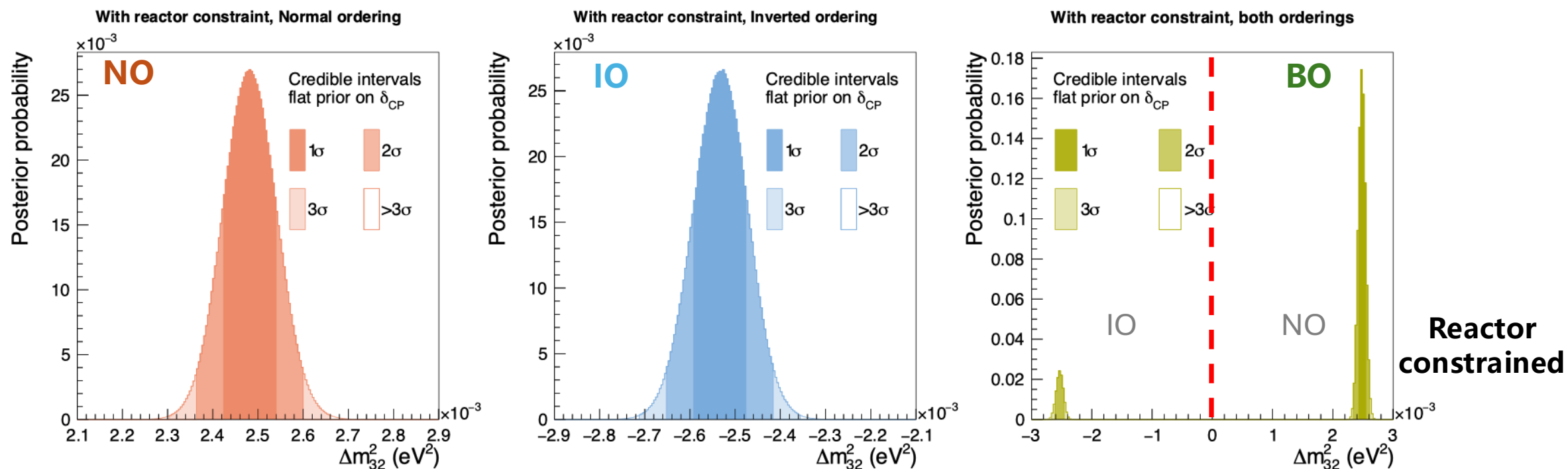


Accelerator



Atmospheric

Δm^2_{32} Posterior Probability Distribution



- Preferences for normal ordering

For two competing hypothesis H_1 and H_2 :

$$B\left(\frac{H_1}{H_2}\right) = \frac{P(H_1|\text{data})}{P(H_2|\text{data})}$$

Bayes factor	T2K+SK	T2K(OA20)
B(NO/IO)	7.20	4.21

Joint fit results

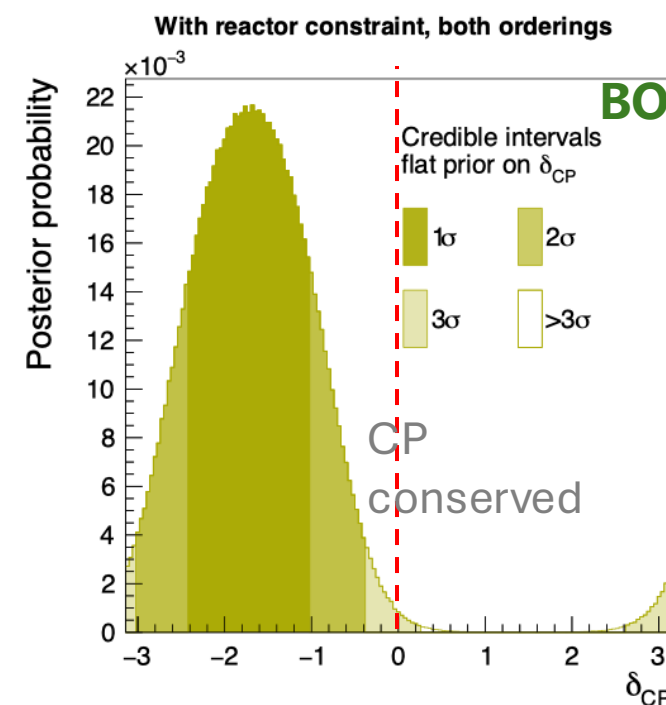
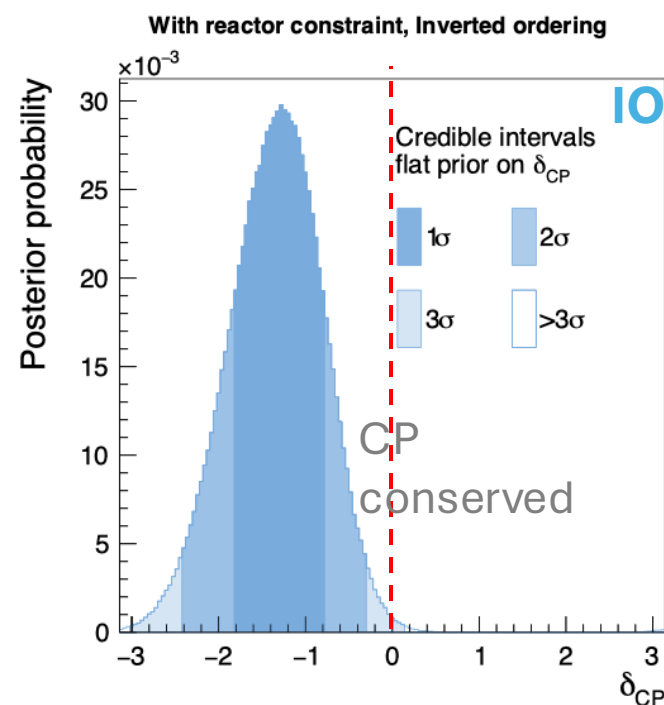
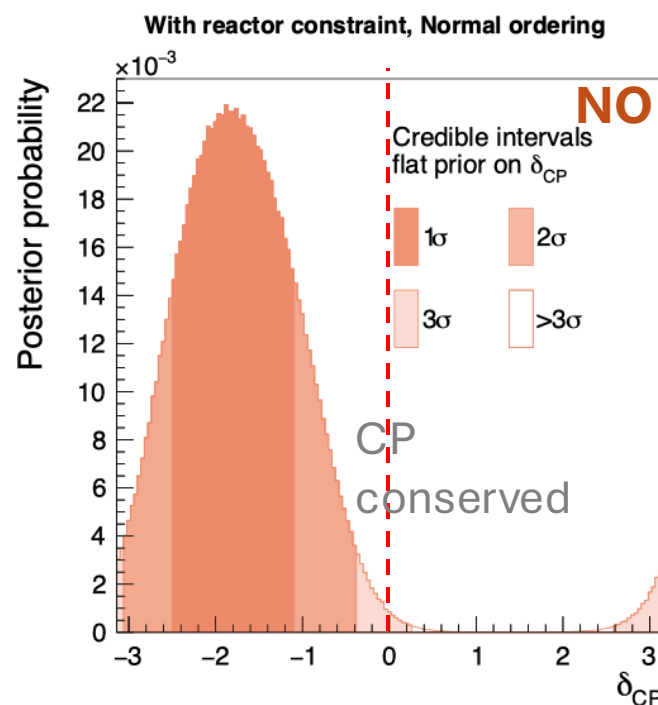


Accelerator



Atmospheric

δ_{CP} Posterior Probability Distribution



Reactor
constrained

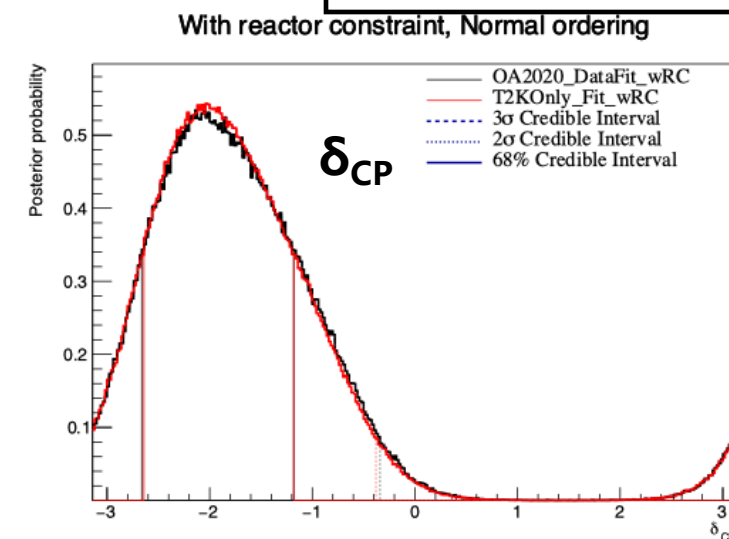
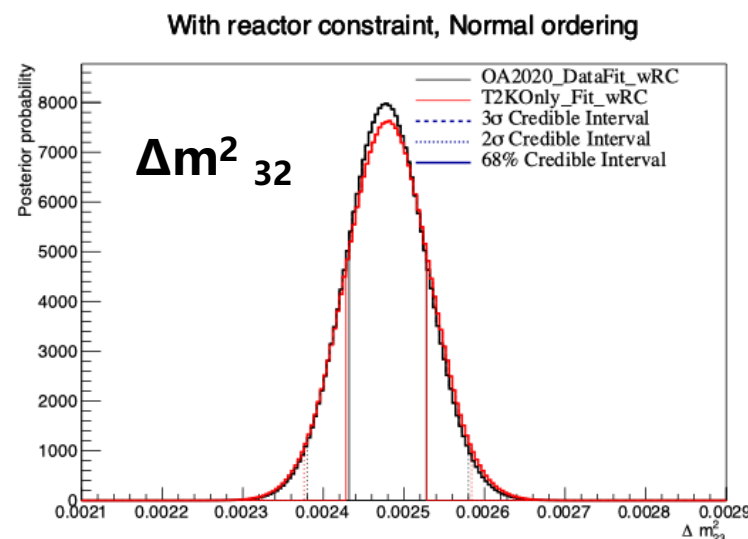
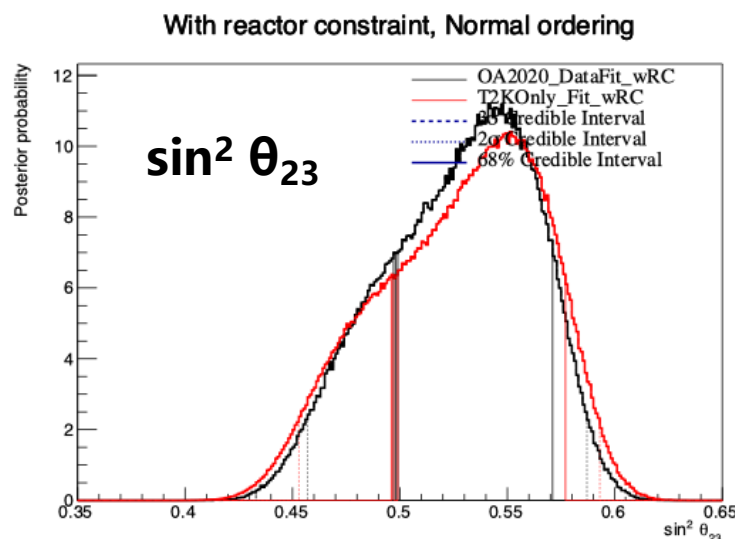
- For flat prior in δ_{CP} , the CP-conserving value can be excluded with a significance of 2σ



Accelerator

Normal Ordering
Reactor constrained
Black: T2K OA20
Red: T2K-only

Single experiment fit results: T2K-only



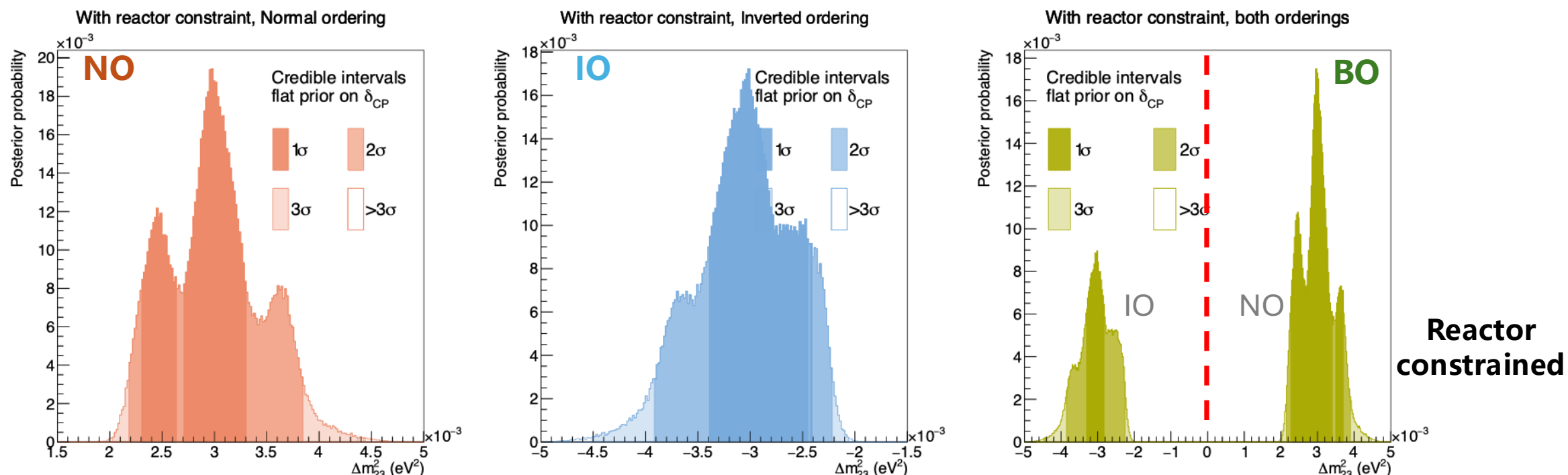
- The two fits agree to each other in the overall shapes, despite some slight differences due to the changes in the joint analysis to T2K OA20
- This consistency with the T2K analysis validates the implementation of the joint framework

Single experiment fit results: SK(ND)-only



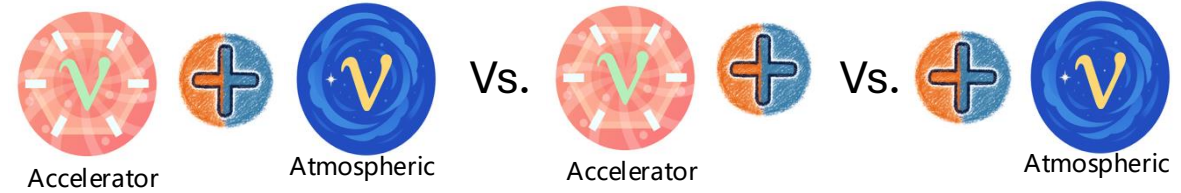
Atmospheric

Δm^2_{32} Posterior Probability Distribution

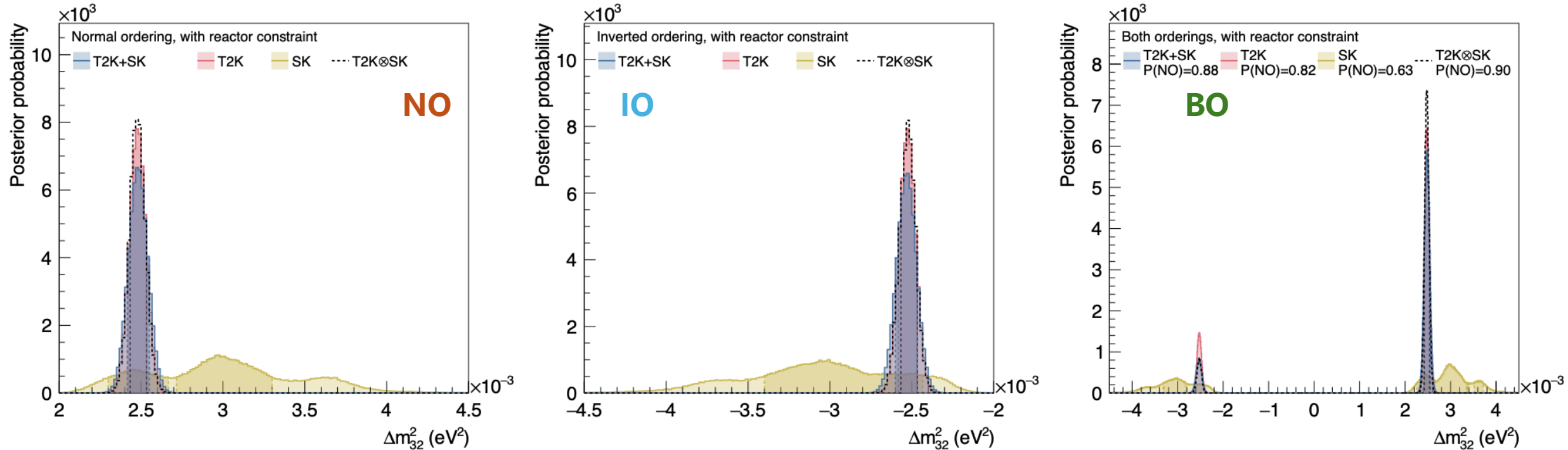


- A tri-modal shape appears for all mass orderings!
- What is its impact?
- Why?

Cross fits comparison



Δm^2_{32} Posterior Probability Distribution



- The tri-modality of the SK-only Δm^2_{32} posterior has no impact on the overall result due to the T2K constraint in the region
- But still, why?

(Potential) Upgrades

1. Atm. neutrino oscillation
probability

2. Far detector uncertainty

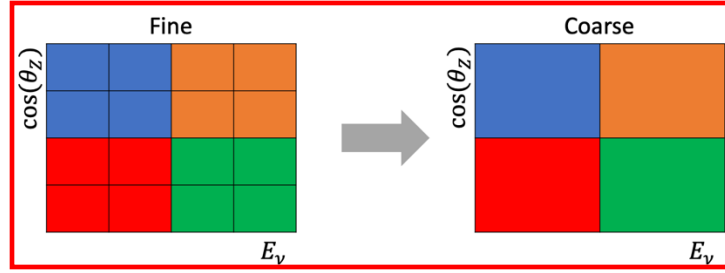


Atmospheric

Atmospheric neutrino oscillation

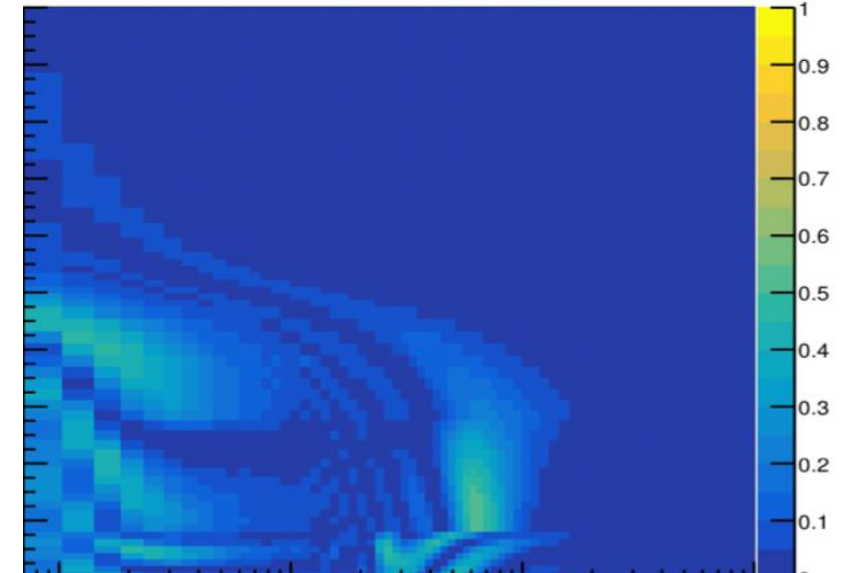
- ❖ Fast oscillation smearing leads to SK-only Δm^2_{32} tri-modality
- ❖ Production height averaging

$$\nu_\mu \rightarrow \nu_e$$



"Sub-sampling
averaging"

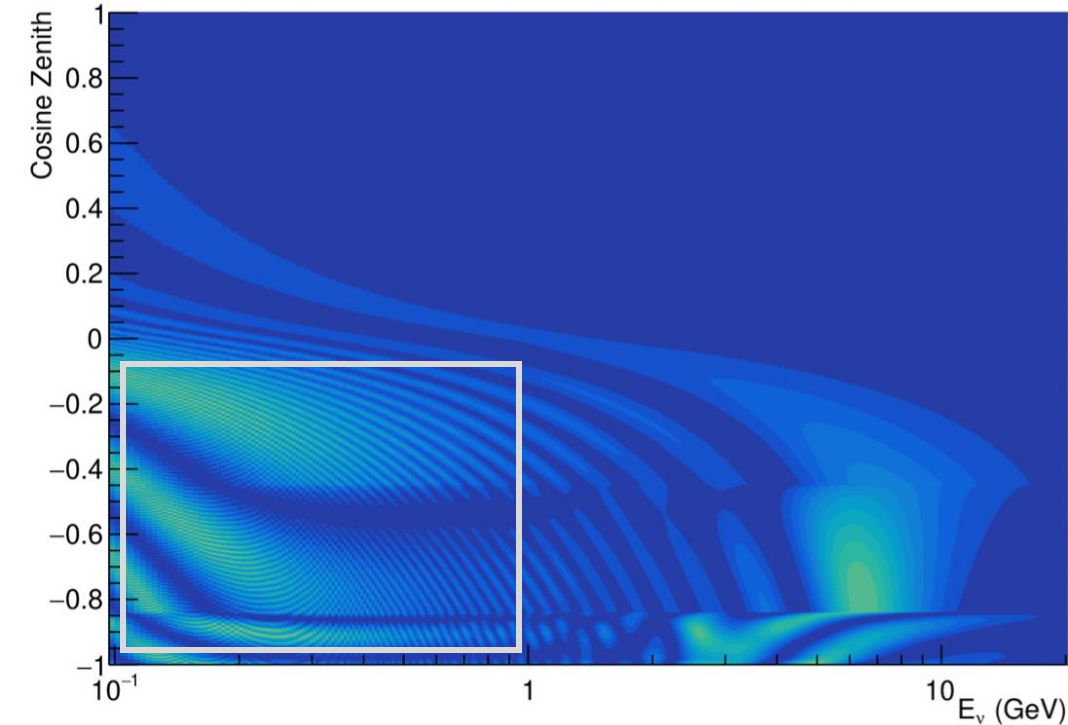
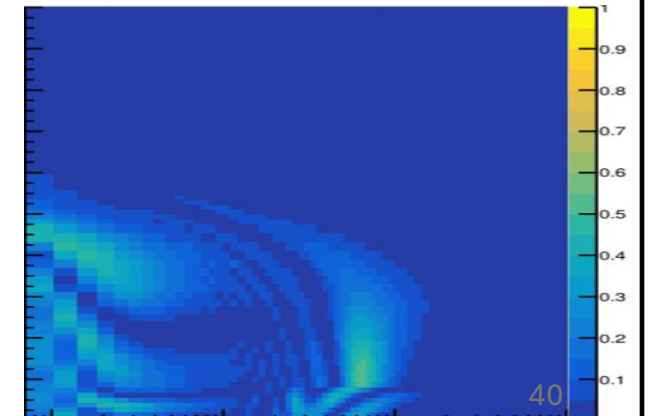
smeared



Credit to: Dan Barrow

Production height averaging

- Atm. neutrinos are produced at heights from 0 km to 50 km
- Analytically average the osc. probability over heights



- Fast oscillation for up-going neutrinos with sub-GeV energies
- Computationally expensive for precise calculation

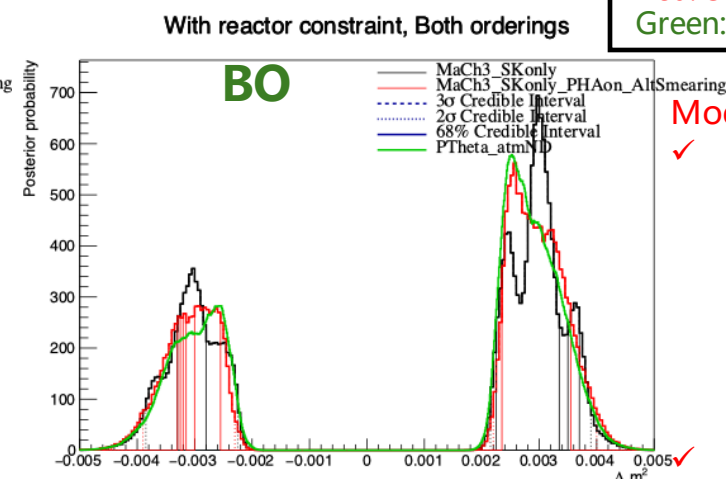
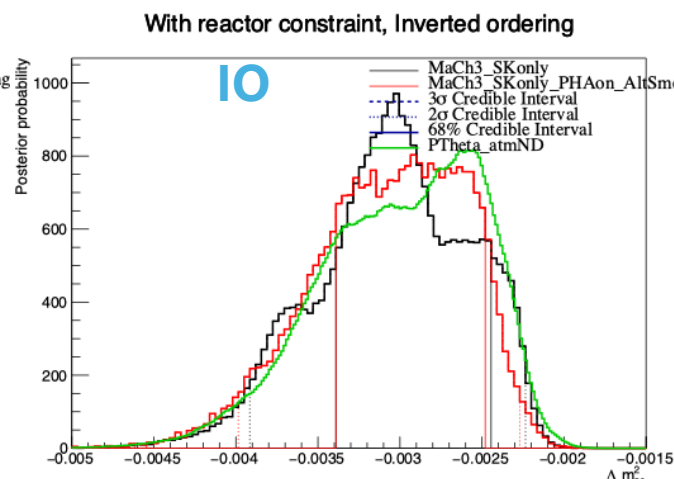
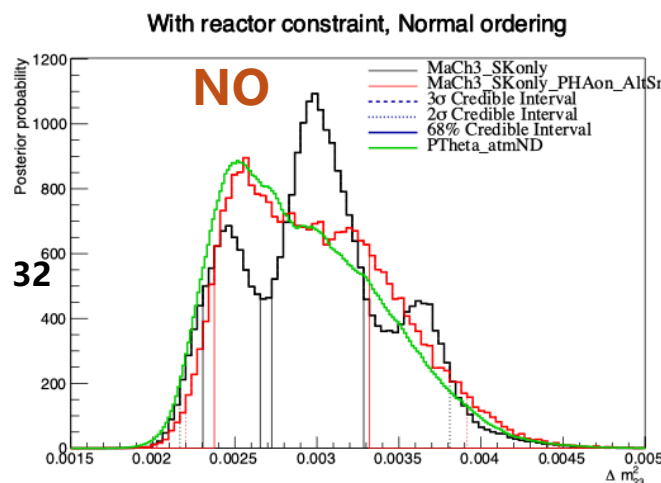
Single experiment fit results: SK(ND)-only



Atmospheric

Black: SK-only
Red: SK-only-Osc.Mod
Green: P-theta SK-only

Δm^2_{32}

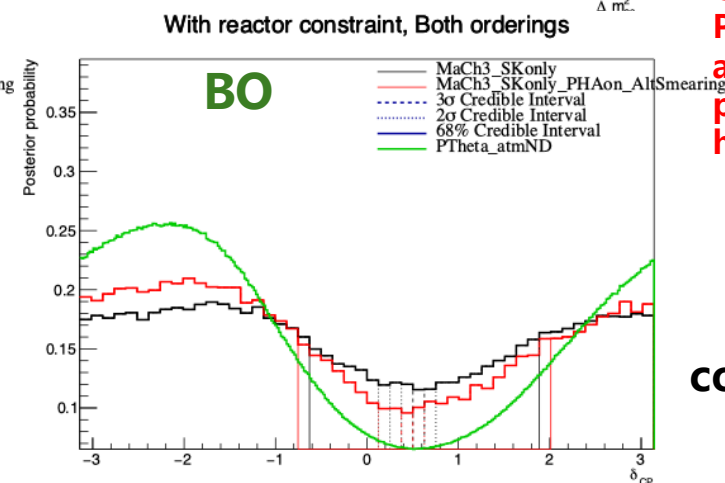
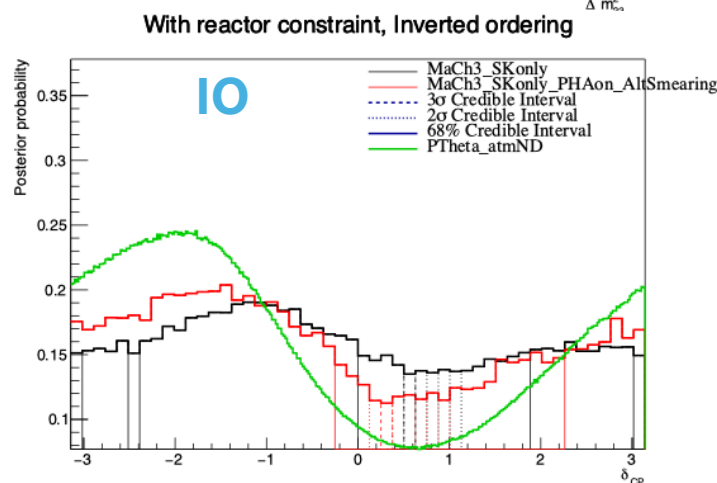
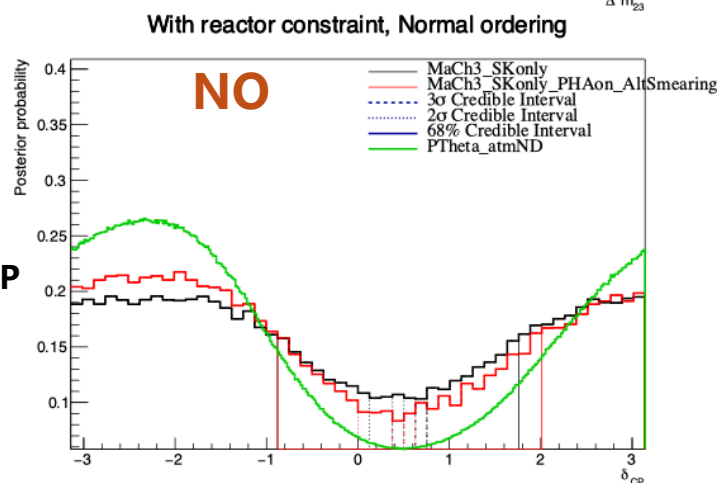


Modifications:

✓ Switching to another coarse grid with wider binning in fast oscillation treatment

Osc. Probability averaged over production height

δ_{CP}



Reactor constrained

- After modifying atm. nu oscillation probability, there are improvements on oscillation parameters for SK-only fits

Joint fit results



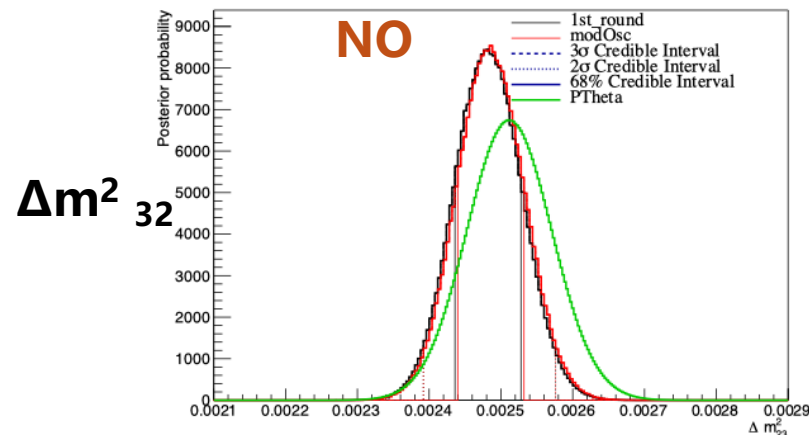
Accelerator



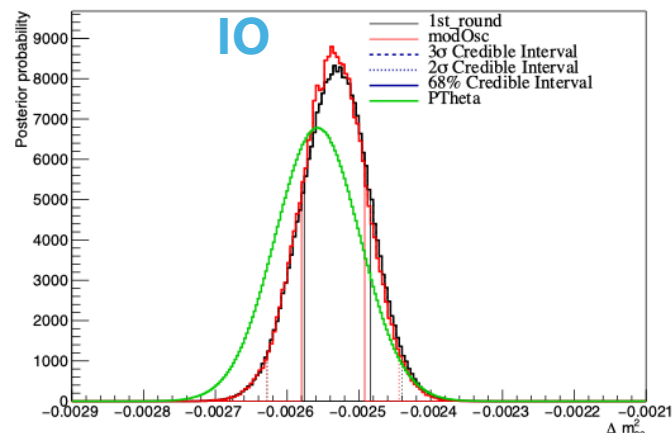
Atmospheric

Black: Joint fit
Red: Joint fit-Osc.Mod
Green: P-theta Joint fit

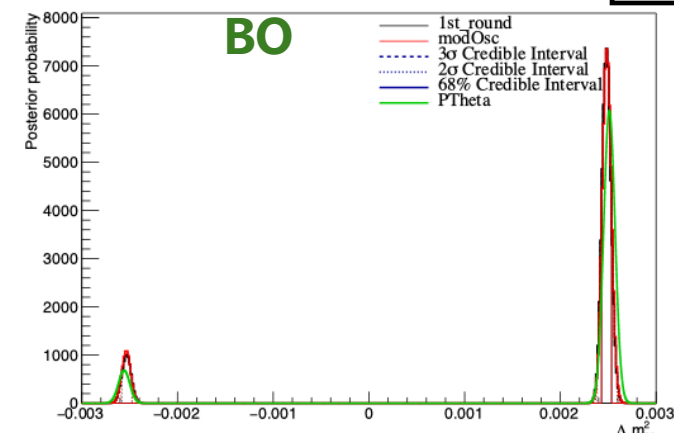
With reactor constraint, Normal ordering



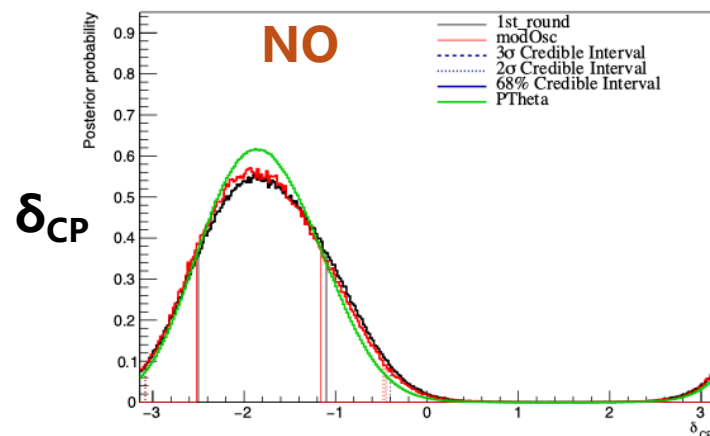
With reactor constraint, Inverted ordering



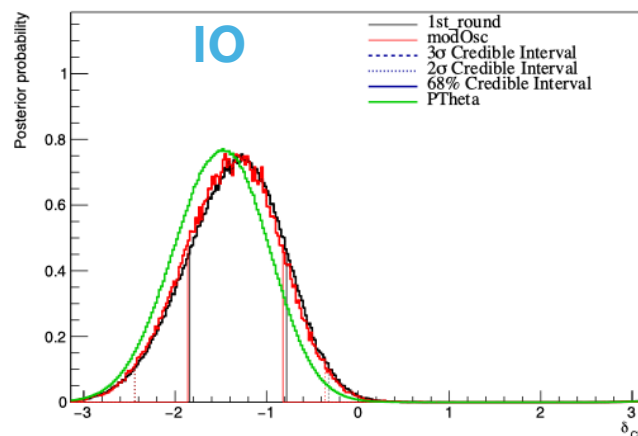
With reactor constraint, Both orderings



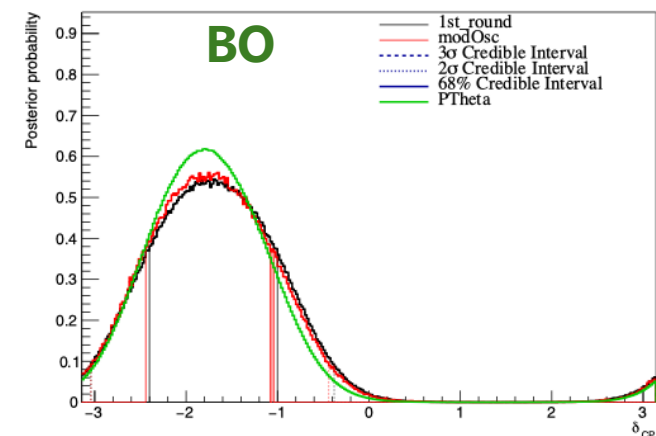
With reactor constraint, Normal ordering



With reactor constraint, Inverted ordering



With reactor constraint, Both orderings



Reactor
constrained

Joint fit results



Accelerator

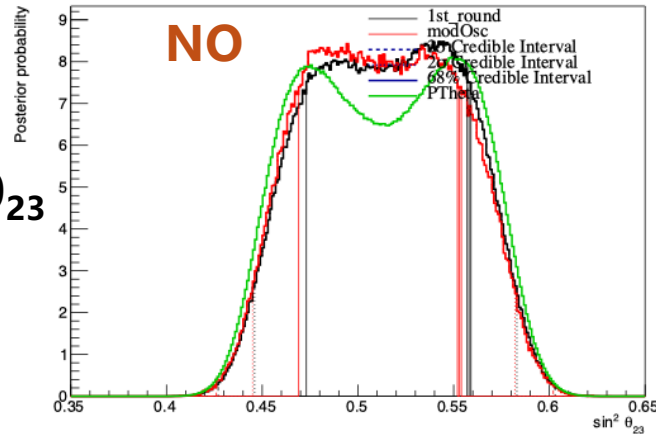


Atmospheric

Black: Joint fit
Red: Joint fit-Osc.Mod
Green: P-theta Joint fit

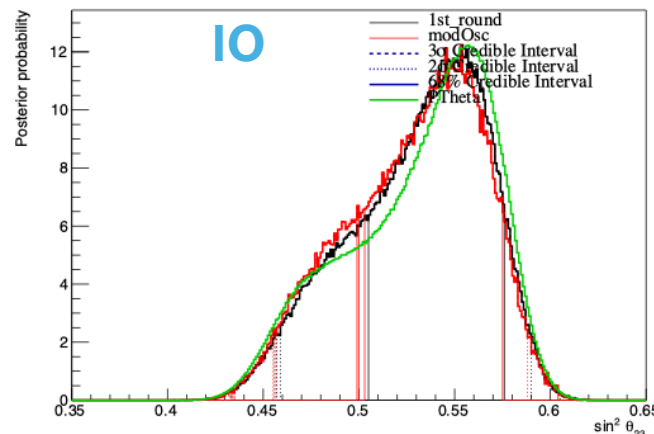
With reactor constraint, Normal ordering

NO



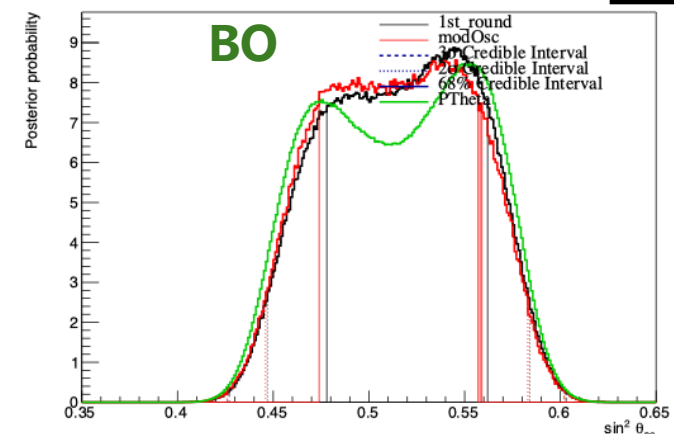
With reactor constraint, Inverted ordering

IO



With reactor constraint, Both orderings

BO



Reactor
constrained

□ The modifications' impacts:

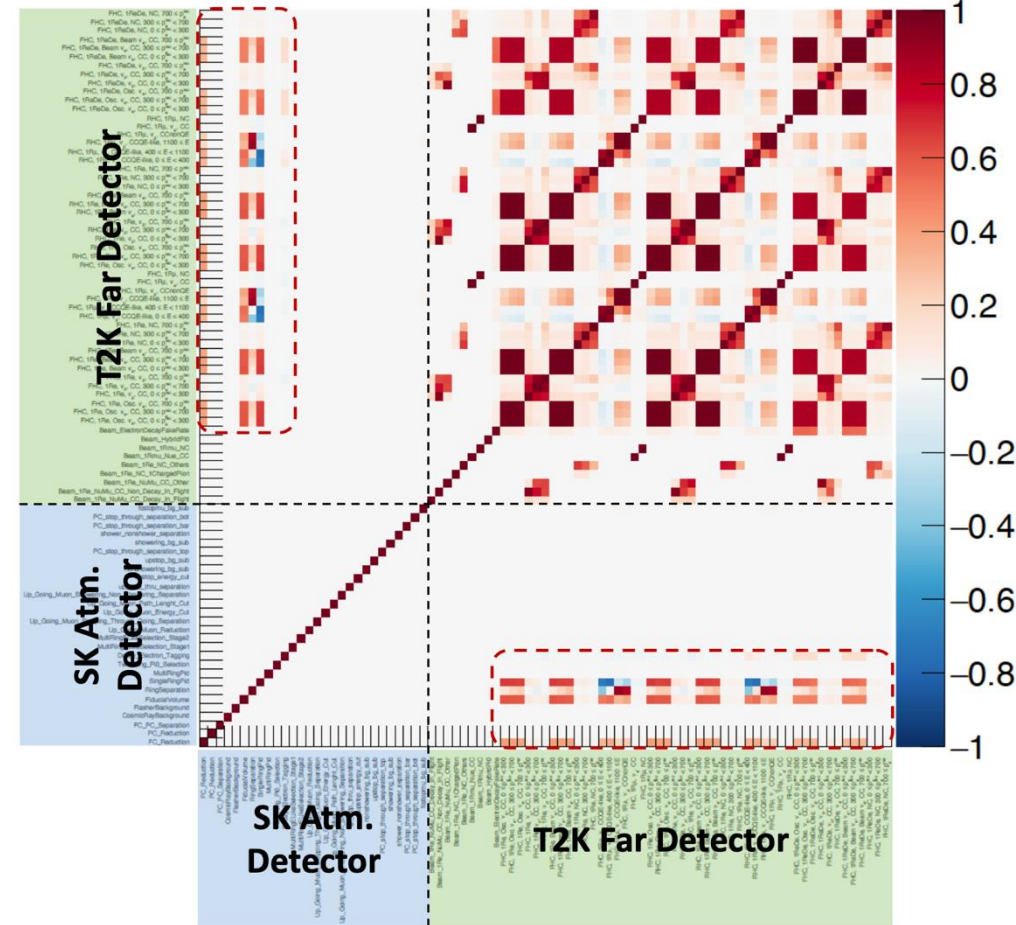
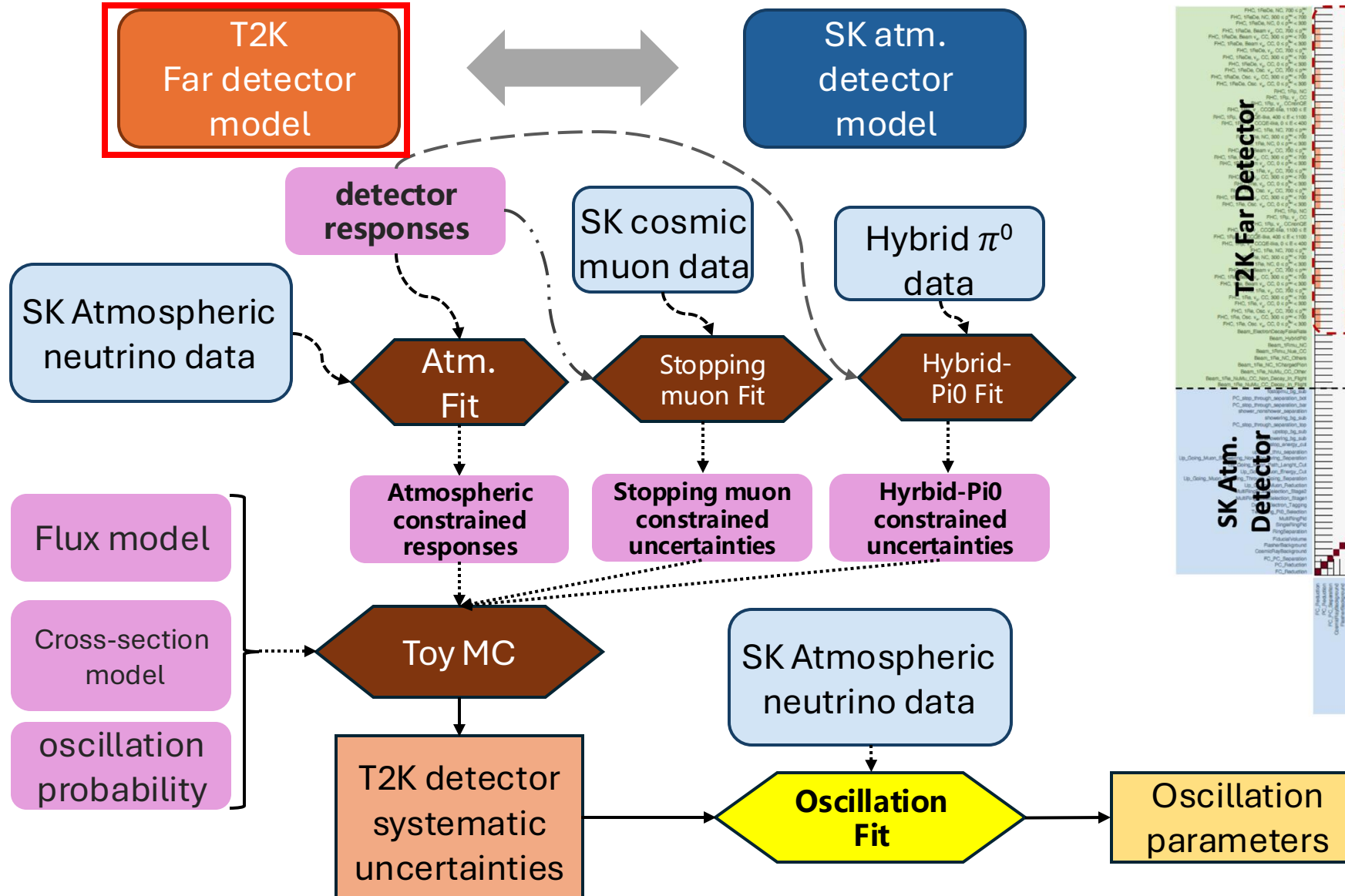
- a small deviation of the peaks of Δm^2_{32} shifting towards larger absolute value
- very slight tighter constraining on δ_{CP}
- more equivalent preference for the upper and lower octant of $\sin^2 \theta_{23}$

□ The P-theta's official Bayesian results are also put into overlay as a cross-fitter comparison

Detector Uncertainty

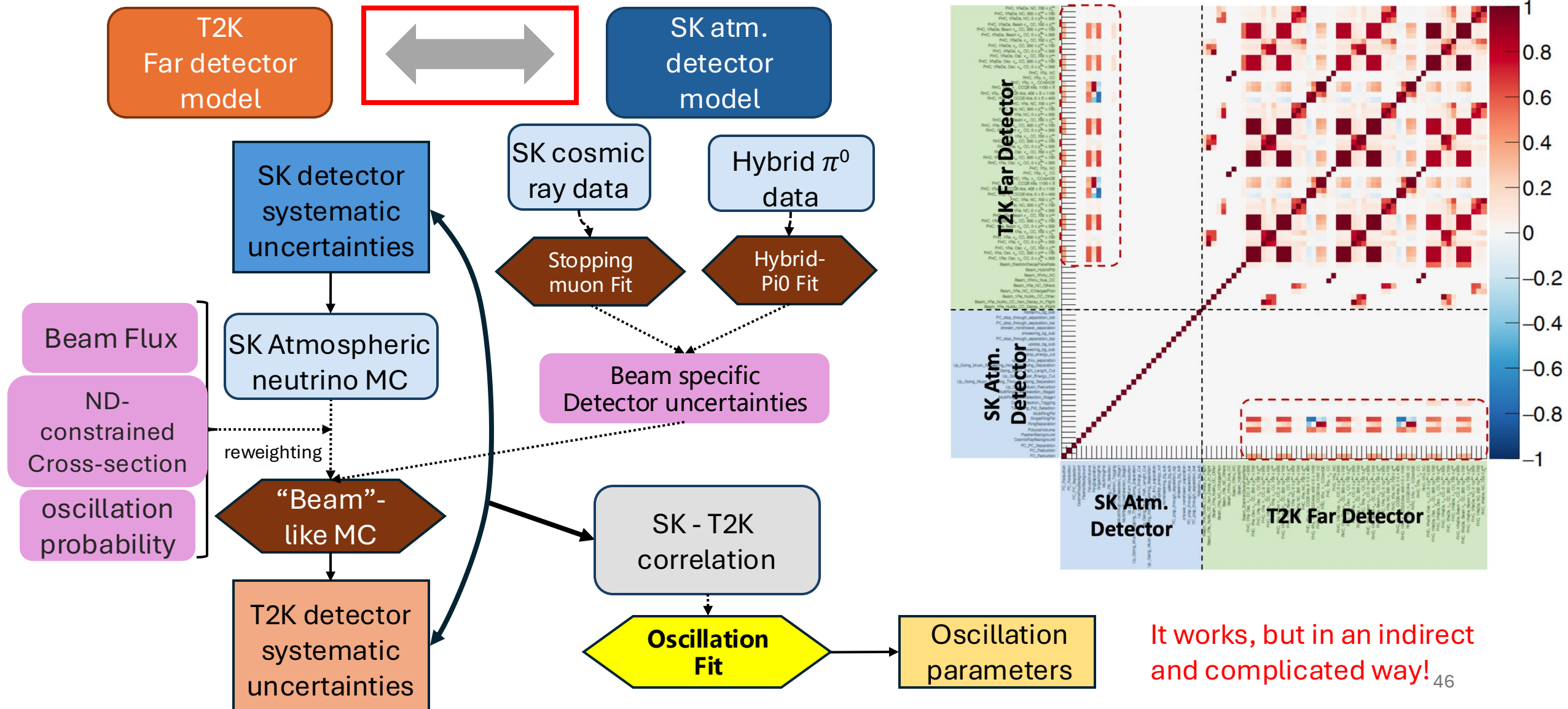
- Why we need an alternative detector model?
- What is the smearing/shifting parameterization of detector uncertainty?
- Any benefits to the oscillation sensitivity?

Revisit the standard detector model

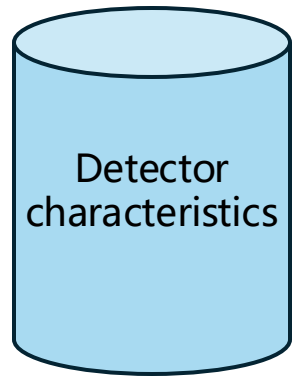


Biased by double-counting SK atm. samples!

Revisit the standard detector model



A more robust option: Smearing/Shifting parameterization



Distortions on event selection

Event selection
variable

smear

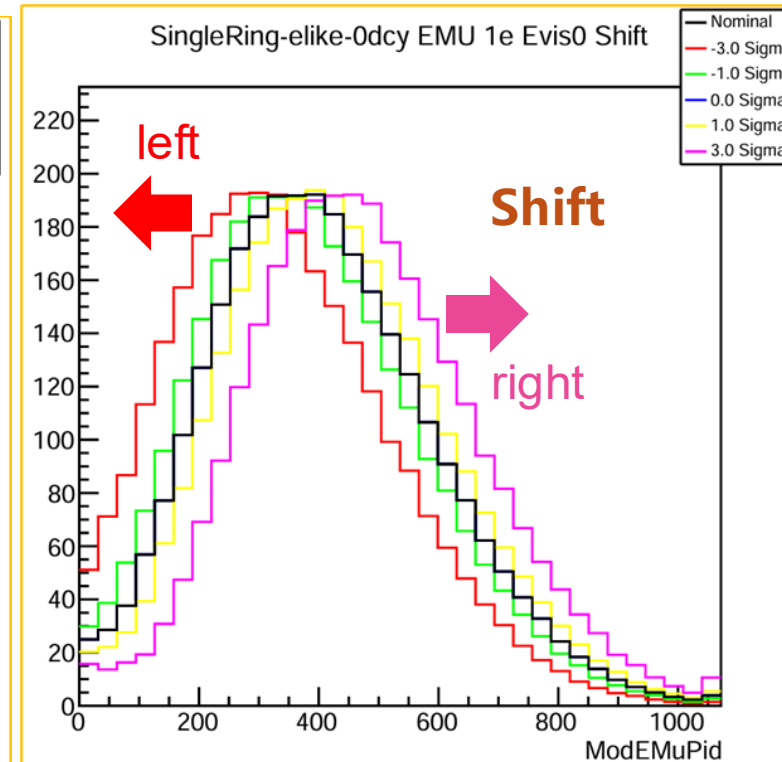
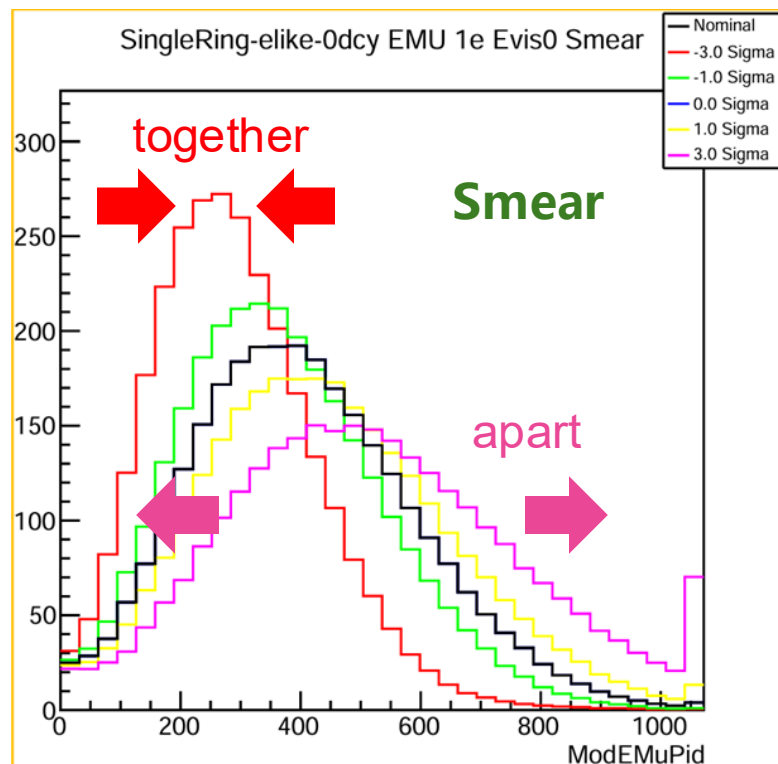
shift

$$\mathcal{X} \rightarrow \mathcal{X}' = \alpha\mathcal{X} + \beta$$

Example:

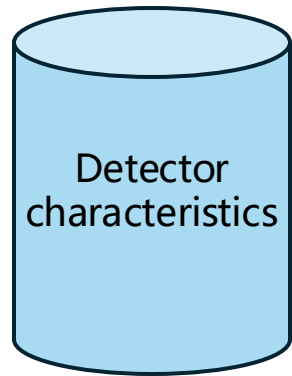
Distorting the e/μ PID

For 1R e-like events of 1e topology with visible energy < 0.3GeV



Credit to: Michael Reh

A more robust option: Smearing/Shifting parameterization



Distortions on event selection

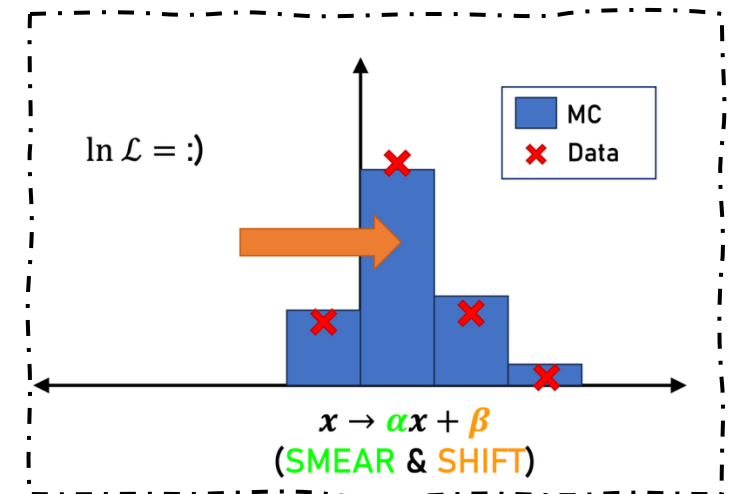
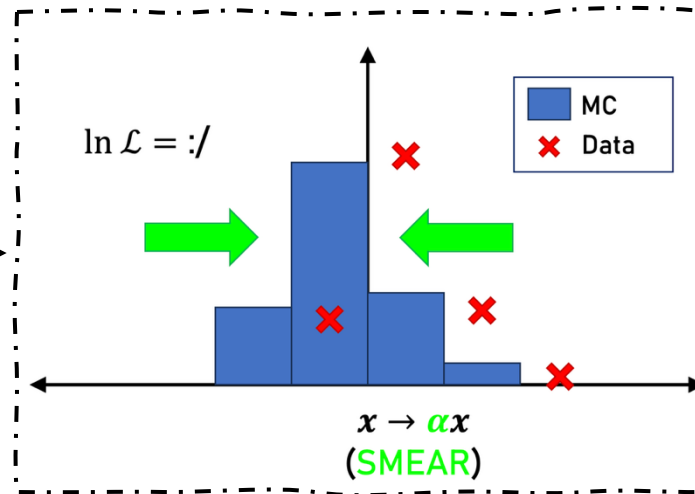
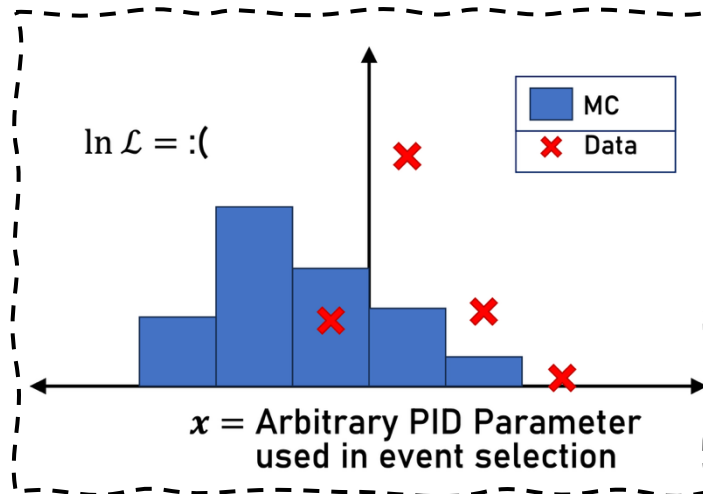
Event selection variable

smear

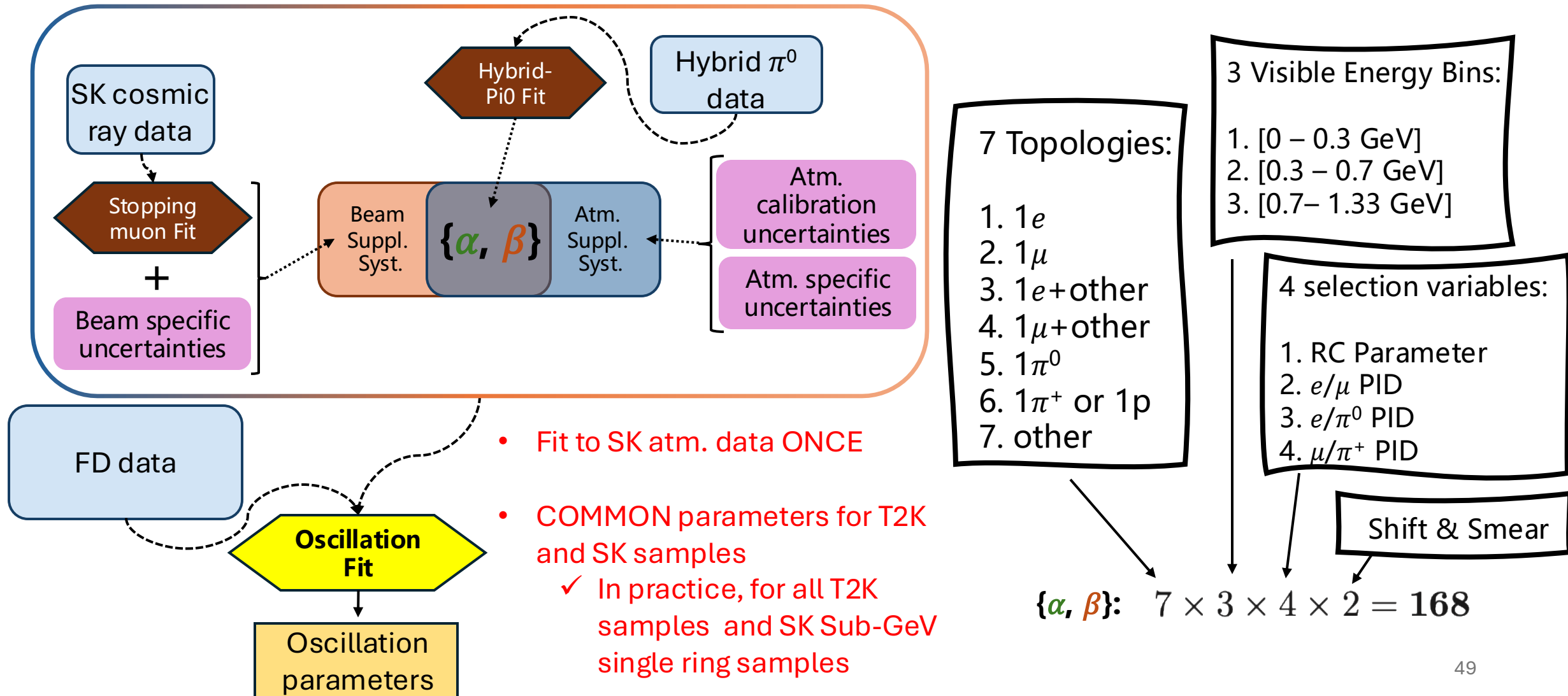
shift

$$\mathcal{X} \rightarrow \mathcal{X}' = \alpha\mathcal{X} + \beta$$

- Compare data and MC in histograms binned in the event selection variables
- The set of $\{\alpha, \beta\}$ encodes detector's uncertainties on event selections



A more robust option: Smearing/Shifting parameterization



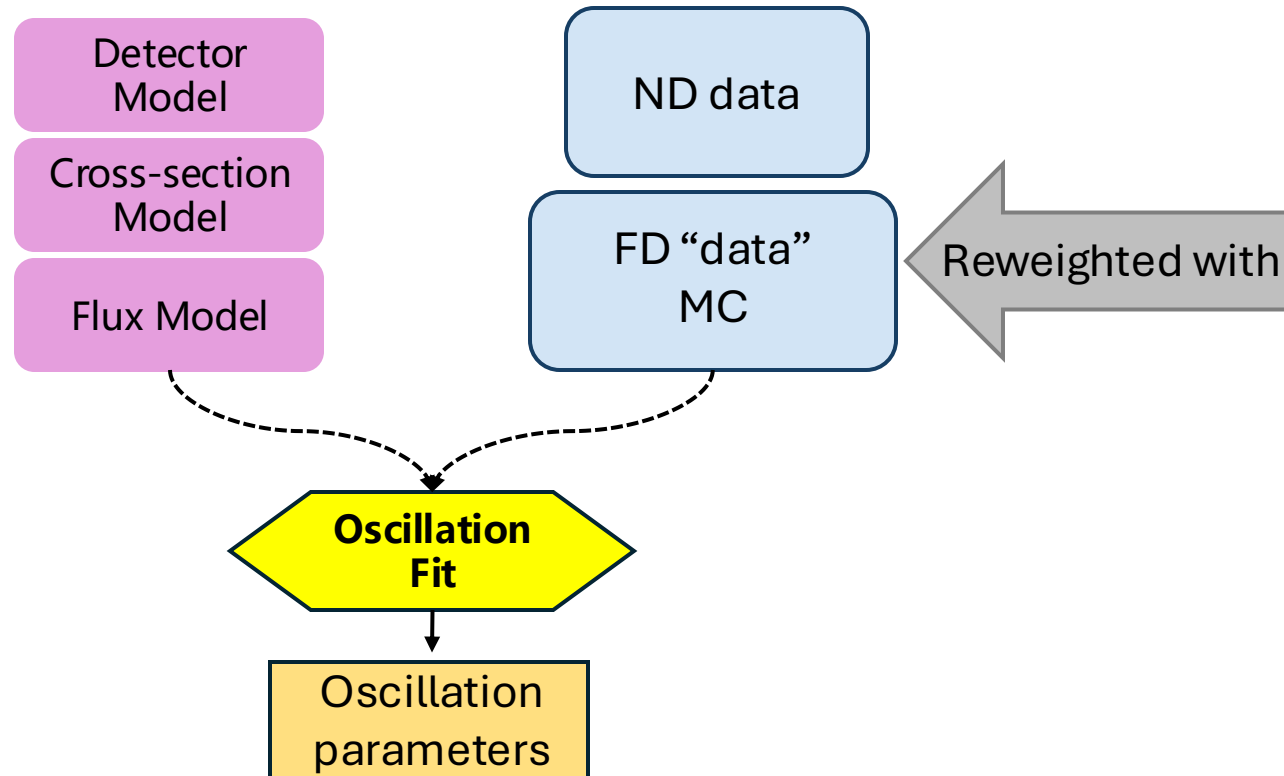
Sensitivity study: Asimov fit



Accelerator



Atmospheric



Oscillation Parameter	Asimov A value
$\sin^2 \theta_{12}$	0.304
$\sin^2 \theta_{13}$	0.0218
$\sin^2 \theta_{23}$	0.528
Δm^2_{21}	$7.53 \times 10^{-5} \text{ eV}^2$
Δm^2_{32}	$2.509 \times 10^{-3} \text{ eV}^2$
δ_{CP}	-1.601

&

ND Constraints
For
Beam flux and
cross-sections

Joint Asimov fit results

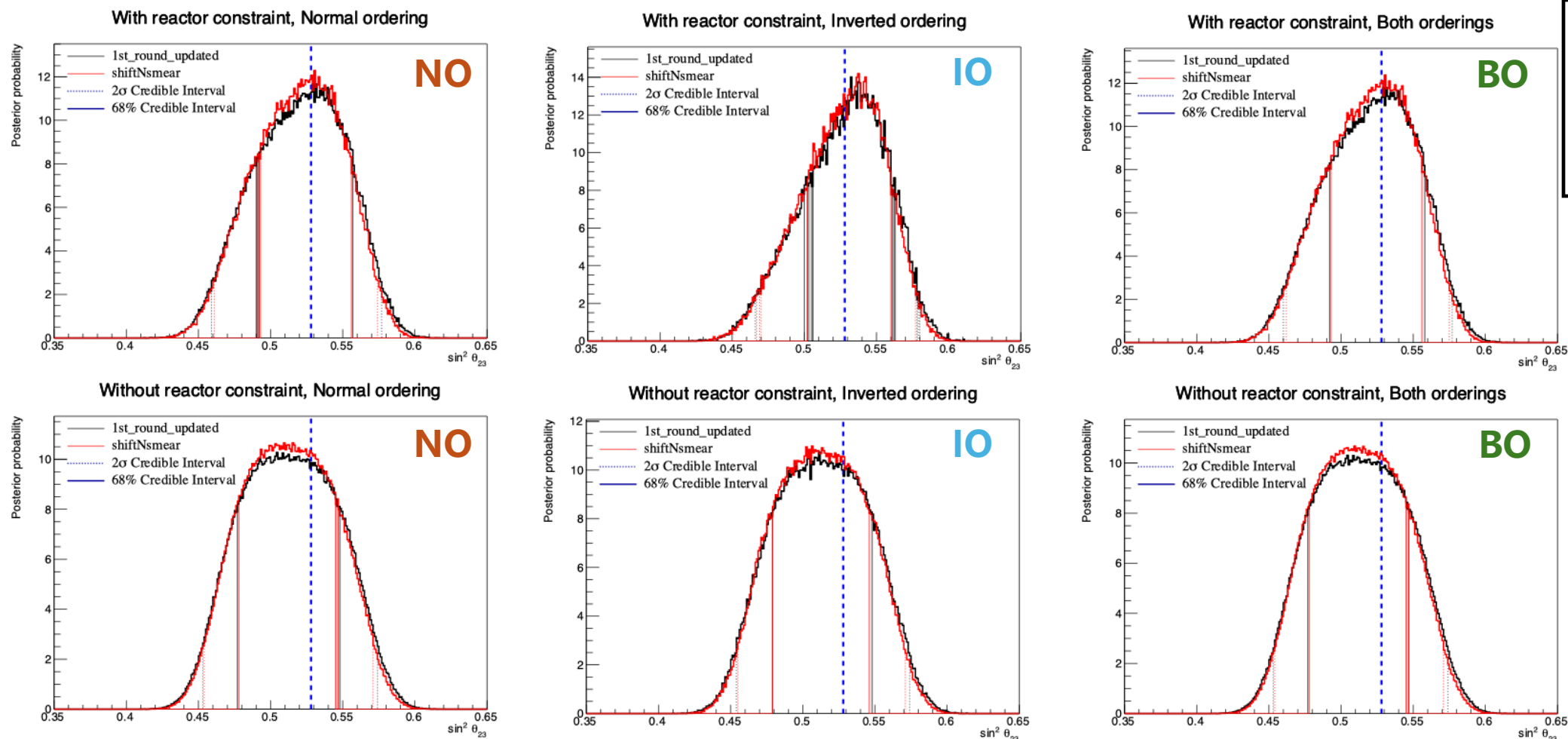
$\sin^2 \theta_{23}$ Posterior Probability Distribution



Accelerator



Atmospheric



Black: Standard joint fit
Red: Joint fit - Alternative Detector model
Blue: Asimov truth

Reactor constrained

No reactor constraints

- Small improvements on constraining $\sin^2 \theta_{23}$

Joint Asimov fit results

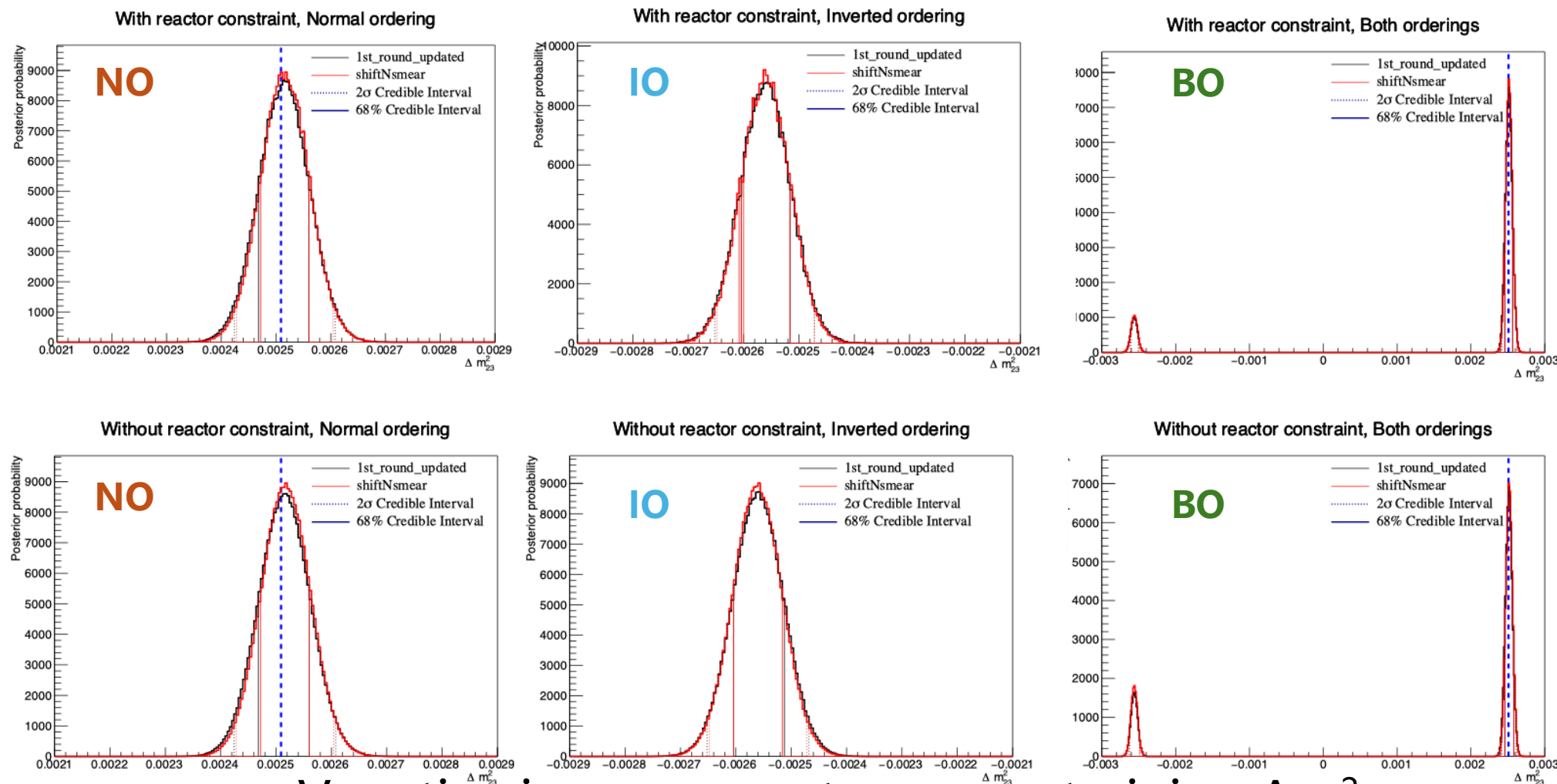
Δm^2_{32} Posterior Probability Distribution



Accelerator



Atmospheric



Black: Standard joint fit
Red: Joint fit - Alternative Detector model
Blue: Asimov truth

Reactor constrained

No reactor constraints

- Very tiny improvements on constraining Δm^2_{32}

Joint Asimov fit results

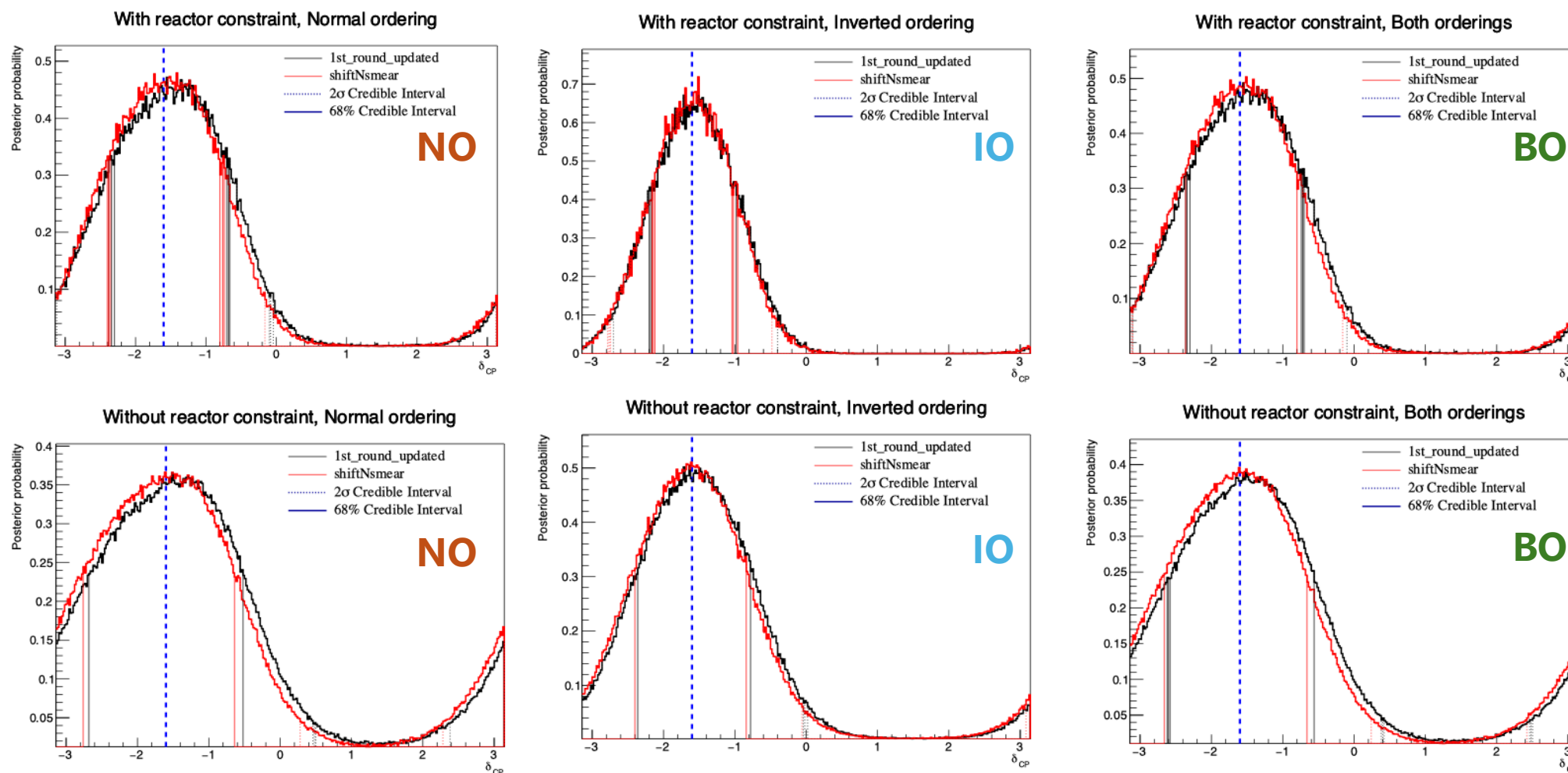
δ_{CP} Posterior Probability Distribution



Accelerator



Atmospheric



Black: Standard joint fit
Red: Joint fit - Alternative Detector model
Blue: Asimov truth

Reactor constrained

No reactor constraints

- Small but clear reduced bias in δ_{CP}

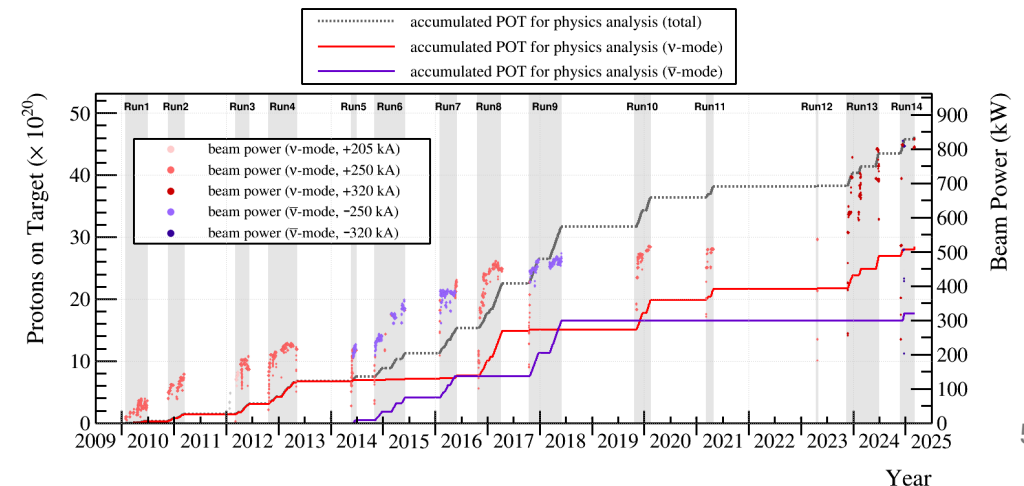
Let me wrap up.

Summary

- Neutrino oscillation experiments are entering the precision era, aiming at θ_{23} , Δm^2_{32} , δ_{CP} and determining neutrino **mass ordering**.
- Besides the upcoming experiments with larger detectors, combined analysis of existing experiments can exploit the potential of current data and provide constraints with increased statistics, e.g., **T2K-SK joint analysis**
 - Official results of T2K-SK joint analysis has been released:
 - T2K-SK joint analysis can exclude CP conservation at 2σ around;
 - It favors normal mass ordering,
 - and weakly prefers upper octant of θ_{23} , still within the maximal mixing consistence.
- The T2K-SK joint analysis is moving on for the second round; among the many possible options to **update the joint analysis**, we test and study:
 - Atmospheric neutrino **oscillation probability calculator**;
 - Alternative far **detector model with smearing/shifting parameters**
- Preliminary sensitivity studies demonstrate the **compatibility** and show **improvements** on oscillation parameters with the smearing/shifting parameterization model
 - Further studies are required before fitting data

Outlooks

- Future T2K-SK joint analysis would include:
 - More samples from both experiments
 - Correlating the SK atmospheric and T2K beam neutrino flux
 - Updating cross-section uncertainty model and developing extra specific systematics
 -
- The MaCh3 fitter is under upgrades to a new version with more speed and flexibility
- The two experiments have updated their apparatus recently, which can provide exciting physics
 - SK: Gd-doping
 - T2K: beamline upgrades; ND upgrades

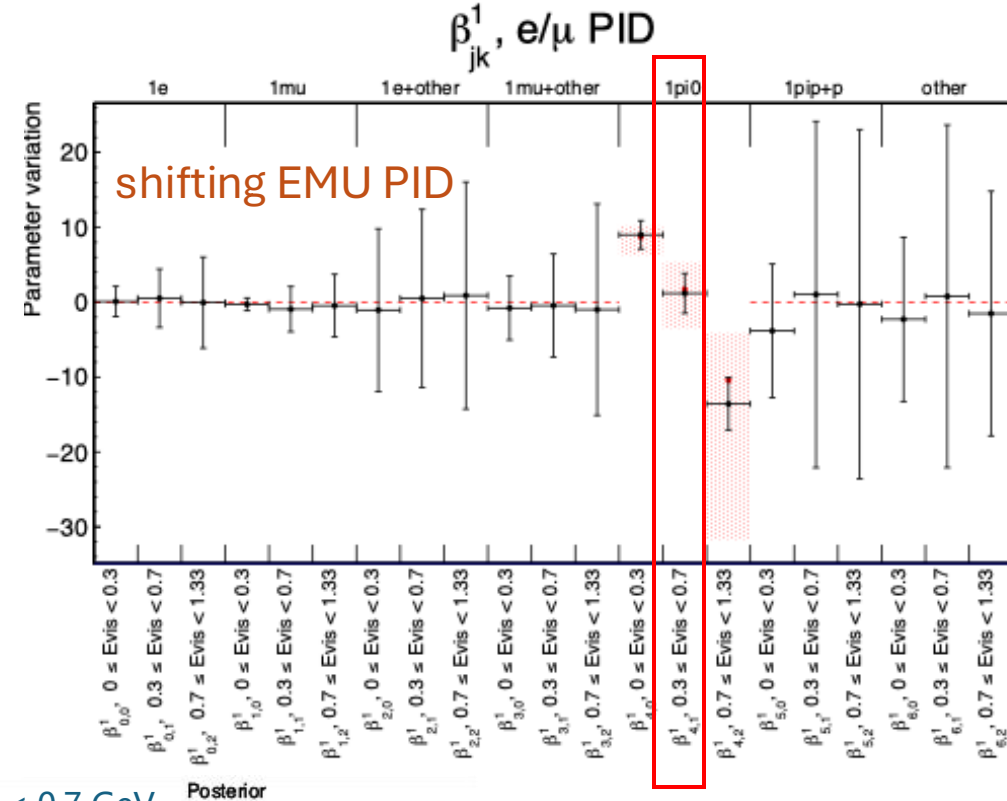
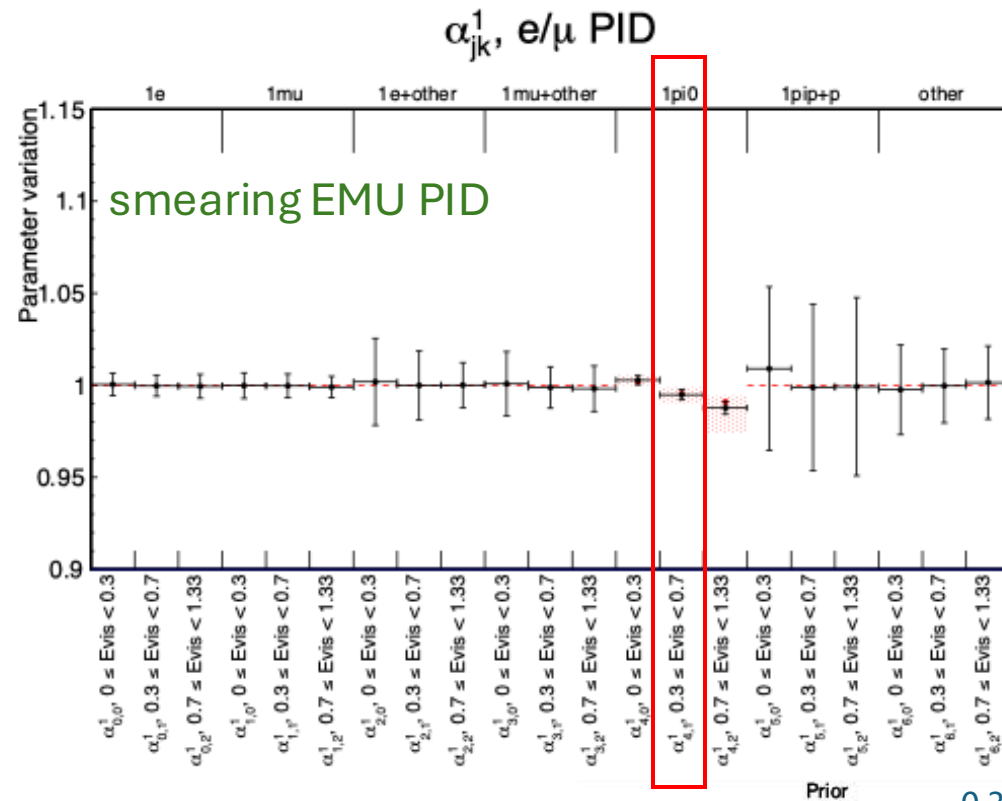


Thank You!

Questions?

Backup slides

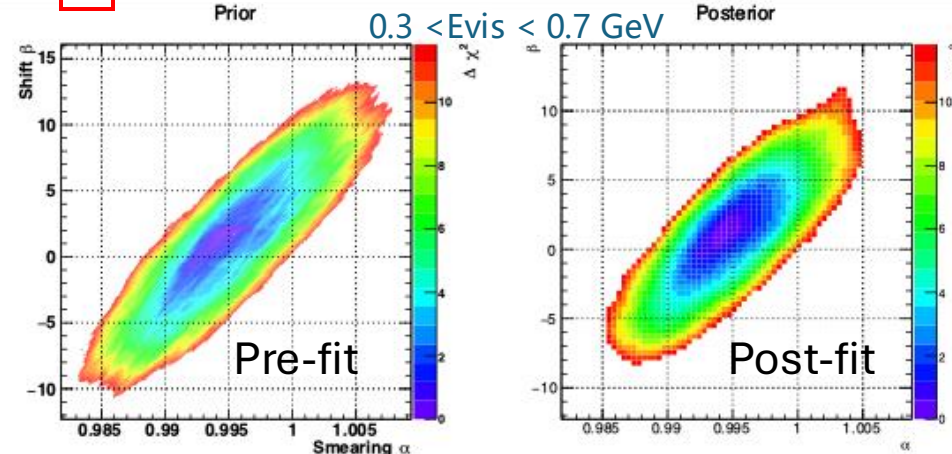
Joint Asimov fit results: Smearing/Shifting Param.



Example

Hybrid π^0
data

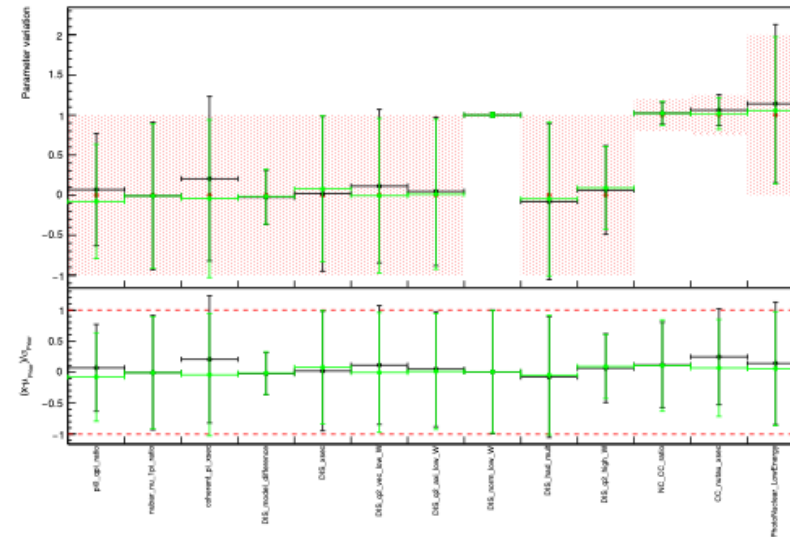
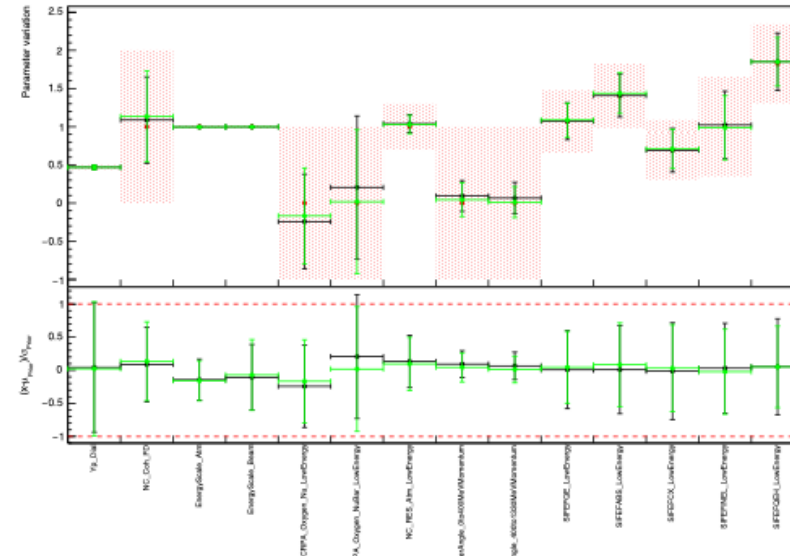
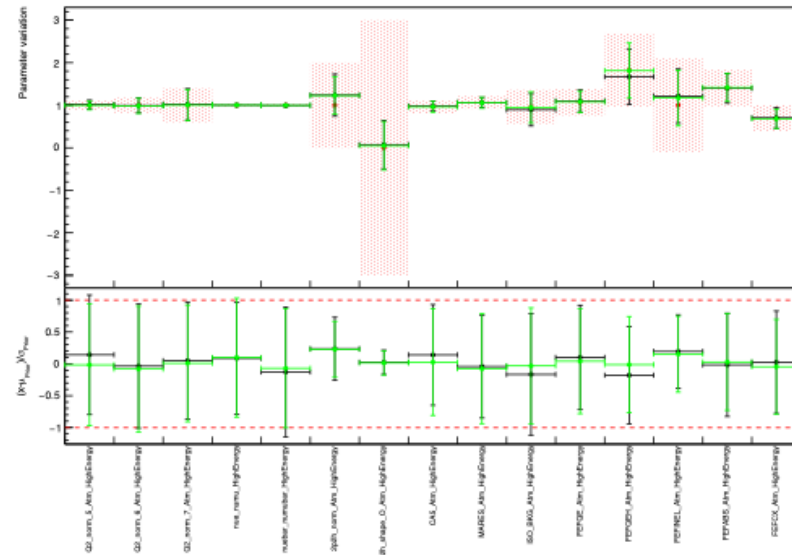
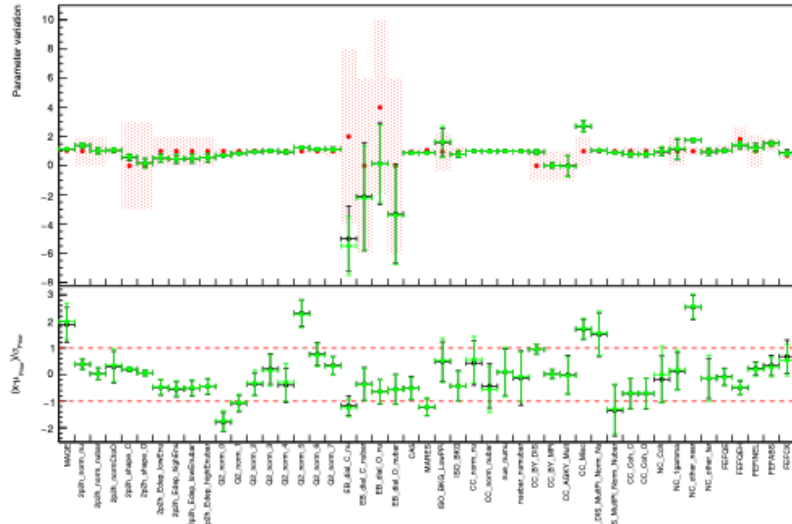
Hybrid-
Pi0 Fit



Black: Joint fit -
Alternative Detector
model
Posterior HPD

Red: Prior values

Joint Asimov fit results: Cross-section Param.



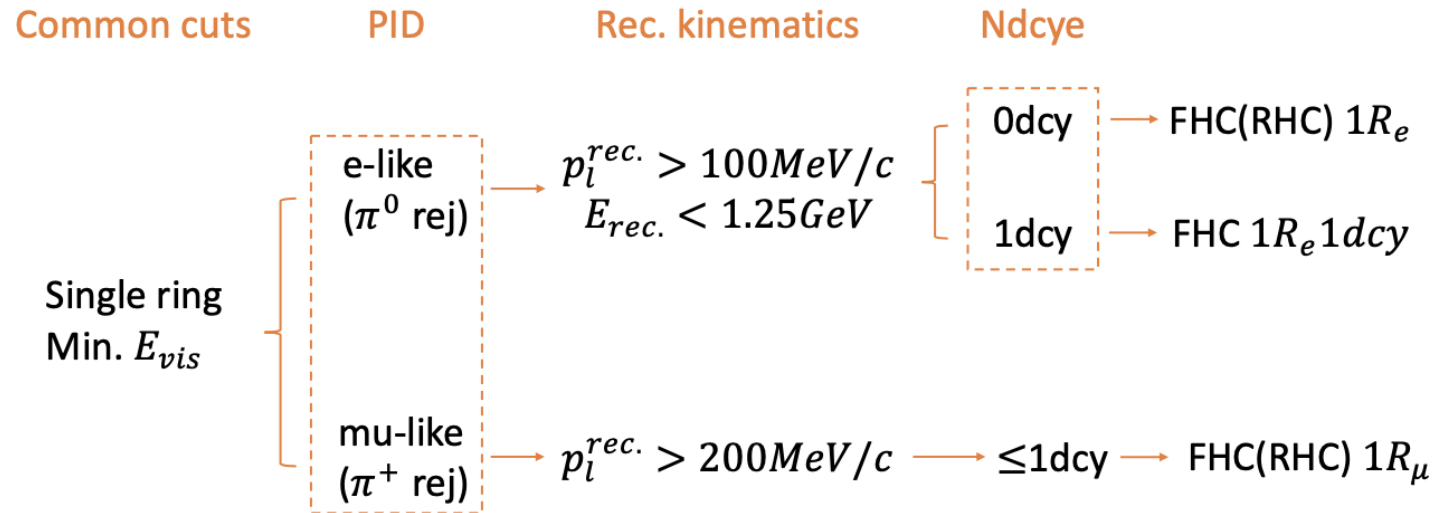
Black: Joint fit -
Alternative Detector
model Posterior

Red: Prior values

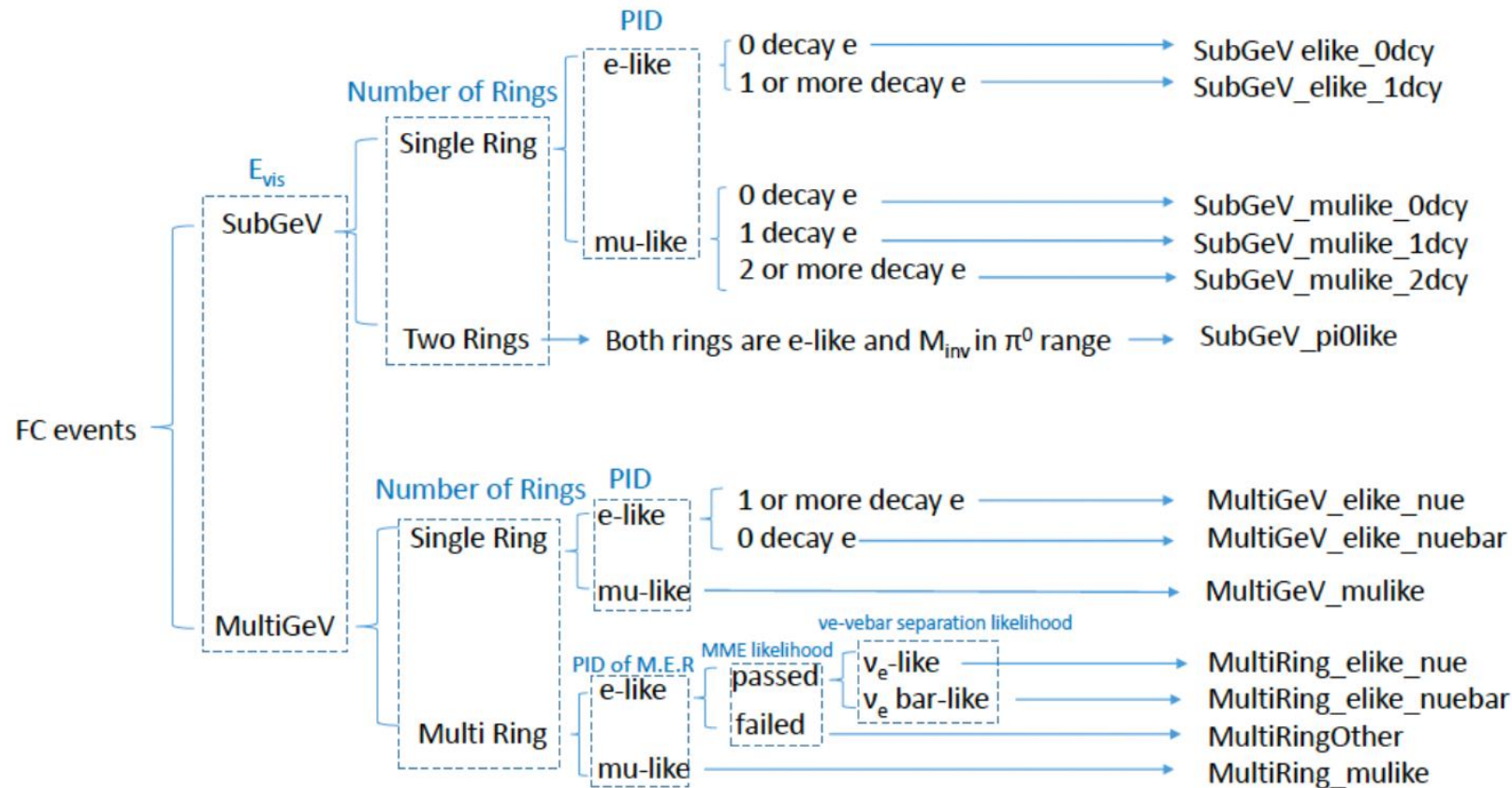
Green: Standard joint
fit Posterior

- No large deviations from standard joint fit posteriors

T2K beam samples' event selections

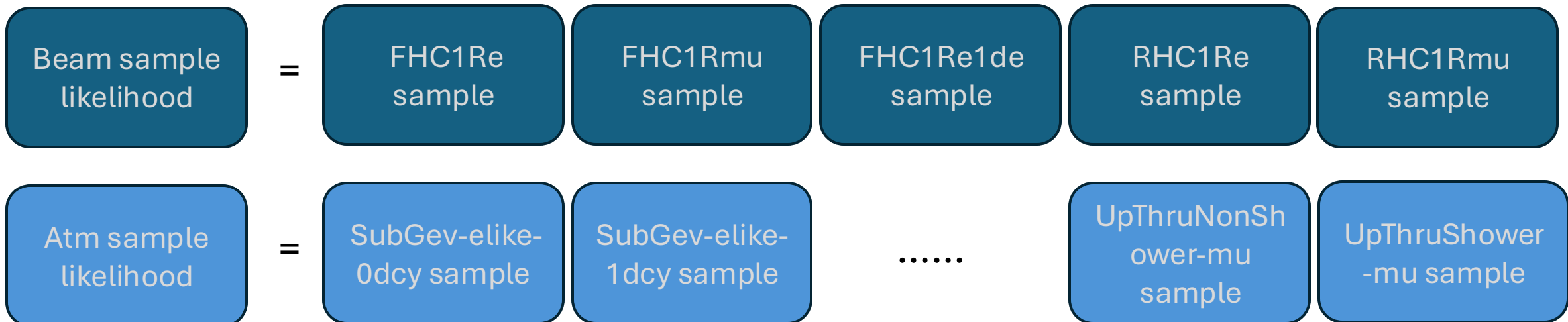


SK atmospheric Full Contained samples' event selections



Total likelihood

$$\begin{aligned}
 L(\mathbf{D}|\boldsymbol{\theta}, \boldsymbol{\alpha}) &= \prod_i^{\text{MC events}} L(\mathbf{D}_i|\boldsymbol{\theta}, \boldsymbol{\alpha}) \\
 &= \prod_i^{\text{ND}} L(\mathbf{D}_i|\boldsymbol{\theta}, \boldsymbol{\alpha}) \prod_i^{\text{beam}} L(\mathbf{D}_i|\boldsymbol{\theta}, \boldsymbol{\alpha}) \prod_i^{\text{atm}} L(\mathbf{D}_i|\boldsymbol{\theta}, \boldsymbol{\alpha})
 \end{aligned}$$



Binning events to histograms

Sample
likelihood

$$\prod_i^{\text{events}} L(\mathbf{D}_i | \boldsymbol{\theta}, \boldsymbol{\alpha}) = \prod_b^{\text{bin}} L_b(N_b^{\text{obs}} | \boldsymbol{\theta}, \boldsymbol{\alpha})$$

- Likelihood as a Poisson distribution:

$$L_b(N_b^{\text{obs}} | \boldsymbol{\theta}, \boldsymbol{\alpha}) = N_b^{\text{pred}}(\boldsymbol{\theta}, \boldsymbol{\alpha})^{N_b^{\text{obs}}} \frac{e^{-N_b^{\text{pred}}(\boldsymbol{\theta}, \boldsymbol{\alpha})}}{N_b^{\text{obs}}!}$$

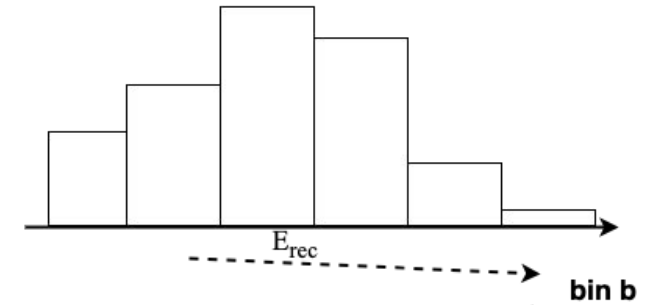
- The prediction is modeled as functions of the oscillation and nuisance parameters

$$N_b^{\text{pred}}(\boldsymbol{\theta}, \boldsymbol{\alpha}) \sim f^{\text{beam}}(\overrightarrow{\text{xsec}}, \overrightarrow{\text{b_flux}}, \overrightarrow{\text{corr_fd}})$$

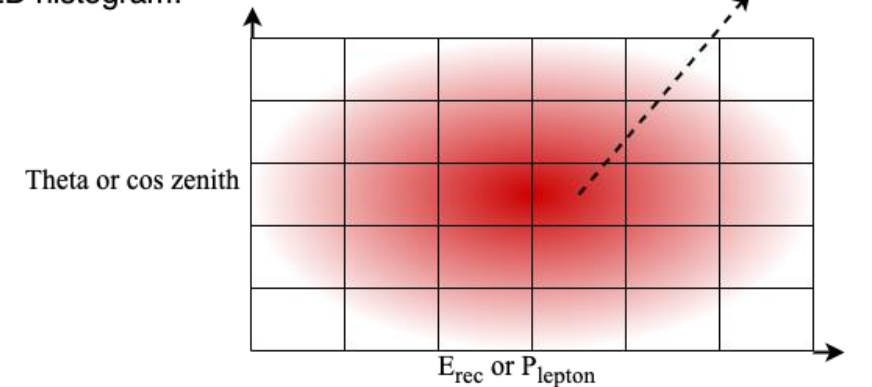
$$N_b^{\text{pred}}(\boldsymbol{\theta}, \boldsymbol{\alpha}) \sim f^{\text{atm}}(\overrightarrow{\text{xsec}}, \overrightarrow{\text{atm_flux}}, \overrightarrow{\text{corr_fd}})$$

Nuisance parameters
implemented in MaCh3

1D histogram:

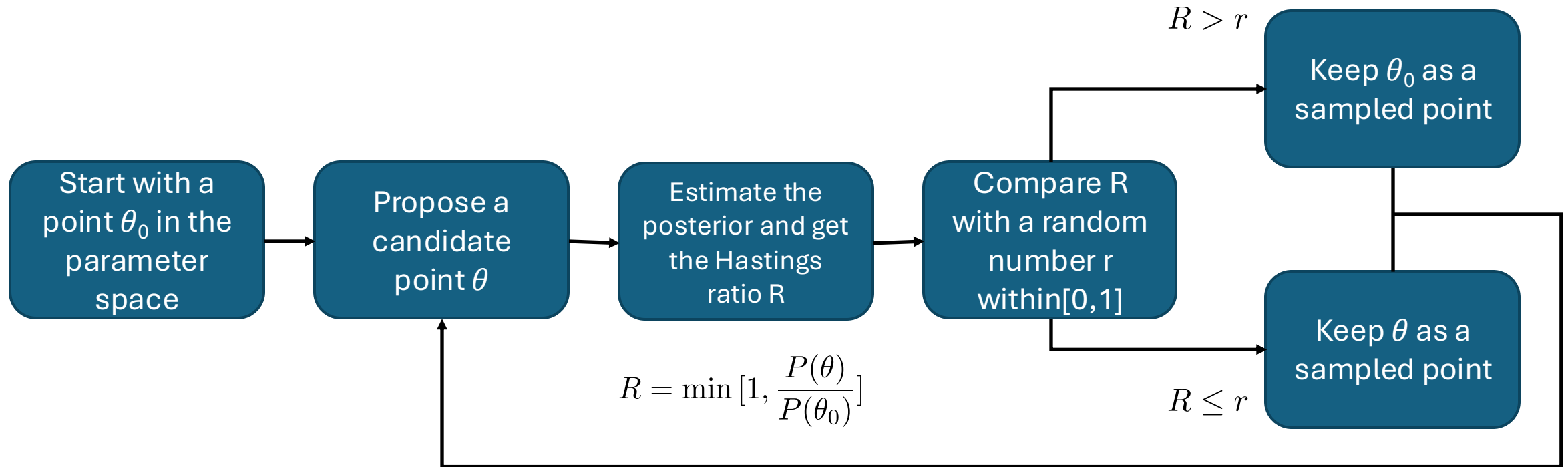


2D histogram:

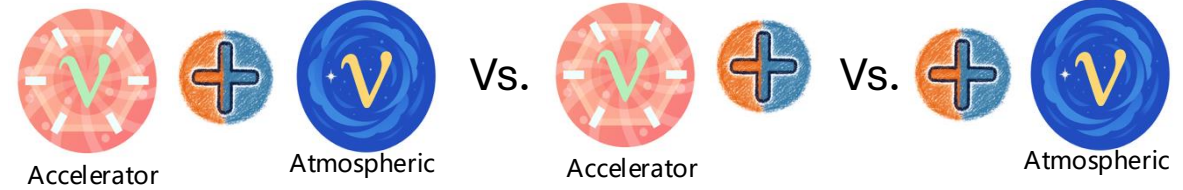


Metropolis-Hastings Algorithm for MCMC

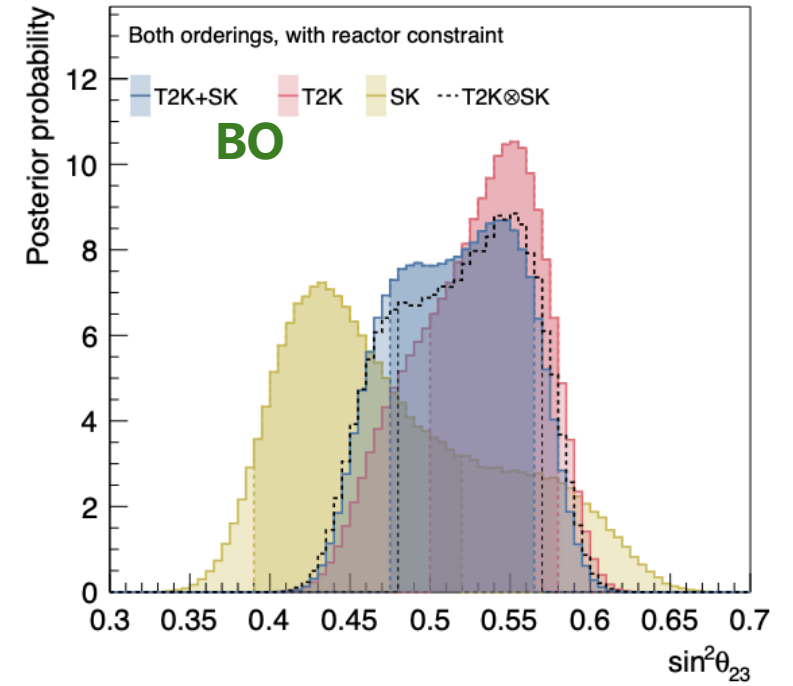
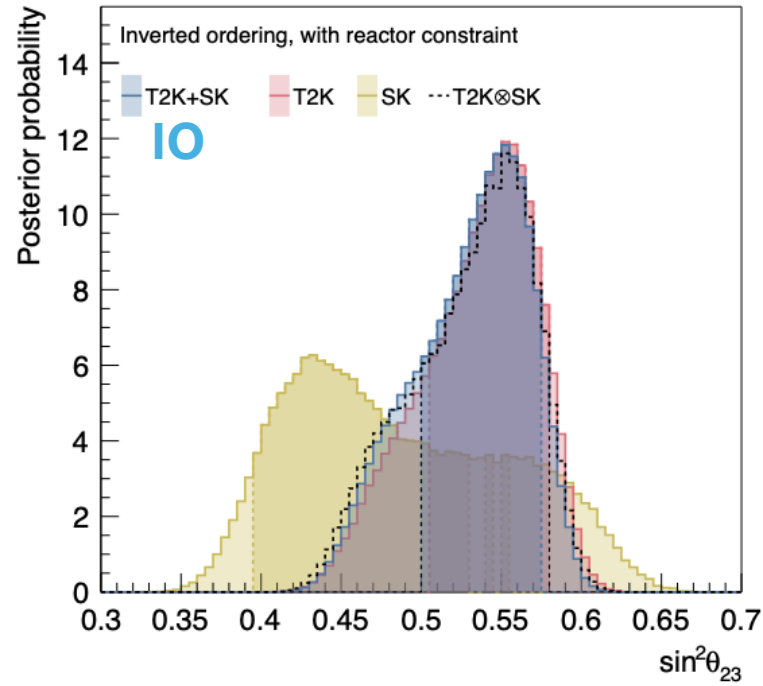
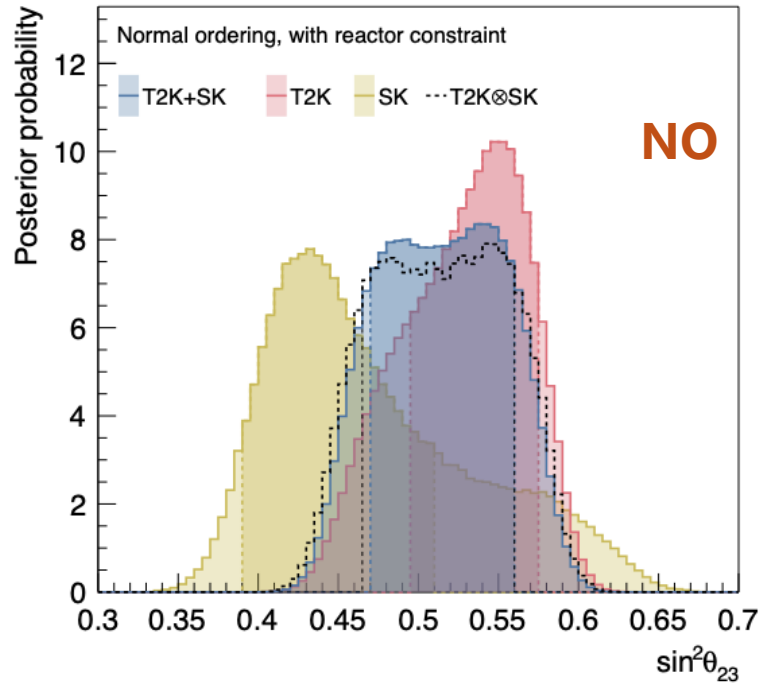
- a semi-random walk through the allowed parameter space



Cross fits comparison



$\sin^2 \theta_{23}$ Posterior Probability Distribution



T2K OA2020 detector systs

- The detector systematic model for T2K beam neutrino sample

FHC 1Re
fraction

FHC 1Rmu
fraction

RHC 1Re
fraction

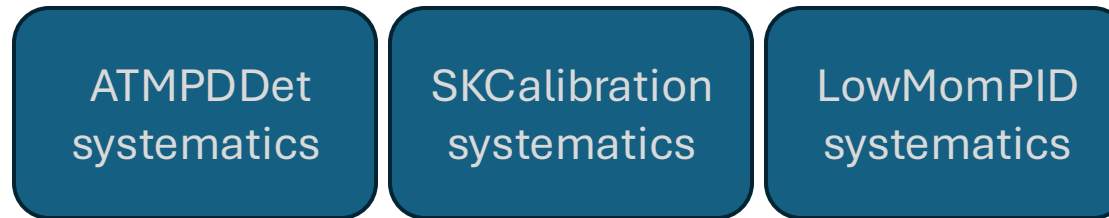
RHC 1Rmu
fraction

FHC 1Re1De
fraction

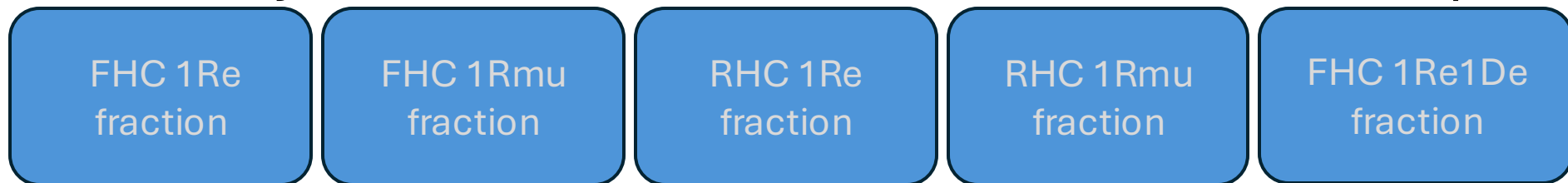
FHC, 1Re, Osc. ν_e , CC, $0 \leq E < 350$	FHC, 1R μ , ν_{μ} , CCQE-like, $0 \leq E < 400$	RHC, 1Re, Osc. ν_e , CC, $0 \leq E < 350$	RHC, 1R μ , ν_{μ} , CCQE-like, $0 \leq E < 400$	FHC, 1ReDe, Osc. ν_e , CC, $350 \leq E < 800$
FHC, 1Re, Osc. ν_e , CC, $350 \leq E < 800$	FHC, 1R μ , ν_{μ} , CCQE-like, $400 \leq E < 1100$	RHC, 1Re, Osc. ν_e , CC, $350 \leq E < 800$	RHC, 1R μ , ν_{μ} , CCQE-like, $400 \leq E < 1100$	FHC, 1ReDe, Osc. ν_e , CC, $800 \leq E < 1250$
FHC, 1Re, Osc. ν_e , CC, $800 \leq E < 1250$	FHC, 1R μ , ν_{μ} , CCQE-like, $1100 \leq E$	RHC, 1Re, Osc. ν_e , CC, $800 \leq E < 1250$	RHC, 1R μ , ν_{μ} , CCQE-like, $1100 \leq E$	FHC, 1ReDe, ν_{μ} , CC, $350 \leq E < 800$
FHC, 1Re, ν_{μ} , CC, $0 \leq E < 350$	FHC, 1R μ , ν_{μ} , CCnonQE	RHC, 1Re, ν_{μ} , CC, $0 \leq E < 350$	RHC, 1R μ , ν_{μ} , CCnonQE	FHC, 1ReDe, ν_{μ} , CC, $800 \leq E < 1250$
FHC, 1Re, ν_{μ} , CC, $350 \leq E < 800$	FHC, 1R μ , ν_e , CC	RHC, 1Re, ν_{μ} , CC, $350 \leq E < 800$	RHC, 1R μ , ν_e , CC	FHC, 1ReDe, Beam ν_e , CC, $350 \leq E < 800$
FHC, 1Re, ν_{μ} , CC, $800 \leq E < 1250$	FHC, 1R μ , NC	RHC, 1Re, ν_{μ} , CC, $800 \leq E < 1250$	RHC, 1R μ , NC	FHC, 1ReDe, Beam ν_e , CC, $800 \leq E < 1250$
FHC, 1Re, Beam ν_e , CC, $0 \leq E < 350$		RHC, 1Re, Beam ν_e , CC, $0 \leq E < 350$		FHC, 1ReDe, NC, $350 \leq E < 800$
FHC, 1Re, Beam ν_e , CC, $350 \leq E < 800$		RHC, 1Re, Beam ν_e , CC, $350 \leq E < 800$		FHC, 1ReDe, NC, $800 \leq E < 1250$
FHC, 1Re, Beam ν_e , CC, $800 \leq E < 1250$		RHC, 1Re, Beam ν_e , CC, $800 \leq E < 1250$		
FHC, 1Re, NC, $0 \leq E < 350$		RHC, 1Re, NC, $0 \leq E < 350$		
FHC, 1Re, NC, $350 \leq E < 800$		RHC, 1Re, NC, $350 \leq E < 800$		
FHC, 1Re, NC, $800 \leq E < 1250$		RHC, 1Re, NC, $800 \leq E < 1250$		

The Correlated Far Detector Uncertainties

- The detector systematic model for SK atmospheric neutrino sample

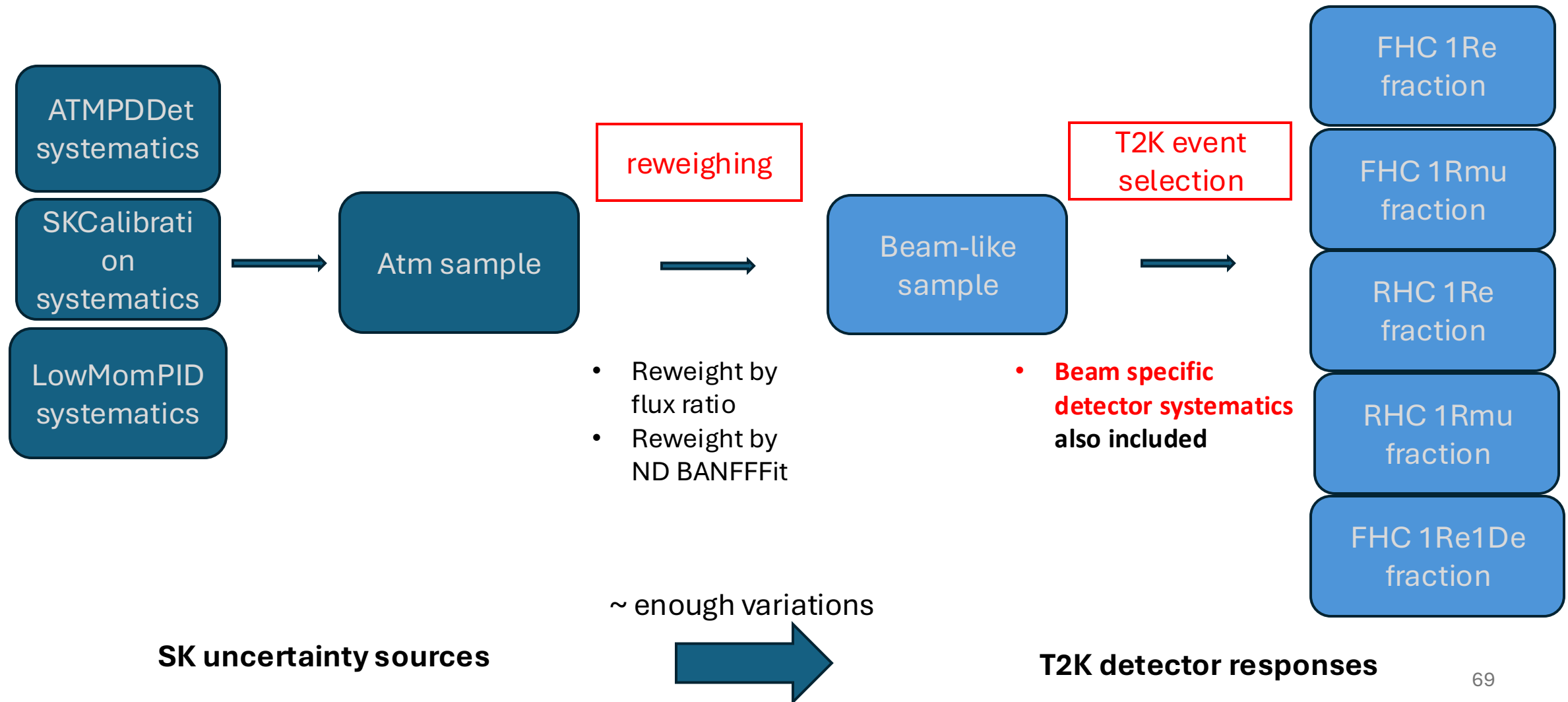


- The detector systematic model for T2K beam neutrino sample



- The correlation between the two models

The correlation of SK and T2K detector uncertainties



Likelihood incorporated with topology cuts

$$\mathbf{D} : \sim \left\{ E_{\text{rec}}, P_{\text{lep}}, \cos z, z, \text{RC}, P^{e/\mu}, P^{e/\pi^0}, P^{\mu/\pi^+} \right\}$$

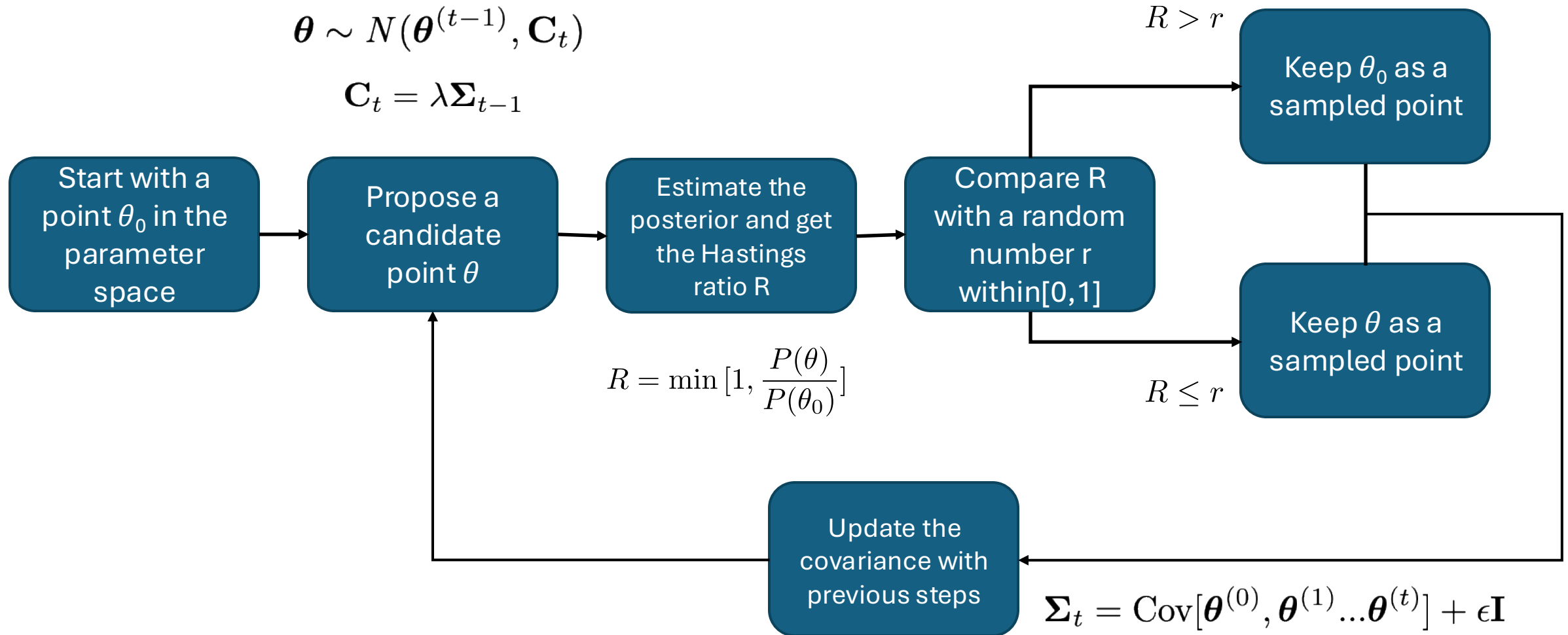
Sample
likelihood

$$\prod_i^{\text{events}} L(\mathbf{D}_i | \boldsymbol{\theta}, \boldsymbol{\alpha}) \approx \prod_b^{\text{bin}} L_b(N_b^{\text{obs}} | \boldsymbol{\theta}, \boldsymbol{\alpha}) L(\text{RC} | \{E_{\text{rec}}\}_b, \boldsymbol{\theta}, \boldsymbol{\alpha}) L(P^{e/\mu} | \{E_{\text{rec}}\}_b, \boldsymbol{\theta}, \boldsymbol{\alpha}) \\ \downarrow L(P^{e/\pi^0} | \{E_{\text{rec}}\}_b, \boldsymbol{\theta}, \boldsymbol{\alpha}) L(P^{\mu/\pi^+} | \{E_{\text{rec}}\}_b, \boldsymbol{\theta}, \boldsymbol{\alpha})$$

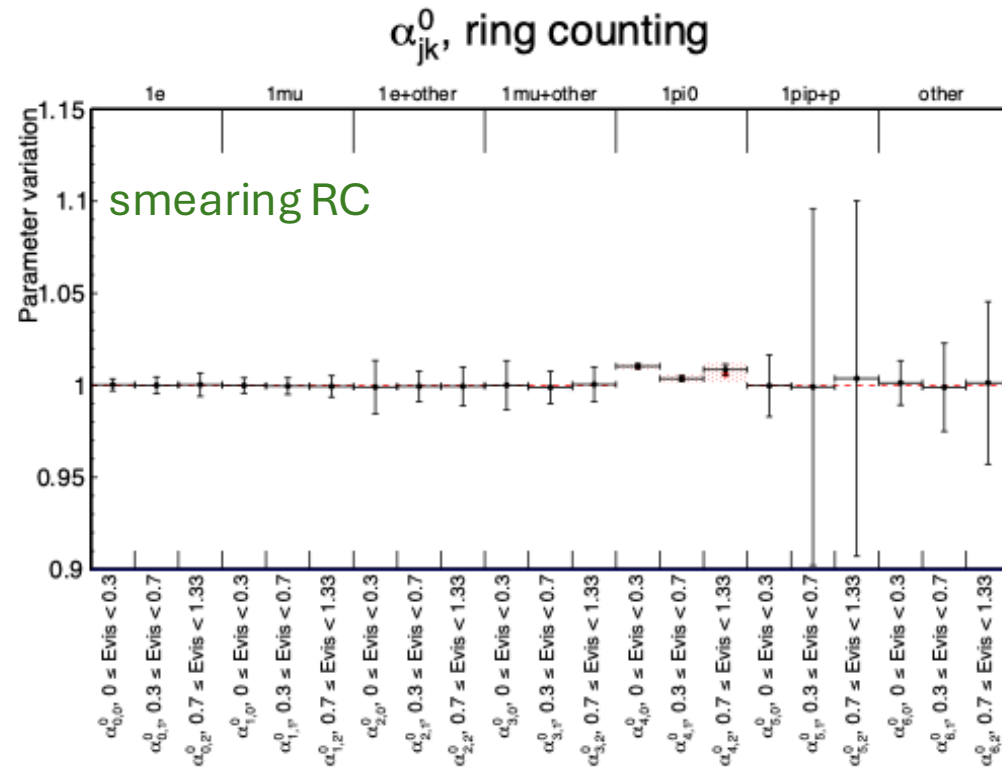
- Take the assumption that the reconstructed variables are uncorrelated or weakly correlated
- For example, the shape likelihood of ring counting parameter is

$$\prod_b^{\text{bin}} L(\text{RC} | \{E_{\text{rec}}\}_b, \boldsymbol{\theta}, \boldsymbol{\alpha}) = \prod_k^{E_{\text{vis}}} L(\text{RC} | E_{\text{vis}}^k, \boldsymbol{\theta}, \boldsymbol{\alpha}) \\ = \prod_k^{E_{\text{vis}}} \prod_l^{\text{Cut Val}} \left(\frac{N_{kl}^{\text{pred}}(\boldsymbol{\theta}, \boldsymbol{\alpha})}{\sum_l^{\text{Cut Val}} N_{kl}^{\text{pred}}(\boldsymbol{\theta}, \boldsymbol{\alpha})} \right)^{N_{kl}^{\text{obs}}}$$

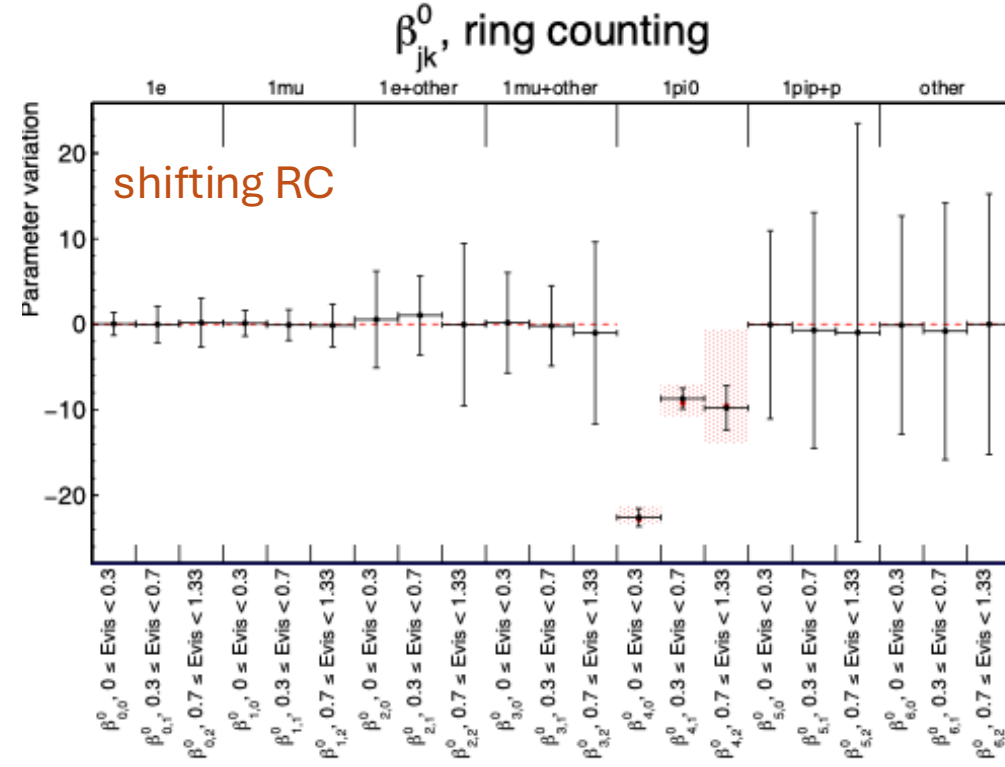
Adaptive Metropolis-Hastings Algorithm



Joint Asimov fit results: Smearing/Shifting Param.

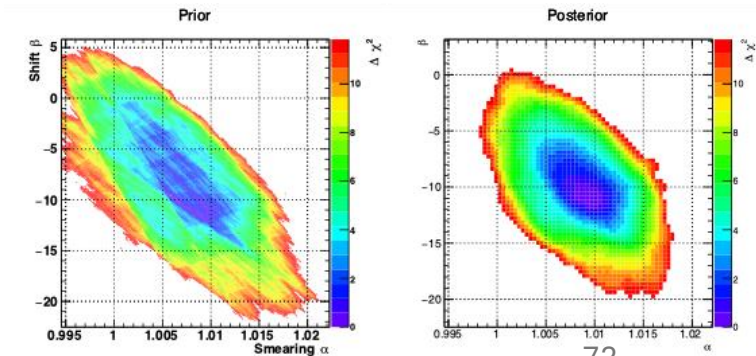
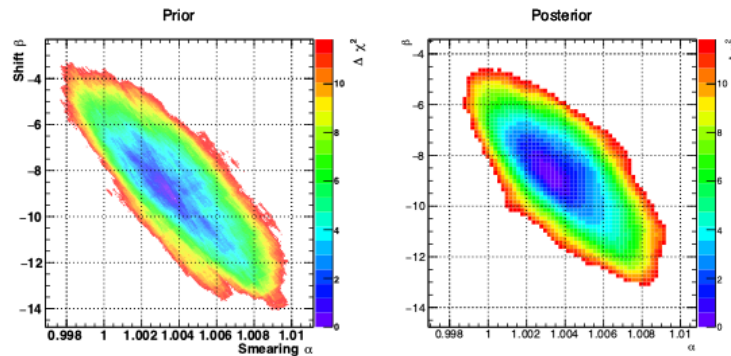
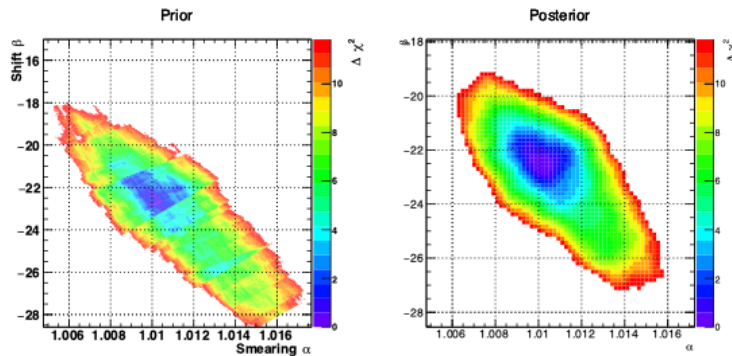


Evis < 0.3 GeV

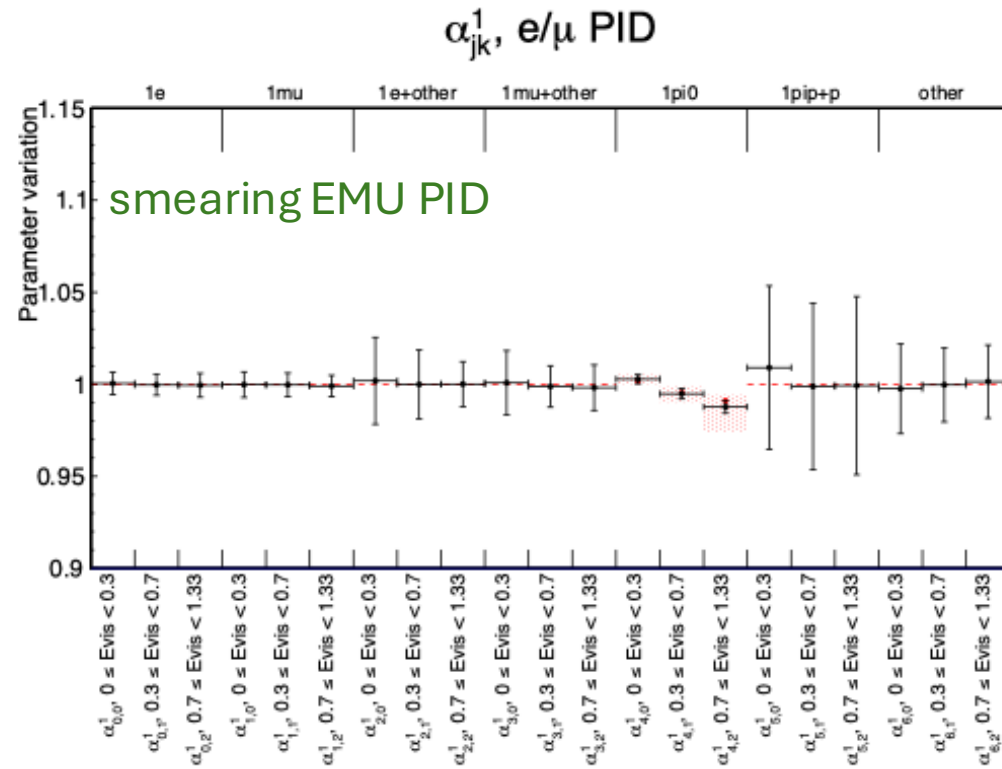


0.3 < Evis < 0.7 GeV

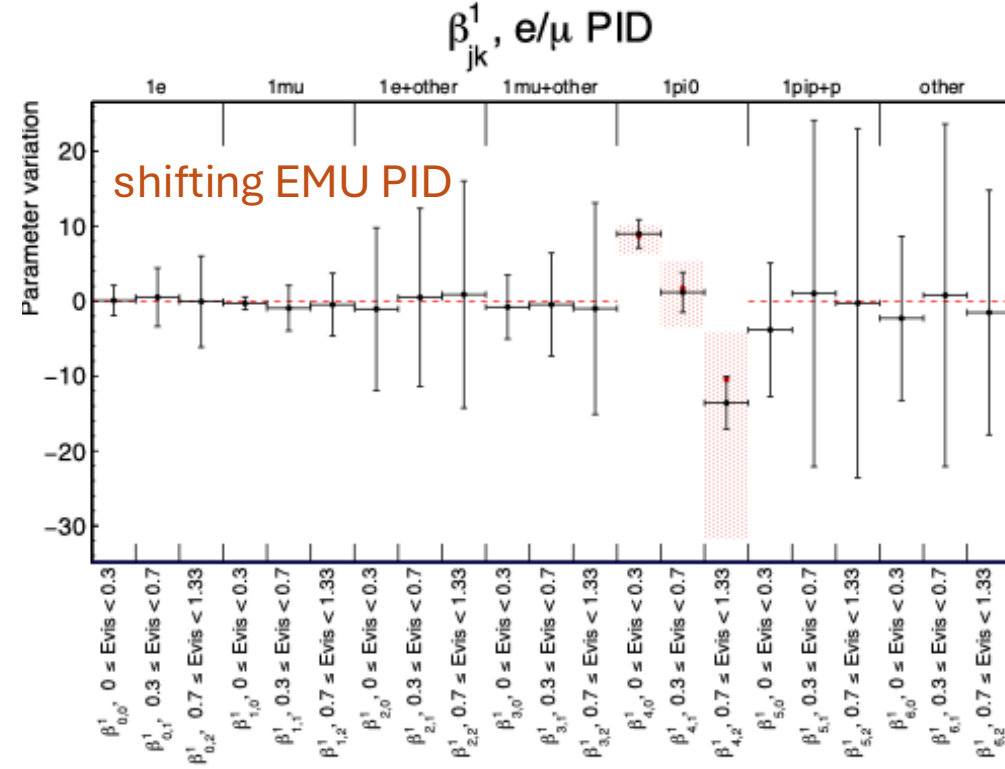
0.7 < Evis < 1.33 GeV



Joint Asimov fit results: Smearing/Shifting Param.

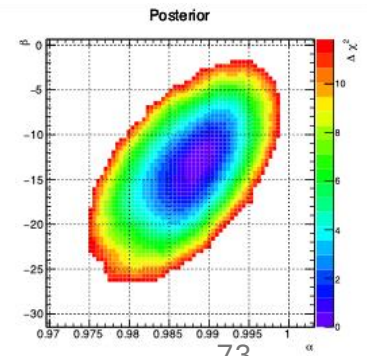
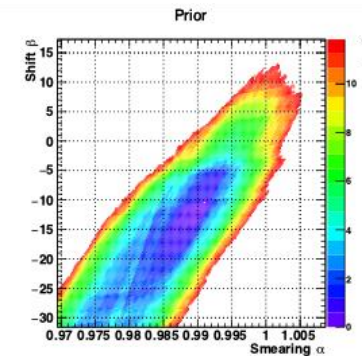
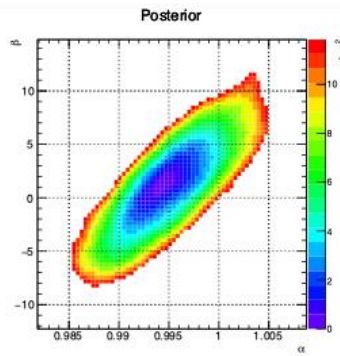
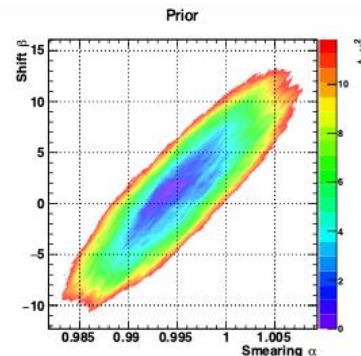
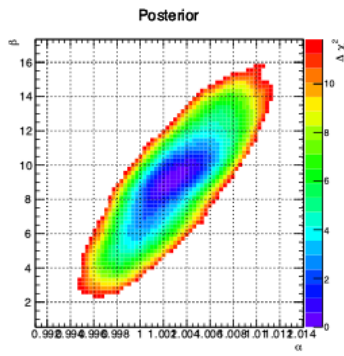
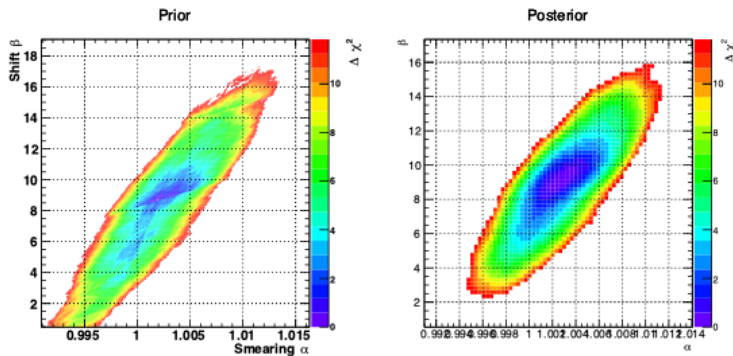


Evis < 0.3 GeV

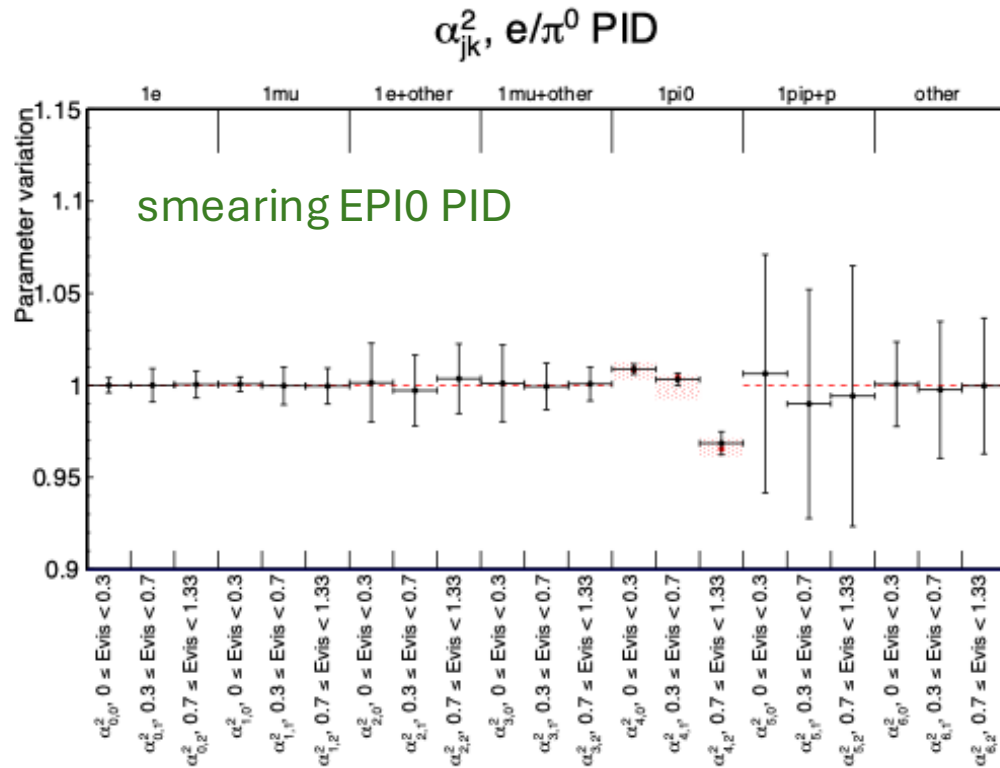


0.3 < Evis < 0.7 GeV

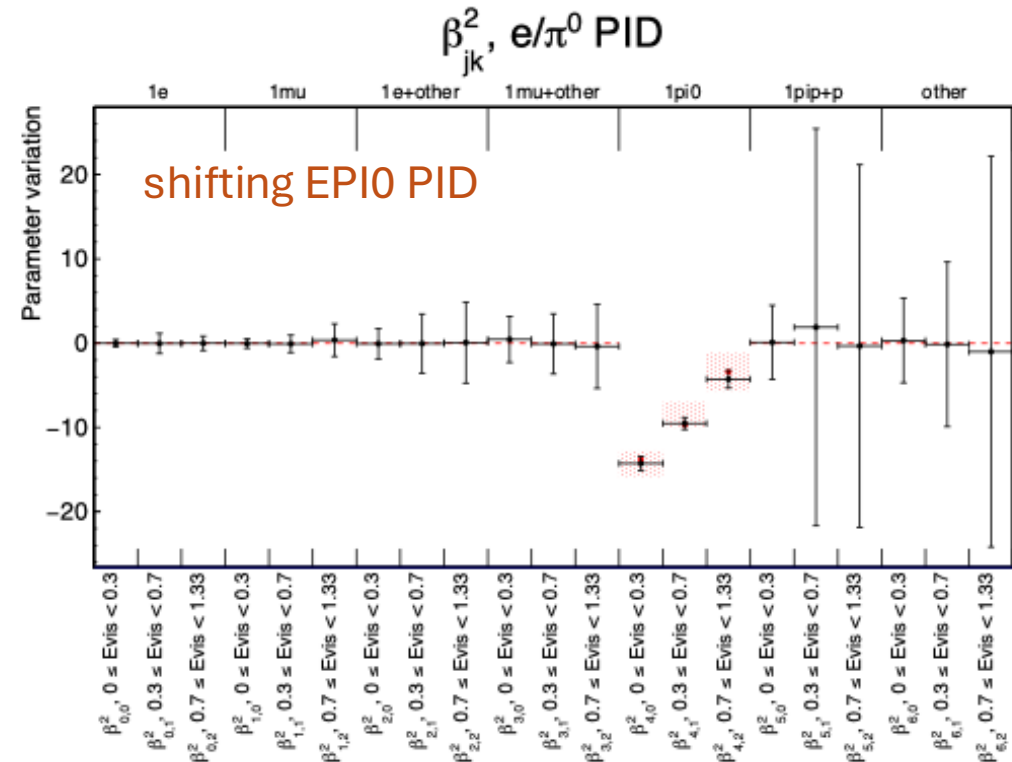
0.7 < Evis < 1.33 GeV



Joint Asimov fit results: Smearing/Shifting Param.

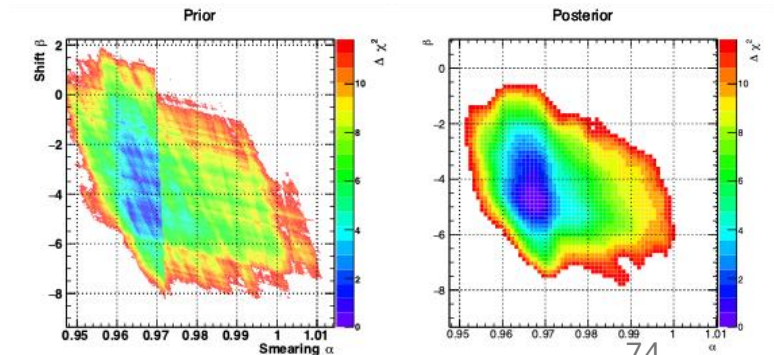
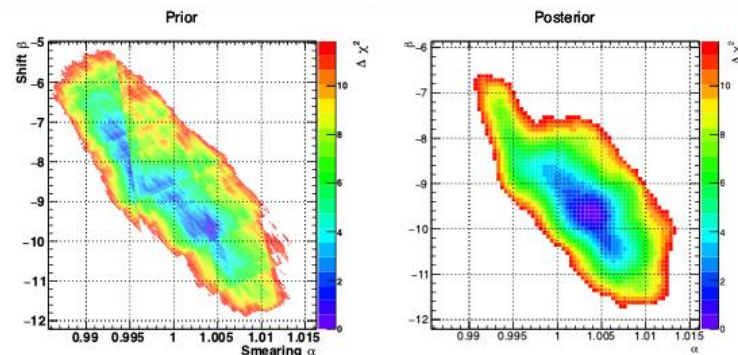
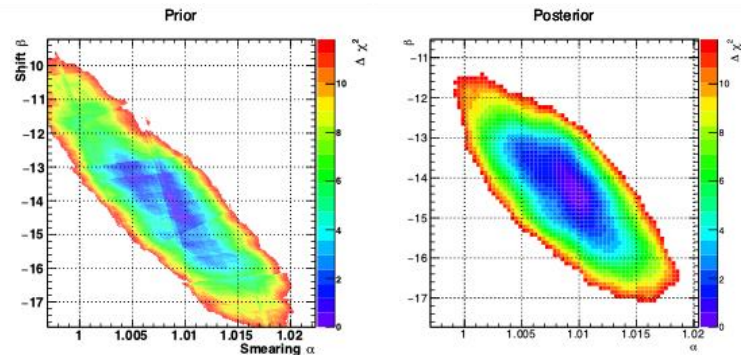


Evis < 0.3 GeV

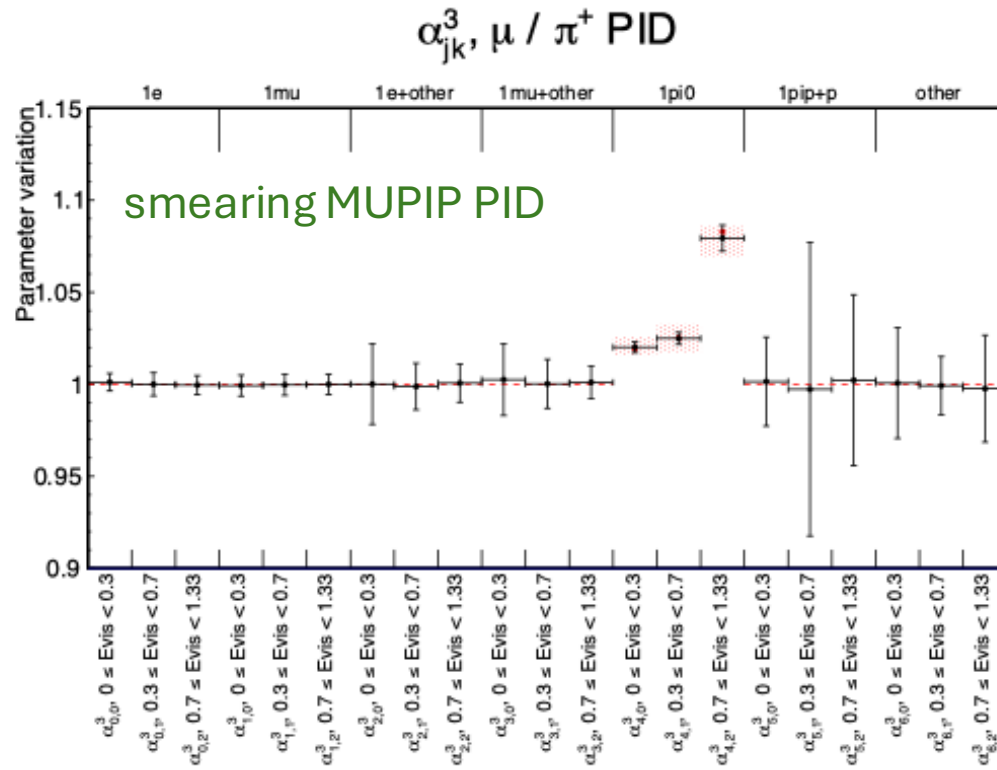


0.3 < Evis < 0.7 GeV

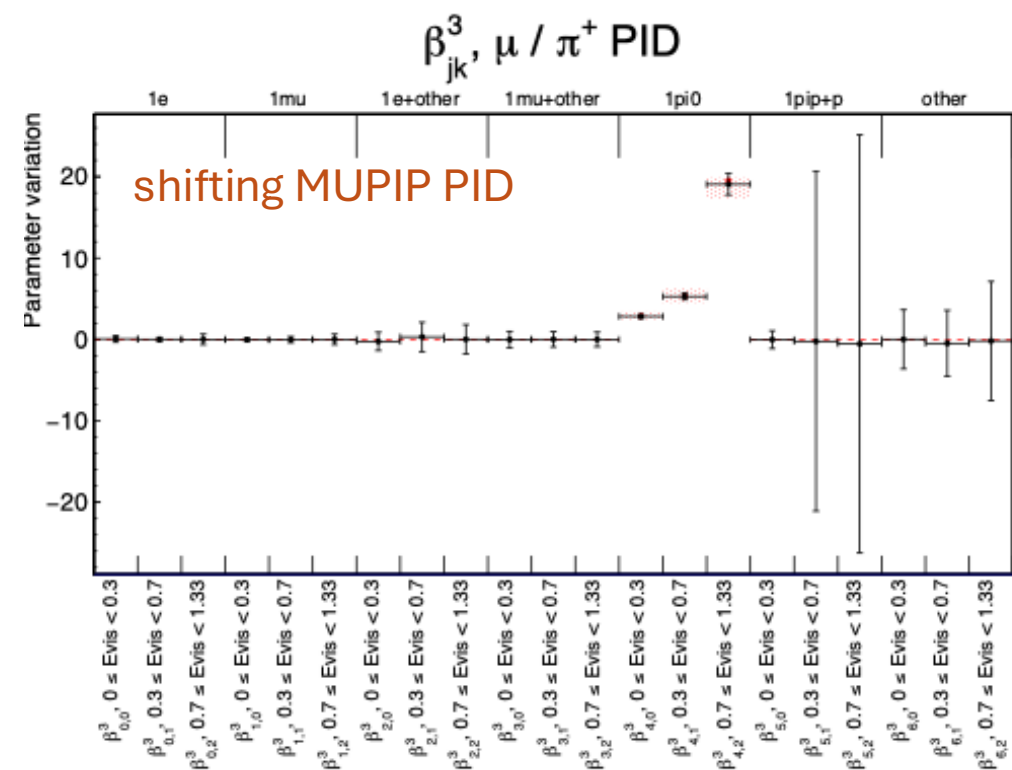
0.7 < Evis < 1.33 GeV



Joint Asimov fit results: Smearing/Shifting Param.

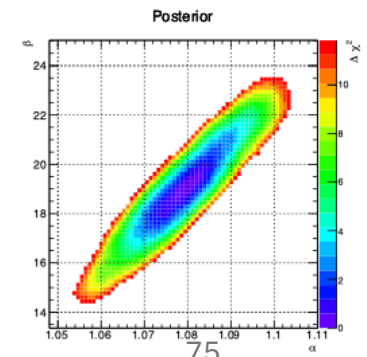
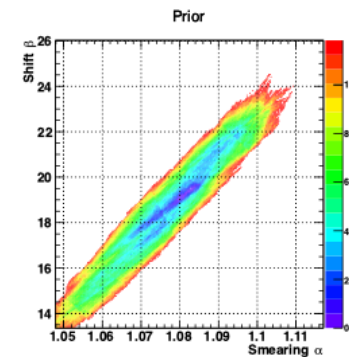
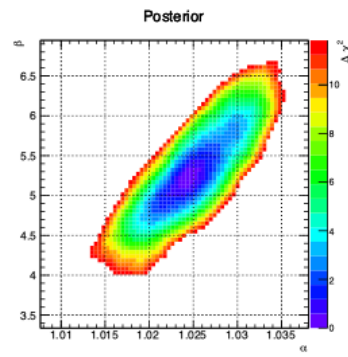
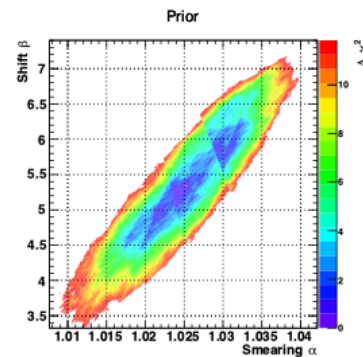
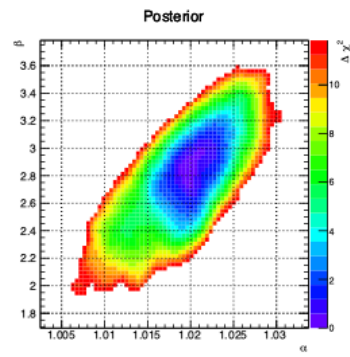
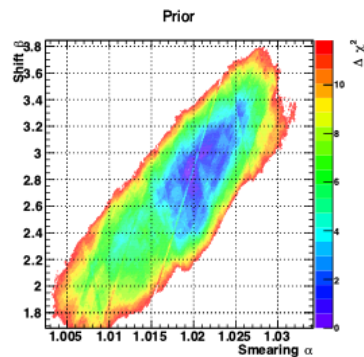


Evis < 0.3 GeV



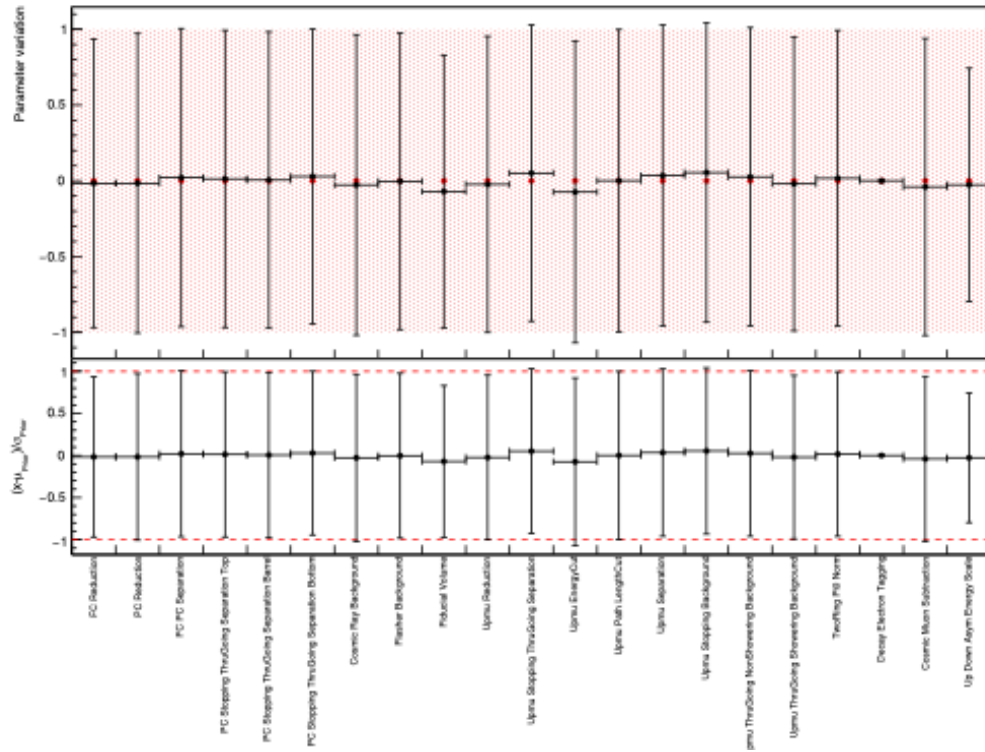
0.3 < Evis < 0.7 GeV

0.7 < Evis < 1.33 GeV

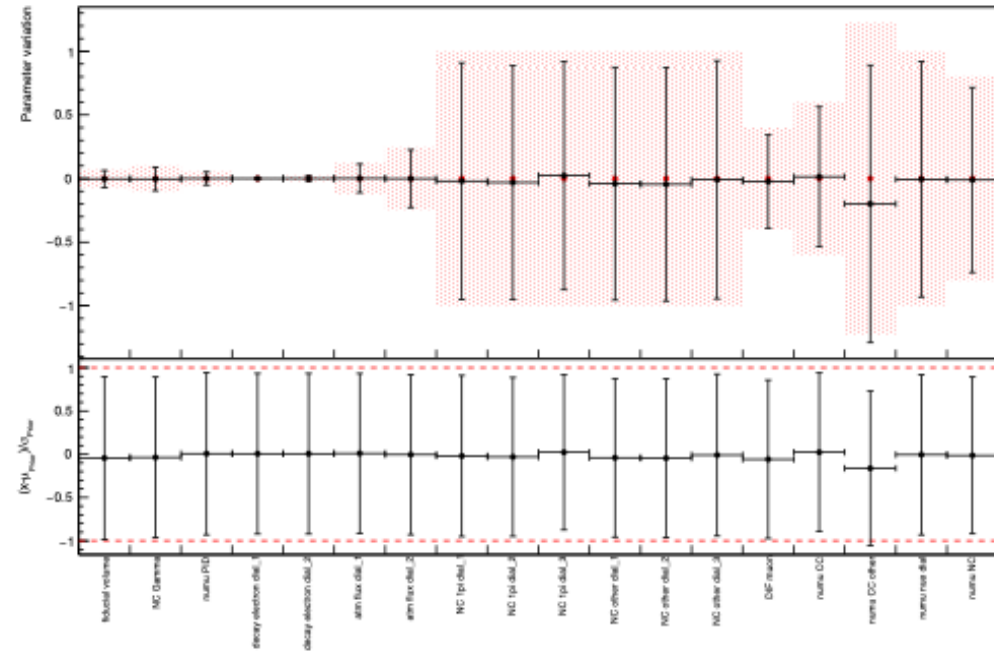


Joint Asimov fit results: Supplementary Detector Param.

SK: SKCalibration Sysys



T2K: SKDetBeam Sysys



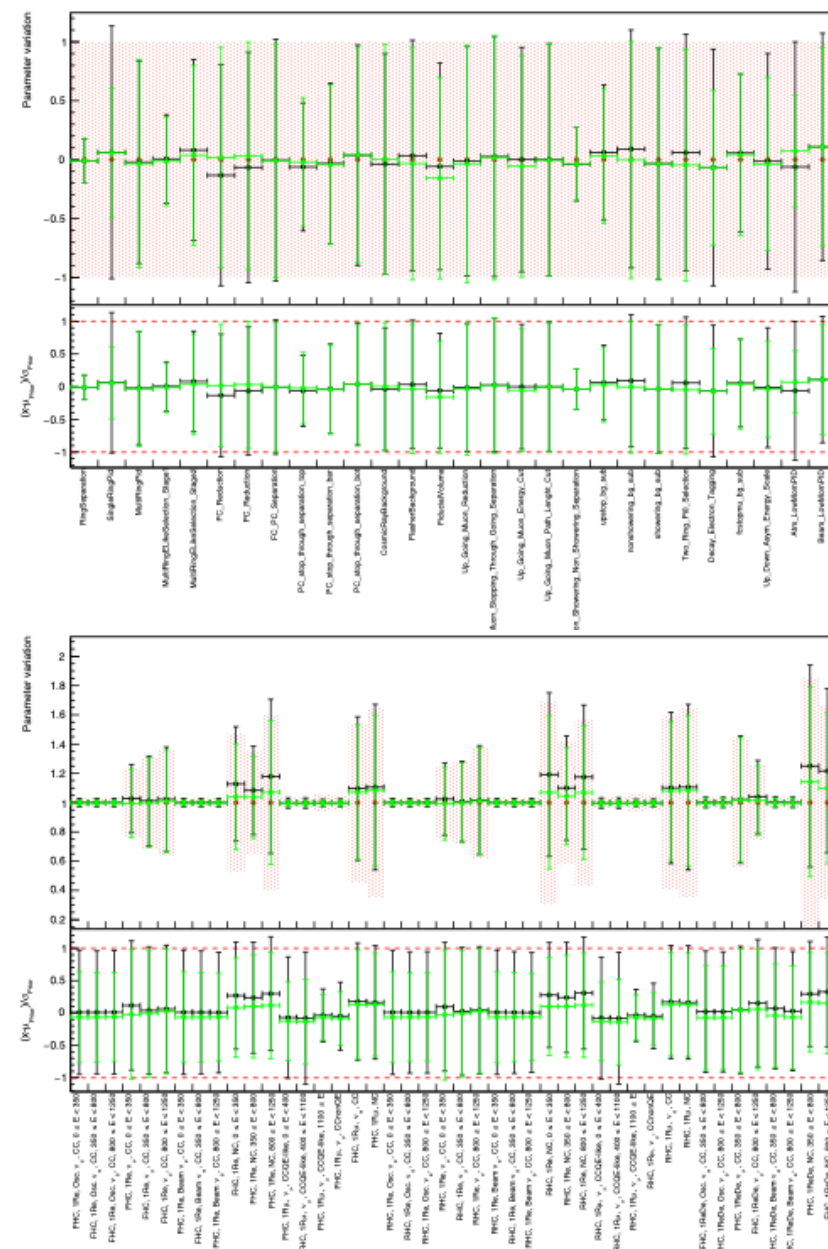
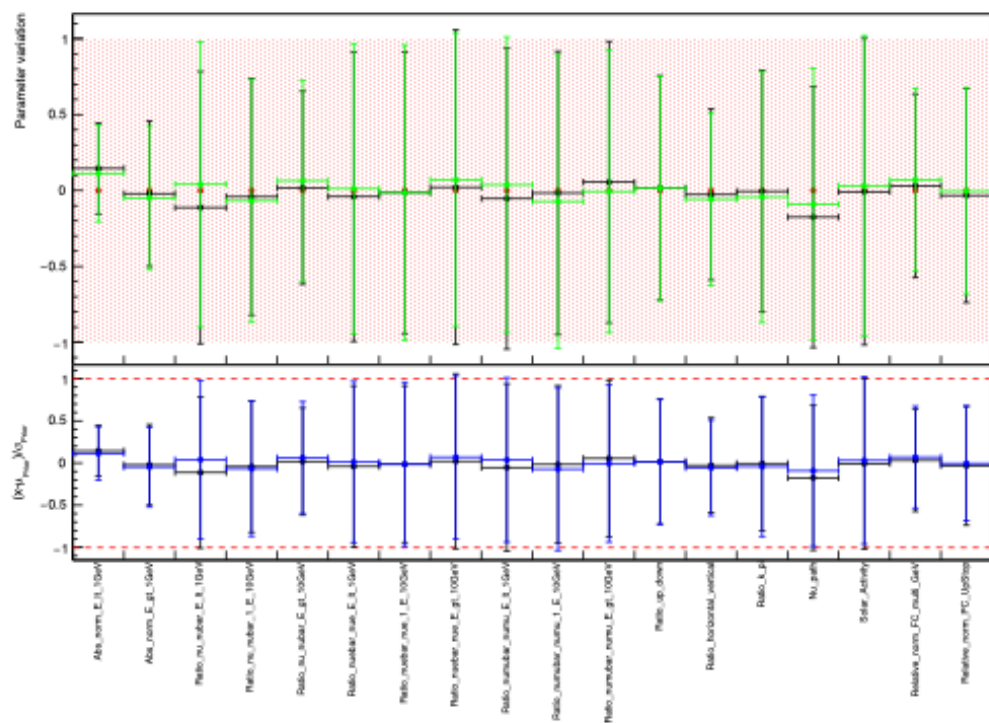
Joint Asimov fit results: Other nuisance Param.

Black: alternative detector model

Green: standard

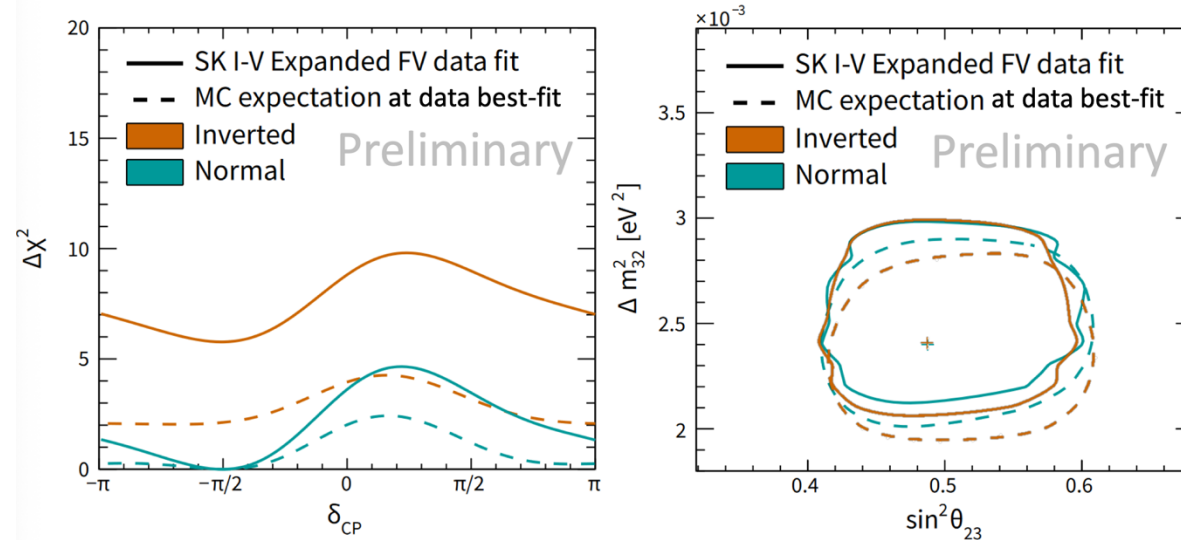
Correlated FD:

Atm Flux



SK recent results:

Oscillation Measurements (SK only)



SK atmospheric neutrino data favors:

- maximal mixing
- $\delta_{CP} \approx -\frac{\pi}{2}$
- NO ($\Delta\chi^2 = 5.8$)

930 bins	χ^2	δ_{CP}	$\sin^2\theta_{23}$	Δm^2_{23}
SK NO	1000.42	4.71	0.49	$2.4 \times 10^{-3} \text{ eV}^2$
SK IO	1006.19	4.71	0.49	$2.4 \times 10^{-3} \text{ eV}^2$

*Results on MO and δ_{CP} exceed sensitivity.

$$\sin^2\theta_{13} = 0.0220 \pm 0.0007$$

2022/06/02

Linyan WAN @ NEUTRINO 2022

10

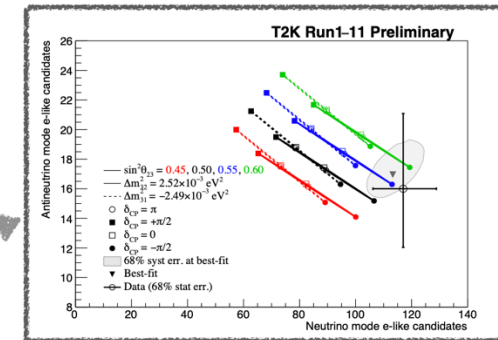
<https://indico.kps.or.kr/event/30/contributions/888/attachments/221/440/Linyan%20Wan.pdf>

T2K recent results:

Oscillation analysis results

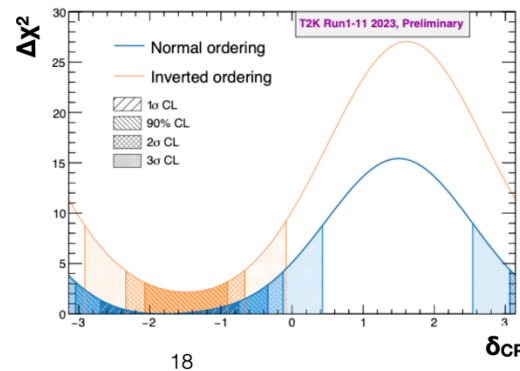


Sample	$\delta_{CP}=-\pi/2$	$\delta_{CP}=0$	$\delta_{CP}=\pi/2$	$\delta_{CP}=\pi$	Data
ν -mode 1R μ	417.2	416.3	417.1	418.2	357
ν -mode MR	123.9	123.3	123.9	124.4	140
$\bar{\nu}$ -mode 1R μ	146.6	146.3	146.6	147.0	137
ν -mode 1Re	113.2	95.5	78.3	96.0	102
$\bar{\nu}$ -mode 1Re+d.e.	10.0	8.8	7.2	8.4	15
$\bar{\nu}$ -mode 1Re	17.6	20.0	22.2	19.7	16

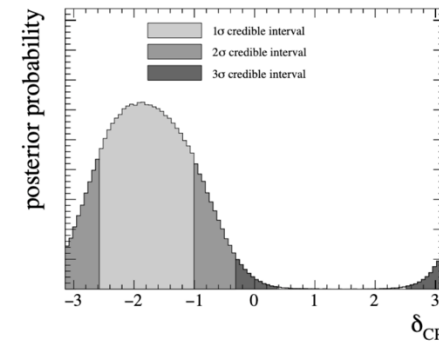


- Preference for $\delta_{CP} \sim -\pi/2$ but CP conserving values are within the 2σ interval

D. Carabadjac poster



Credible intervals marginalized over both hierarchies



T2K recent results:

Mass ordering and θ_{23} octant

- Slight preference for normal ordering and upper octant but none of them is significant
 - Bayes factor NO/IO = 3.3
 - Bayes factor $(\theta_{23} > 0.5) / (\theta_{23} < 0.5) = 2.6$

	$\sin^2 \theta_{23} < 0.5$	$\sin^2 \theta_{23} > 0.5$	Sum
NH ($\Delta m_{32}^2 > 0$)	0.23	0.54	0.77
IH ($\Delta m_{32}^2 < 0$)	0.05	0.18	0.23
Sum	0.28	0.72	1.00

